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The Event Argument and the Semantics of Verbs

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A note on these chapters

The chapters posted here are the first 4 chapters of a book manuscript entitled **The Event Argument and the Semantics of Verbs** that I have been working on for more than 10 years. The remaining chapters are close to completion as well, and I will post them as they become available. The chapters posted so far deal with the relation between verbs and their arguments (chapters 1 to 3) and with verbal plurality (chapter 4). The remaining chapters address issues related to voice alternations (actives versus passives, middles, reflexives) and certain transitivity alternations, in particular resultatives. An earlier chapter on adjectival passives (“Building Statives”) appeared in the 2000 Proceedings of the Berkeley Linguistics Society. Another earlier chapter (“Telicity and the Meaning of Objective Case”) will be published separately in a forthcoming book on the **Syntax of Tense** edited by Jacqueline Guéron and Jacqueline Lecarme (MIT Press) and will not be included in the present book.

I will keep changing the current manuscript. Any feedback will be greatly appreciated.

Chapter 1

Verb meanings and argument structure

When children have acquired the meaning of a verb, they know the possible states or events it describes. But is that all there is to know about the meaning of a verb? When verbs appear in phrases and sentences, they rarely come alone. Mastering the use of verbs requires knowledge of the company they ask for. Most verbs come with arguments, some obligatory, others optional. Do we know what arguments a verb takes when we know the verb's meaning?

We will see in the chapters to come that in order to answer this question, we have to distinguish between the meanings of the verb stems themselves and the semantic contribution of verbal inflection. With many detours I will explore the sometimes surprising consequences of the idea that voice inflection is responsible for the introduction of external arguments, and that as a result, external arguments are not true arguments of their verbs. All by itself, this proposal is by now not news anymore. But so far, it hasn't developed much beyond a mere idea in linguistic theorizing. We still do not know how external arguments are matched up with the right kind of verbs, for example, nor what the repercussions on theories of argument structure are. Shouldn't such an idea at least lead to an explanation of the major voice alternations we observe - the different kinds of passives and middles, that is? Most importantly, crucial properties of verbal projections are now expected to be different, $\square$ and dramatically so. For example, there is no longer a problem with quantifier phrases in object position. That oldest of all puzzles in logical

semantics has quietly disappeared. Subjects and direct objects are now expected to have sister constituents of the same semantic types. No longer can a type mismatch force direct objects to leave their base positions, then. If direct objects move to higher positions at all, a different force must be invoked to drive them. That kind of conceptual shift will take time to get used to. Too much capital has already been invested on apparent type mismatches of quantifier phrases in direct object positions. Too many apparent consequences have already been derived.

In syntax, a verb's arguments are tracked by verbal agreement morphology. Chomsky 2000 has suggested that verbal agreement features provide syntactic scaffolding for meaningful inflectional heads. In the context of the present essay, Chomsky's proposal would make us expect a tight connection between voice inflection and overt or non-overt verbal agreement morphology. What exactly would force such a connection? And what would it imply? Is it true that verbal agreement features merely provide meaningless syntactic scaffolding for notionally significant inflectional heads? Or could they have meanings of their own?

If we want to even start thinking about such issues, we have to be willing to think about the meanings of verbs in non-traditional ways. The verbs we see – surrounded by their arguments and with all their inflections tucked on – might not be the verbs that are ultimately fed to the semantic interpretation component. As semanticists, we have to find a way to unveil the meanings of bare verb stems. Only then do we have a chance to gauge the semantic contribution of verbal inflection. We have to solve what Sandro Zucchi called

“the problem of indirect access”¹. If we, like Zucchi were interested in the semantics of English tense and aspect, for example, we would have to formulate hypotheses about the meaning of uninflected, tense- and aspectless forms, even though we might never encounter those forms in reality. We have to develop experimental techniques, then, that allow us to infer the properties of bare verb roots. Some of the chapters in this book will rely on the use of such techniques. They will allow us to observe the properties of verbs during the very early stages of a syntactic derivation, crucially before any functional structure is built. We’ll find that during those early stages, verbs still lack external arguments.

The first formal semanticists inherited their beliefs about verb denotations from their logician ancestors. Events had no part in those beliefs. The denotation of a verb like **purchase**, for example, was taken to be a binary relation between individuals. The relation would be true of the pair consisting of us and those slippers, for example, just in case we purchase them. According to this view, information about the number and kind of verb arguments is part of a verb’s denotation, but the traditional insight that verbs describe events or states is gone. The importance of events and states for the semantics of verbs was eventually given theoretical recognition in the work of Donald Davidson. With the return of states and events, the question whether or not verb denotations include information about the number and kind of a verb’s arguments came back as well. It was now no longer true that, by their very nature, verb denotations had to carry all information about a verb’s argument structure.

1 . Zucchi 1999.

According to Davidson 1967, a sentence like (1) below makes a statement about the existence of a past event which is a purchase of those slippers by us, and which takes place in Marrakesh.

(1) **We purchased those slippers in Marrakesh.**

The logical form of (1) would be (1'):

(1') $\exists e[\text{purchase}(\text{those slippers})(\text{we})(e) \ \& \ \text{in}(\text{Marrakesh})(e)]$.

In (1'), 'purchase' is a three-place predicate. Apart from an event argument, it has an argument denoting the agent and another one denoting what was purchased, the 'theme' or 'object' of the purchase. That is, the subject and the direct object in (1) correspond to arguments of the main predicate in the logical representation (1'), while the locative argument 'Marrakesh' is introduced by a secondary predicate, the preposition 'in'. The way Davidson reconciled the traditional view that verbs describe events or states with the logician's way of thinking about verb denotations is by adopting the logician's view and add an extra argument, the event argument. As a result, verb meanings determine a verb's arguments while also characterizing a set of events or states. Bringing events back into semantic theory immediately solved some longstanding puzzles with adverbial modification. Since then,

events have been shown crucial for the semantics of perception reports, causal constructions, aspectual operators, and plurality².

Most of the past successes of event based analyses of linguistic phenomena do not depend on Davidson's particular view on the relation between verbs and their arguments. Davidson's distinction between arguments and adjuncts was questioned by Castañeda right after Davidson's talk³, and has been given up in the work of Parsons⁴. Both Castañeda and Parsons let agent and theme arguments be introduced by independent predicates as well. Consequently, the verbs themselves express mere properties of events or states. For Parsons, the independent predicates are two-place predicates denoting general thematic relations. On this proposal, inspired by Panini and Fillmore's case grammar ⁵, (1) has the logical form (1''):

(1'') $\exists e[\text{purchase}(e) \ \& \ \text{agent}(\text{we})(e) \ \& \ \text{theme}(\text{those slippers})(e) \ \& \ \text{in}(\text{Marrakesh})(e)]$

(1'') says that there is an event that is a purchase, whose agent is us, whose theme are those slippers, and which takes place in Marrakesh. Some terminology will be useful. (1') uses what Dowty 1989 calls the 'ordered-

2. Higginbotham 1983; Hinrichs 1985; Parsons 1985, 1990; Bach 1986; Krifka 1986, 1987, 1992, 1998; Schein 1986, 1993; Lasnik 1988, 1990, 1995, 1998; Landman 1996, 1998, 2000, and many others.

3. Castañeda 1967.

4. Parsons 1990 and earlier work.

5. Fillmore 1968.

argument method' for the association of arguments with their verb. In (1") we see what Dowty labels the 'neo-Davidsonian method'. On the neo-Davidsonian method, verbs describe events or states and arguments are associated with their verbs via secondary predicates that denote general thematic relations like 'agent of', 'theme of', etc.

Which mode of argument association is right for verbs in natural languages? Are there empirical reasons for picking one or the other option? Before trying to answer these questions, we have to reflect on what kind of questions we are asking. The theories of Davidson and Parsons are meant to be theories of logical form, where a logician's or philosopher's logical form is not the same as a syntactician's Logical Form or LF, a level of syntactic representation. For a logician or philosopher, the logical form of a sentence is an expression of a logical language that captures those features of the sentence that account for certain types of inferences. Neglecting differences of opinion relating to assumptions about the psychological reality of logical representations, Davidson's and Parsons' logical forms are akin to the semantic or conceptual structures of Bierwisch, Jackendoff or Levin and Rappaport⁶.

Parsons 1995 emphasizes that the theory presented in Parsons 1990 is a "proposal for the logical forms of sentences, unsupplemented by an account of how those forms originate by combining sentence parts"⁷. In other words, Parsons' theory is a theory of logical form that is not committed to particular claims about argument association in the syntax. Scholars who agree that a

6. See Bierwisch 1983, 1989, Jackendoff 1983, 1990, Levin & Rappaport-Hovav 1996, Rappaport-Hovav & Levin 1998, Speas 1990, Wunderlich 1997, and many others.

7. Parsons 1995, 650.

verb's arguments are associated by the ordered argument method in the syntax, for example, might still disagree about the mode of association for the arguments of the logical or conceptual counterpart of that very same verb. Here is an illustration of what such a disagreement would be about.

Quarrels without syntactic consequences		
a.	purchase	$\lambda x \lambda y \lambda e [\text{purchase}(e) \ \& \ \text{theme}(x)(e) \ \& \ \text{agent}(y)(e)]$ 'e is a purchase with theme x and agent y'
b.	purchase	$\lambda x \lambda y \lambda e [\text{purchase}(x)(e) \ \& \ \text{agent}(y)(e)]$ 'e is a purchase of x with agent y'
c.	purchase	$\lambda x \lambda y \lambda e [\text{purchase}(y)(e) \ \& \ \text{theme}(x)(e)]$ 'e is a purchase by y with theme x'
d.	purchase	$\lambda x \lambda y \lambda e [\text{purchase}(x)(y)(e)]$ 'e is a purchase of x by y'

Figure 1

In the table above, I displayed four different lexical meaning assignments for the English verb **purchase**⁸. All four proposals agree that **purchase** is a three-place predicate in the syntax. The event argument is the highest argument, the agent argument comes next, and the theme argument is at the bottom. What distinguishes the four views is their assumptions about the arguments of the logical-conceptual predicate 'purchase'. For a classical

8. If necessary, I will use boldface to distinguish between the English verb **purchase** and its logical or (lexical) conceptual counterpart 'purchase'.

Davidsonian, it would be a three-place predicate, as in option (d). For a neo-Davidsonian, it would be a predicate with just one argument, the event argument, as in option (a). Options (b) and (c) have mixed modes of association in logical-conceptual structure, a theoretical possibility. In all four cases, the syntactic behavior of the verb is expected to be the same, assuming that syntactic operations have no access to logical-conceptual representations.

Following up on earlier work (Kratzer 1994, 1996), I will argue in this essay that some neo-Davidsonian argument association is present in the syntax of verbs. Not all of a verb's arguments are syntactically associated by the ordered argument method. Most importantly, I am going to present evidence that agent arguments and other external arguments are associated by the neo-Davidsonian method in the syntax. On this proposal, the English (inflectionless) verb (stem) **purchase**, for example, is a predicate without external argument. Here are the major options for lexical meaning assignments that remain after the agent argument has gone:

Severing the external argument from its verb		
a.	purchase	$\lambda x \lambda e [\text{purchase}(e) \ \& \ \text{theme}(x)(e)]$ 'e is a purchase with theme x'
b.	purchase	$\lambda x \lambda e [\text{purchase}(x)(e)]$ 'e is a purchase of x'
c.	purchase	$\lambda e [\text{purchase}(e)]$ 'e is a purchase'

Figure 2

On proposals (a) and (b), syntactic argument association is asymmetric. The theme argument is, but the agent argument is not a syntactic argument of the verb stem **purchase**. On proposal (c), **purchase** only has an event argument. Proposal (c) might be right for certain types of verbs - those whose transitivity is derived in the syntax - but **purchase** doesn't seem to be one of those.

As for proposal (a) in figure 2, the question arises whether there is ever any genuine lexical decomposition at all, that is, decomposition of lexical items at logical-conceptual structure that is not matched by parallel decomposition in the morphology or syntax. The issue is controversial and divides the major approaches to lexical semantics. Hale and Keyser 1993, for example, argue for a lexical theory where all decomposition of lexical items is reduced to syntactic decomposition. What this might mean is stated most radically in Hale and Keyser 1999, 2002, where the argument structure of a verb is

literally identified with the syntactic structure it projects. An approach like Hale and Keyser's is incompatible with possibility (a) in figure 2 above, which has genuine thematic decomposition at logical-conceptual structure. Interestingly, in the following chapters, we'll see evidence that speaks against the existence of a general, all-purpose, thematic role like 'theme' or 'object', hence against possibility (a) as spelled out in figure 2. All by itself this does not mean that there couldn't be other, more specialized, thematic roles that introduce direct object arguments at logical-conceptual structure, of course. That is, all by itself, evidence against a general 'theme' or 'object' relation does not imply that there couldn't be any kind of logical-conceptual predicate introducing theme arguments in a neo-Davidsonian way. So other possible object relations will eventually have to be considered, too, and I will start to do so in chapter 3.

Compelling evidence for properties of logical-conceptual representations is very hard to obtain, a point that Jerry Fodor and Ernest Lepore have been making over the years⁹. I don't think the task is impossible, though. For example, I consider Schein's Argument, which I will discuss in the following chapter, a compelling argument for neo-Davidsonian association of external arguments at logical-conceptual structure. However, Schein's argument is also an excellent illustration for just how much sweat it takes to construct such arguments. They do not come your way on a leisurely walk.

A rich source of evidence for logical-conceptual decomposition is cross-linguistic comparison of lexicalization patterns, another laborious technique, which is used in Hale and Keyser's work, in Talmy 2000, and in many other

⁹. Fodor & Lepore 1998, 1999, and references mentioned there.

studies on lexical semantics. There is no good reason to think that it shouldn't be possible to reconstruct hypothesized universal logical-conceptual representations by comparing the ways they are sounded out in typologically diverse natural languages. The methodology followed is not too different in spirit from the comparative reconstruction techniques developed in historical linguistics. An unknown form is inferred by thinking about the general principles that would be able to produce the observed variety of forms from a common ancestor. If the grammars of different natural languages are different ways of sounding out universal logical-conceptual structures, a view implicit in theories ranging from Generative Semantics to Minimalism, we expect there to be principles for how to sound out those structures, possibly involving intermediate levels of syntactic representation. Be this as it may, there must be a point on the path from logical-conceptual representations to their pronounced counterparts where linguistic variation comes in. Complex predicates at an earlier stage of the derivation might surface as single lexical items in some languages, for example. Two or more adjacent inflectional heads might be merged into a single affix. Some stretches of an input structure might not have to be sounded out at all. This last possibility has fascinated syntacticians since the very beginnings of generative grammar, and has led to theories of empty pronouns, ellipsis, and zero-affixation.

Properties of logical-conceptual representations might also be unveiled by exploring the expected consequences of hypothetical constraints for the predicates that can appear at that level of mental representation. Constraints that have promise here, I believe, are not constraints whose violation would shut down our minds and produce mental blackouts. I am not thinking of a calculus of thinkable concepts. I am interested in constraints for 'natural denotations', denotations that children might posit for

basic lexical items without being given explicit definitions or instructions. I will explore the consequences of one such hypothetical constraint, the Cumulativity Universal, which was originally proposed by Manfred Krifka and has since been pursued extensively by Fred Landman. The version of the Cumulativity Universal that is relevant here has it that the denotations of basic predicates at logical-conceptual structure are cumulative from the very start. If there is a basic logical-conceptual predicate ‘red’, for example, that is true of my hat and your scarf (two singularities), then the Cumulativity Universal says that that very same predicate is also true of the sum of my hat and your scarf (a plurality). Cumulativity extends to relational predicates. If the predicate ‘buy’, for example, is cumulative, then whenever it relates, say, some action of yours to your scarf, and some action of mine to my hat, it also relates the sum of our buying actions to the sum of what we each bought. Suppose there was independent support in favor of the Cumulativity Universal. This would be of great importance for us since, as we will see shortly, the Cumulativity Universal immediately disqualifies the ‘theme’ or ‘object’ relation and a significant number of other object related thematic relations from being possible denotations of thematic role predicates at logical-conceptual structure, and thus at any level of syntactic representation. I will use this fact to argue that, in all likelihood, there are verbs whose direct objects are not neo-Davidsonian at any level of mental representation.

The Cumulativity Universal is far from being uncontroversial. I will thus launch a detailed defense of it in chapters 4 and 5. It will be a long and complicated argument, fed by the work of many of the key players in this popular field of investigation. The argument will be of interest beyond the issues of argument association and semantic universals, however, revealing

many not at all obvious consequences of Davidsonian event semantics including some relating to the placement possibilities of quantifier phrases and adverbials within a verb's extended projection. Last not least, the discussion of cumulativity might shed some light on the nature of verbal number agreement, and thus ultimately help us understand the connection between voice and verbal agreement. Like many longwinded arguments, then, that one too, will open up unexpected vistas into new and uncharted terrains that will guide subsequent excursions.

Returning to argument association in the syntax, what kind of facts could give us information about the way arguments are linked to their heads in that central component of the grammar? Optionality of arguments is a possible diagnostic for neo-Davidsonian association, as pointed out in Dowty 1989. Dowty notes that some event nouns do not show any real subcategorization, and suggests that this could be explained by assuming that in contrast to verb arguments, the arguments of those nouns are associated by the neo-Davidsonian method in the syntax. Here is the example he considers:

- (2)
- a. **Gifts of books from John to Mary would surprise Helen.**
 - b. **Gifts of books from John would surprise Helen.**
 - c. **Gifts of books to Mary would surprise Helen.**
 - d. **Gifts from John to Mary would surprise Helen.**
 - e. **Gifts from John would surprise Helen.**
 - f. **Gifts of books would surprise Helen.**
 - g. **Gifts to Mary would surprise Helen.**
 - h. **Gifts would surprise Helen.**

Not all event nouns behave like eventive **gift**. A number of authors, including Lebeaux 1986 and Grimshaw 1990, have observed that certain deverbal event nouns must realize the verb's direct object, as long as they preserve the original eventive interpretation of the verb. Example (3) is Grimshaw's, with minor adjustments¹⁰.

- (3) a. * **The constant assignment is to be avoided.**
b. **The constant assignment of unsolvable problems is to be avoided.**

What is it that distinguishes the noun **gift** from the noun **assignment**? It seems that the crucial difference is that in contrast to eventive **gift**, the meaning of eventive **assignment** is predictably related to the meaning of the verb it is derived from. An assignment of unsolvable problems is no more and no less than an event of assigning unsolvable problems. On the other hand, gifts of books to Mary are not just events of giving books to Mary. The books have to be gifts for Mary to keep. The noun **gift**, then, is not compositionally derived from **give**, and I suspect that this is why eventive **gift** does not have to preserve the argument structure of **give**.

If verbs have the arguments they do because they mean what they do, preservation of a verb's meaning entails preservation of its 'argument structure'. If lexical items must syntactically realize whatever arguments they have (except possibly the event argument¹¹), verbs have to realize their

¹⁰. Grimshaw 1980, p. 50.

¹¹. Whether event arguments do or do not have to be realized in the syntax is an interesting empirical question that I will leave open in this essay. We would have to

arguments, even when they are part of a nominalization - as long as the nominalization is compositional. The fact that **assign** must realize a direct object when it is part of a compositional event nominalization, then, suggests that having such an object is part of its meaning. If this is the correct explanation for the ungrammaticality of 3(a), there are consequences for external arguments as well. If external arguments were true arguments of their verbs, they, too, should have to be obligatorily realized in compositionally derived deverbal constructions. Compositional event nominalizations do not seem to have to realize the verb's external argument, however. Not overtly, that is. The situation is more complicated. In (3), the verb's external argument could still be realized as an implicit impersonal pronoun, as in the case of verbal passives:

- (4) **Too many unsolvable problems were assigned.**

According to Baker, Johnson and Roberts 1989, the presence of a verb's external argument in verbal passives is diagnosed by the fact that verbal passives do not allow a self-action interpretation of the verb. We have, for example:

- (5) a. **The children are being dressed.**
 b. **The climbers are being secured with a rope.**

5(a) is not compatible with a situation where the children are dressing themselves, and 5(b) excludes a scenario where the climbers are securing

investigate anaphoric possibilities and the interaction between event, time, and situation arguments. Since I will be moving in the lower regions of the hierarchy of inflectional heads in this essay, I cannot seriously address this very important issue.

themselves with a rope. Suppose, as some have claimed, that verbal passive constructions include an implicit external argument that functions like an impersonal pronoun. 5(a) and (b) could now be likened to 6(a) and (b) respectively, and the sentences under (5) could be argued to lack a self-action interpretation for the same reasons as the sentences under (6) do.

- (6) a. **They are dressing the children.**
- b. **They are securing the climbers.**

If this kind of reasoning is accepted¹², we should investigate whether compositional event nominalizations exclude a self-action interpretation of the verbs they are derived from. To find the right test cases, we have to look for nominalizations of verbs that readily admit a self-action interpretation¹³. Verbs of caring for the body¹⁴ are prime candidates. Those verbs only seem to have event nominalizations ending on **-ing**, however. As illustrated in 7(a) and (b), nominalizations ending on **-ing** do not permit a self-action reading:

- (7) a. **The report mentioned the painfully slow dressing of the children.**
- b. **The article praised the expeditious securing of the climbers.**

¹². See chapter XX for more discussion of this point.

¹³. Note that mere compatibility with a reflexive pronoun is not sufficient to diagnose a self-action interpretation. See chapter XXX for extensive discussion.

¹⁴. See Levin 1993, p. 35 for a list of ‘caring for the body’ verbs. In English, some of those verbs do not require a reflexive pronoun when used to describe self-actions.

Since the embedded sentences in 7(a) and (b) are incompatible with self-action, we can conclude that event nominalizations of the kind illustrated in (7) require the implicit realization of the verb's external argument, just as verbal passives do. We should look for other deverbal constructions, then. There is at least one where the verb's external argument does not seem to be even implicitly present. Compare the following two sentences:

- (8) a. **The climbers are secured with a rope.**
 b. **The climbers are being secured with a rope.**

8(a), which can be interpreted as an adjectival passive, is compatible with the climbers having secured themselves. On the other hand, 8(b), which must be a verbal passive, requires the climbers to be secured by somebody else. Following our earlier reasoning, it looks like adjectival passives are deverbal constructions where the verb's external argument can be missing. The crucial question, then, is whether or not this construction is compositional. In chapters XXX, we will look at adjectival passives in more depth and will see that, contrary to common beliefs, the meanings of adjectival passives are derived in a completely compositional fashion¹⁵.

If the verb's external argument is not obligatorily present in adjectival passives, we might be tempted to weaken the requirement that lexical items must syntactically realize all of their (non-event) arguments. Maybe there is an operation that eliminates or 'suppresses' external arguments under certain conditions, a mechanism that has traditionally been assumed to be part of 'passivization'. Within an event semantics, another possible

¹⁵ . See also Kratzer 2000.

explanation is available, however. External arguments might be neo-Davidsonian in the syntax, hence might not be true arguments of their verbs at all. If this was so, we might be able to account for the occasional absence of a verb's external argument without having to give up the hypothesis that lexical items must realize their (non-event) arguments wherever they occur.

I will slowly build a case for neo-Davidsonian association of external arguments in the syntax. We will then have to figure out what it is that forces the presence of a verb's external argument in just the right range of constructions. What will gradually emerge in the course of our investigation is a new perspective on voice alternations and passives. It's not that some mishap befalls external arguments that makes them fade or disappear when their verb is passivized. When external arguments are missing, I suggest, we are at a stage of the syntactic derivation where they are not yet there. We just haven't yet projected enough functional structure. The same kind of explanation will be given for cases where external arguments have to be there, but aren't yet able to be fully there. They might have to be realized by an unpronounced impersonal pronoun or a prepositional phrase, for example. In those cases, the story will go, not enough functional structure has been projected to give those arguments what they need to appear as full-fledged DPs.

Reduced relatives can serve as a first illustration motivating the suggestion that external arguments enter as we are projecting more structure. In German, we have contrasts like the following:

- (9) a. **Die schön gekämmten Kinder**
The nicely combed children
(Compatible with self-action)

- b. **Die gestern gekämmten Kinder**
The yesterday combed children
'The children who were combed yesterday'
(Incompatible with self-action)

8(a) is, but 8(b) is not compatible with the children having combed themselves. Using Baker, Johnson, and Roberts' diagnostic, we can conclude that an external argument must be present in 8(b), but not necessarily in 8(a). 8(a) has a manner adverb, 8(b) comes with a temporal adverb. Temporal adverbs must precede manner adverbs in German, showing that they occupy a higher position than manner adverbs:

- (9) a. **Ich hab' dich gestern schön gekämmt.**
 I have you yesterday nicely combed.
 I combed you nicely yesterday.
- b. * **Ich hab' dich schön gestern gekämmt.**
 I have you nicely yesterday combed.

The same conclusion can be drawn from the topicalization contrasts in 10(a) and (b):

- (10) a. **Schön gekämmt** **hat er dich nicht.**
 Nicely combed has he you not.
 'He didn't comb you nicely yesterday'.
- b. * **Gestern gekämmt** **hat er dich nicht.**
 Yesterday combed has he you not.
 'He didn't comb you yesterday.'

Manner adverbs can, but temporal adverbs cannot be topicalized along with the verb. The adverb in 8(b), then, seems to require more verbal structure to

be projected than the adverb in 8(a). Whatever that extra structure is, it obligatorily brings along the external argument.

Schein 1993 presents data of a very different kind in support of neo-Davidsonian association of verb arguments. In a nutshell, what Schein does is confront us with a reading of a type of sentence for which, according to him, appropriate logical representations can only be derived if verb arguments are neo-Davidsonian not only at logical-conceptual structure, but also in the syntax. In the following chapter, I will critically examine Schein's case. The result of my investigation will be that Schein's examples do indeed provide evidence for neo-Davidsonian association of verb arguments, but only for external arguments, and only for logical-conceptual structure, not for the syntax. Arguments for the syntactic independence of external arguments have to come from a different source: the properties of nominalizations and participles, and the range of possible voice alternations, for example.

Chapter 2

Schein's Argument

Schein's argument is very sophisticated and laid out in great detail in his book, so I can do no more than sketch what I take to be its essence for our current concerns. In particular, I will neglect foundational issues, and rely on a mereological account of plural individuals, which goes right against what Schein actually says. My strategy will be to look at a fairly simple example that illustrates the phenomenon that Schein is after, and take the reader step by step through the process of first finding a suitable logical representation for it, and then deriving that logical representation from an appropriate syntactic structure. For consistency, the whole discussion will be placed within the semantic framework assumed here. Look at sentence (1), a variation of Schein's more complicated (2):

- (1) **Three copy editors caught every mistake in the manuscript.**
- (2) **Three video games taught every quarterback two new plays.**

(1) can have a 'cumulative' interpretation, in which case it would be true, say, in a situation where we hired three copy editors to look at a manuscript independently of each other, and between them, they caught all the mistakes in the manuscript¹. Some mistakes might have been caught by one of the

¹. By stipulating that the three copy editors work independently of each other, I tried to discourage an understanding of the scenario where we might be inclined to grant 'team

copy editors, some by two and the rest by all three. A cumulative reading for (1) is surprising, since **every** is an otherwise distributive quantifier. (2) is especially interesting since it has a reading that can be paraphrased as ‘three video games were responsible for the fact that every quarterback learned two possibly different plays’. On this reading, **every quarterback** and **three video games** are related cumulatively: between them, a total of three video games taught all the quarterbacks. On the very same reading, however, **every quarterback** behaves just like an ordinary distributive quantifier phrase in its relation with **two new plays**: every quarterback learned two possibly different plays. That standard formalizations have difficulties with the interaction of distributive and ‘cumulative’ quantifiers had already been observed in Roberts 1997, 1990, who discussed sentence (3)²:

- (3) **Five insurance associates gave a \$25 donation to several charities.**

The problematic reading is one where taken together, the contributions of five insurance associates amounted to a gift of \$25 to each of several charities. Let us explore the thorny issue of interacting cumulative and distributive quantifiers for the simpler sentence (1). The most straightforward formalization of the reading of (1) that we are interested in doesn’t yield a satisfying result:

credit’ (in the sense of Lasnik 1988, 1990, 1995) to the group of three copy editors. I do not think, however, that my attempt is fully successful. For the three copy editors to be considered a team, it is not necessary that they work in a coordinated fashion. Schein’s own examples are constructed in such a way that a team credit interpretation is more clearly ruled out.

². Cited from Roberts 1990, 86.

$$(4) \quad \exists x [3 \text{ copy editors}(x) \ \& \ \forall y [\text{mistake}(y) \rightarrow \exists e [\text{catch}(y)(x)(e)] \] \]$$

In (4) and subsequent logical-conceptual representations, variables from the end of the alphabet range over singular or plural individuals, construed as mereological sums. The variables ‘e’, ‘e’’, ‘e’’ range over singular or plural events, also construed as mereological sums, as proposed by Emmon Bach and many others since³. (4) says that each and every mistake was caught by some plurality of three copy editors, the same plurality for every mistake. This is not the reading we want. A formalization along the lines of (5) doesn’t fare much better (‘ $\leq$ ’ stands for the mereological part relation):

$$(5) \quad \exists x [3 \text{ copy editors}(x) \ \& \ \forall y [\text{mistake}(y) \rightarrow \exists e \exists z [z \leq x \ \& \ \text{catch}(y)(z)(e)] \] \]$$

(5) could be true in a situation where, say, a single copy editor found all the mistakes without any other copy editor having been involved in the project at all. Splitting off the agent argument gets us a bit closer to an acceptable reading:

$$(6) \quad \exists e \exists x [3 \text{ copy editors}(x) \ \& \ \text{agent}(x)(e) \ \& \ \forall y [\text{mistake}(y) \rightarrow \exists e' [e' \leq e \ \& \ \text{catch}(y)(e')] \] \]$$

(6) says that three copy editors were the agents of an event in which every mistake was caught. The 3 copy editors are plural agents of a plural event. The paraphrase sounds good, but it is still not quite right. Suppose three copy

³. Bach 1986, but without the bits of processes.

editors spent an afternoon together: Adam built a birdhouse, Bill ironed his shirts, and Chris caught all the mistakes in the manuscript. The sum of Adam, Bill, and Chris would be the plural agent of an event in which all the mistakes were caught. Intuitively, (1) wouldn't be true in such a situation. Maybe it would help to split off the theme argument as well, as in (7):

$$(7) \quad \Box e \Box x [3 \text{ copy editors}(x) \ \& \ \text{agent}(x)(e) \ \& \ \text{catch}(e) \ \& \ \Box y [\text{mistake}(y) \ \Box \ \Box e' [\text{e}' \leq e \ \& \ \text{theme}(y)(e')]]]]$$

(7) is still wrong. As before, suppose that three copy editors spent an afternoon together. But this time, Adam caught fish, Bill caught rabbits, and Chris caught all the mistakes in the manuscript. The sum of Adam, Bill, and Chris is the plural agent of a catching event (a hunt, say) in which every mistake in the manuscript was caught, and this is what (7) requires to be true. Again, (1) wouldn't be true in such a situation. Maybe we should say that three copy editors were the agents of a minimal event in which every mistake was caught. Here is a try ('<' stands for the mereological proper part relation, and 'P' and 'Q' are variables that range over properties of events):

$$(8) \quad \Box e \Box x [3 \text{ copy editors}(x) \ \& \ \text{agent}(x)(e) \ \& \ \text{Min}(P)(e)],$$

where $P = \Box e \Box y [\text{mistake}(y) \ \Box \ \Box e'' [e'' < e \ \& \ \text{catch}(y)(e'')]]$, and

$$\text{Min} = \Box Q_{<se>} \Box e [Q(e) \ \& \ \sim \Box e' [e' < e \ \& \ Q(e')]]^4.$$

We are not there yet. Unfortunately, (8) excludes scenarios we do not want to exclude. Imagine that there were just two mistakes in the manuscript, call them 'Addition' and 'Omission'. One of the copy editors found Addition, but

⁴. See von Stechow 1996 for an analogous minimalization operator.

not Omission. The other two copy editors both found Omission, but not Addition. In such a situation, (1) is true, even though (8) winds up false. The three copy editors are not agents of any minimal event in which every mistake in the manuscript was caught. If there is an event in which every mistake was caught, and which has the plurality consisting of the three copy editors as agents, that event must include two sightings of Omission and one sighting of Addition. But this means that such an event is not a minimal event in which every mistake was caught. What we ultimately want to say is that the three copy editors were the agents of an event that was a completed event of catching every mistake in the manuscript. But what is an event that is a completed event of catching every mistake in the manuscript? It is an event in which every mistake in the manuscript was caught, and which does not contain anything that is irrelevant to the enterprise of catching mistakes in the manuscript. That in turn is an event in which every mistake in the manuscript was caught, and which is a catching of mistakes in the manuscript. This intuition is captured by (9):

- (9) $\exists e \exists x [3 \text{ copy editors}(x) \ \& \ \text{agent}(x)(e) \ \& \ \forall y [\text{mistake}(y) \\ \rightarrow \exists e' [e' \leq e \ \& \ \text{catch}(y)(e')]] \ \& \ \forall y [\text{mistakes}(y) \ \& \ \text{catch}(y)(e)]]$.

In (9), the condition ' $\forall y [\text{mistakes}(y) \ \& \ \text{catch}(y)(e)]$ ' guarantees that e is an event of catching mistakes, not merely an event in which mistakes are caught. This is just how event descriptions are understood in Davidsonian event semantics. 'Catch(y)(e)' says that e is a completed event of catching of y , not that e is an event in which y is caught⁵.

5. See Kratzer 1990, 1998, 2002 for how to relate propositions to event descriptions in a situation semantics. Instead of the condition ' $\forall y [\text{mistakes}(y) \ \& \ \text{catch}(y)(e)]$ ', we could also

Finding an appropriate logical representation for sentence (1) has been a laborious exercise, even without worrying about a compositional derivation. The most pressing question is, of course, whether there isn't an equally insightful formalization of (1) that does not require neo-Davidsonian association of the agent argument. There is much discussion of this point in Schein's work. To consider additional options, we might also try to follow Montague and assume that the direct object positions of verbs have the semantic type of generalized quantifiers. (1) could then be formalized as (10):

$$(10) \quad \exists x \exists e [3 \text{ copy editors}(x) \ \& \ \text{catch}(\lambda Q \lambda e \lambda y [\text{mistake}(y) \sqsubseteq Q(y)(e)]) (x)(e)]$$

In (10), the variable 'Q' ranges over (Schönfinkelled) relations between individuals and events. ' $\lambda Q \lambda e \lambda y [\text{mistake}(y) \sqsubseteq Q(y)(e)]$ ' is the analogue of a generalized quantifier within an event semantics. It maps relations between individuals and events into properties of events. (10) is unsatisfactory as it stands, as long as we are not told what it means, quite generally, for a generalized quantifier, a plural individual, and an event to stand in the catching relation. What condition is it that the actual world has to satisfy for such a relation to hold? At this point, Schein's sentence (2) should be brought into the discussion as well.

(2) **Three video games taught every quarterback two new plays.**

use Link's $\sqsubseteq$ -operator and posit ' $e = \sum e' \sqsubseteq y [\text{mistake}(y) \ \& \ \text{catch}(y)(e') \ \& \ e' \leq e]$ '. The event e would now be required to be identical to the sum of all its subevents that are catchings of a mistake (in the manuscript).

(2) has an additional quantifier phrase, and we want a scope relationship between **every quarterback** and **two new plays**. That is, we are interested in a reading where if there were, say, four quarterbacks, a total of eight new plays might have been taught. I do not see an analysis of the relevant reading of (2) that follows the model of (10) without seriously straining our assumptions about what logical representations are and can do. I am ready to conclude, then, that if agent arguments are neo-Davidsonian at logical-conceptual structure, an analysis of sentence (1) and its kin becomes available that does not seem to be straightforwardly available otherwise. What we have, then, is a first piece of support for neo-Davidsonian association of agent arguments of verbs at logical-conceptual structure.

It seems that Schein's argument can be extended to at least certain other kinds of external arguments:

- (11) a. **(Between them), three real estate agents own every house in town.**
- b. **(Between them), three high school students from Pelham won every scholarship there was this year.**

Both 11(a) and (b) have the 'cumulative' interpretation we are after. In 11(a), the external argument denotes the possessor of a state, in 11(b) the patient of an event. What about internal arguments, theme arguments, for example? The question is not pursued in Schein's book. In a footnote he notes, however, that for his purposes, it is not necessary to separate off the theme argument in the logical form of the sentence **Brutus stabbed Caesar**. He considers '[λe [agent(Brutus)(e) & stab(Caesar)(e)]]' an acceptable formalization of that sentence. Can we construct a sentence modeled after (1) or (2) that

would force us to sever the theme argument from its verb in the syntax? It seems that we can't. Take (12):

(12) **Every copy editor caught 500 mistakes in the manuscript.**

(12) is like (1), but this time, the distributive quantifier **every** is part of the subject, not the object. (12) does not have a cumulative reading saying that between them, the copy editors caught a total of 500 mistakes in the manuscript. Let's passivize (12) so as to place the theme argument in subject position. (13) still doesn't have the reading we are after.

(13) **500 mistakes in the manuscript were caught by every copy editor.**

Following the spirit of Schein, the non-existent cumulative reading for (12) or (13) would correspond to (14), neglecting the complications relating to minimality that we went through above:

(14) $\exists e \exists x [500 \text{ mistakes}(x) \ \& \ \text{theme}(x)(e) \ \& \ \exists y [\text{copy editor}(y) \ \& \ \exists e' [e' < e \ \& \ \text{agent}(y)(e') \ \& \ \text{catch}(e')]]]]$

(14) says that 500 mistakes were the plural theme of an event in which every copy editor was a catcher. The paraphrase I just gave doesn't sound optimal, and there may be good reasons why, but within those limitations, (14) captures the non-existing reading of (12) or (13) that we are interested in. That (12) and (13) do not have the reading represented by (14) tells us that we cannot rely on representations like (14) to extend Schein's reasoning to theme arguments. That we can't seem to come up with cases that would force

neo-Davidsonian theme arguments doesn't mean that neo-Davidsonian themes are not an option, of course. Both 15(a) and (b) correctly represent one of the readings of (12) and (13), for example, and both 16(a) and (b) correctly represent the other.

- (15) a. $\exists x [500 \text{ mistakes}(x) \ \& \ \exists y [\text{copy editor}(y) \ \& \ \exists e [\text{theme}(x)(e) \ \& \ \text{agent}(y)(e) \ \& \ \text{catch}(e)]]]$
 b. $\exists x [500 \text{ mistakes}(x) \ \& \ \exists y [\text{copy editor}(y) \ \& \ \exists e [\text{catch}(x)(y)(e)]]]$
- (16) a. $\exists y [\text{copy editor}(y) \ \& \ \exists x [500 \text{ mistakes}(x) \ \& \ \exists e [\text{theme}(x)(e) \ \& \ \text{agent}(y)(e) \ \& \ \text{catch}(e)]]]$
 b. $\exists y [\text{copy editor}(y) \ \& \ \exists x [500 \text{ mistakes}(x) \ \& \ \exists e [\text{catch}(x)(y)(e)]]]$

15(a) and 16(a) use the neo-Davidsonian method, 15(b) and 16(b) have ordered argument association for the verb's arguments, including the theme argument. The main difference between the logical-conceptual representation (14) on the one hand, and the representations in (15) and (16) on the other, is the scope of the event quantifier. As long as the event quantifier has narrow scope with respect to the other two quantifiers, the mode of argument association does not matter. We can conclude at this point that the fact that (12) and (13) lack the reading (14) doesn't show that theme arguments are not neo-Davidsonian. However, we also have not yet found any evidence that they are. The kind of argument Schein presented for neo-Davidsonian association of external arguments, then, can't be extended to theme arguments. Moreover, Schein's argument only seems to bear on the issue of argument association in logical-conceptual structure. So far, nothing at all is implied about argument association in the syntax.

To use the properties of sentences like (1) as support for neo-Davidsonian association of agent arguments in the syntax, we have to look at the mapping from syntactic structures to logical-conceptual structures, and show that the kind of logical representations needed for such sentences can only be derived in an empirically plausible way by assuming that neo-Davidsonian association of agent arguments is already present in the syntax. In any such demonstration, a lot depends on what is considered to be ‘empirically plausible’. It is not too difficult to come up with some way of deriving the intended logical representation for sentence (1), while still associating agent arguments via the ordered argument method in the syntax. Here is a proposal deriving the logical representation (6) (just to keep things simple). The proposal also serves as an introduction to the formal framework I will be using in this study. To understand all the technical details of those computations is not essential for understanding the story I will tell, so they may be skipped by readers who don’t worry that the proposals made might not do what they are claimed to do.

(17) a. English sentence:

Three copy editors caught every mistake (in the manuscript).

b. Logical representation to be derived compositionally:

$$\begin{aligned} & \exists e \exists x [3 \text{ copy editors}(x) \ \& \ \text{agent}(x)(e) \ \& \ \exists y [\text{mistake}(y) \ \& \\ & \exists e' [e' \leq e \ \& \ \text{catch}(y)(e')]]] \end{aligned}$$

Derivation 1:

1. $T(\text{every mistake}) = \exists R_{\langle e \langle st \rangle \rangle} \exists e \exists y [\text{mistake}(y) \ \& \ \exists e' [e' \leq e \ \& \ R(y)(e')]]$
2. $T(\text{catch}) = \exists Q_{\langle \langle e \langle st \rangle \rangle \langle st \rangle \rangle} \exists x \exists e [\text{agent}(x)(e) \ \& \ Q(\text{catch}_{\langle e \langle st \rangle \rangle})(e)]$

3. $T(\text{catch}(\text{every mistake})) =$
 $\lambda x \lambda e [\text{agent}(x)(e) \ \& \ T(\text{every mistake})(\text{catch})(e)] =$
 $\lambda x \lambda e [\text{agent}(x)(e) \ \& \ \lambda y [\text{mistake}(y) \ \sqcap \ \lambda e' [e' \leq e \ \& \ \text{catch}(y)(e')]]]$
 From (1), (2), by Functional Application.
4. $T(3 \text{ copy editors}) = \lambda R_{\langle e, \langle st \rangle \rangle} \lambda e \lambda x [3 \text{ copy editors}(x) \ \& \ R(x)(e)]$
5. $T(3 \text{ copy editors}(\text{catch}(\text{every mistake}))) =$
 $T(3 \text{ copy editors})(\lambda x \lambda e [\text{agent}(x)(e) \ \& \ \lambda y [\text{mistake}(y) \ \sqcap \ \lambda e' [e' \leq e \ \& \ \text{catch}(y)(e')]]]) =$
 $\lambda e \lambda x [3 \text{ copy editors}(x) \ \& \ \text{agent}(x)(e) \ \& \ \lambda y [\text{mistake}(y) \ \sqcap \ \lambda e' [e' \leq e \ \& \ \text{catch}(y)(e')]]]$
 From (3), (4), by Functional Application.
6. $\lambda e \lambda x [3 \text{ copy editors}(x) \ \& \ \text{agent}(x)(e) \ \& \ \lambda y [\text{mistake}(y) \ \sqcap \ \lambda e' [e' \leq e \ \& \ \text{catch}(y)(e')]]]$
 From (5), by Existential Closure.

The computation just given is a derivation of the denotation of sentence 17(a) (= (1)) on the intended reading, leaving out some complications. The interpretation process assigns denotations to bracketed strings of lexical items in a type-driven fashion⁶. The interpretation procedure does not need to see syntactic category labels, nor does it care about linear order. For any string $\square$, $T(\square)$ is the denotation of $\square$. Denotations are given through translations into expressions of an extensional type logic. For the time being, we might think of those expressions as logical-conceptual representations, and this is what I will tentatively do in what follows. The type logic assumed here has three basic types: Type e (individuals), type s (events or states, that is, eventualities in the terminology of Bach 1977), and type t (truth-values).

6. Klein and Sag 1985. See also Bittner 1994 and Heim & Kratzer 1998.

When necessary, we will occasionally switch to an intensional semantics that has quantification over possible worlds, and we'll then add a separate type of possible worlds and a corresponding interpretation domain. If not obvious, I indicate the semantic types of variables and constants by subscripts on the first occurrence of the expression. The variable 'R' and the constant 'catch', for example, are both expressions of type $\langle e \langle st \rangle \rangle$, that is, they denote (Schönfinkelled) relations between individuals and events. The variable 'Q' is of type $\langle \langle e \langle st \rangle \rangle \langle st \rangle \rangle$, which is the type of quantifier phrases. As before the variables 'e', 'e' ', 'e' ', etc. range over (possibly plural) events, and variables from the end of the alphabet range over (possibly plural) individuals. The denotations of lexical items are specified as part of their lexical information. To calculate the denotations of complex expressions, there are a handful of composition principles that apply freely whenever they can. In this particular example, the only composition principles used were Functional Application and Existential Closure.

Derivation 1 establishes that examples like (1) do not force us to assume that agent arguments must be associated via the neo-Davidsonian method in the syntax. As we see in step 2 of the computation, all arguments of **catch** are associated via the ordered argument method. Derivation 1 also informs us about the price we have to be willing to pay if we hold on to ordered argument association when facing Schein's examples. We need a complicated semantic type for the direct object position of **catch** and we have to posit different argument structures for **catch** and 'catch'. Let us now compare derivation 1 with one that relies on neo-Davidsonian association of the agent argument in the syntax.

Derivation 2:

1. $T(\mathbf{every\ mistake}) = \lambda R_{\langle e, \langle st \rangle \rangle} \lambda e \lambda y [mistake(y) \wedge \lambda e' [e' \leq e \ \& \ R(y)(e')]]$
2. $T(\mathbf{catch}) = \lambda x \lambda e catch_{\langle e, \langle st \rangle \rangle}(x)(e)$
3. $T(\mathbf{catch\ (every\ mistake)}) = \lambda e \lambda y [mistake(y) \wedge \lambda e' [e' \leq e \ \& \ catch(y)(e')]]$
 From (1), (2), by Functional Application.
4. $T(\mathbf{3\ copy\ editors}) = \lambda R_{\langle e, \langle st \rangle \rangle} \lambda e \lambda x [3\ copy\ editors(x) \ \& \ R(x)(e)]$

We are stuck. The denotations of the subject and the VP cannot be combined. Since the verb lacks an agent argument, the VP does, too. If the association of agent arguments is neo-Davidsonian in the syntax, there must be something that introduces agent arguments in the course of a syntactic derivation. Schein assumes that INFL, the carrier of verbal inflection, introduces external arguments, without being clear, however, about how INFL is able to do so. Suppose there is an inflectional feature **[active]** that is responsible for the introduction of external arguments. For the time being, let us assume that agent arguments are the only kind of external arguments. We'll worry about other types of external arguments later. Following much recent work in syntactic theory, let us assume furthermore that inflectional features may head their own projections. Derivation 2 can now continue as follows:

5. $T([\mathbf{active}]) = \lambda x \lambda e \text{ agent}(x)(e)$
6. $T([\mathbf{active}] (\text{catch } (\mathbf{every\ mistake}))) =$
 $\lambda x \lambda e [\text{agent}(x)(e) \ \& \ \lambda y [\text{mistake}(y) \ \sqcap \ \lambda e' [e' \leq e \ \& \ \text{catch}(y)(e')]]]$
 From (3), (5) by Event Identification.
7. $T(3 \text{ copy editors } ([\mathbf{active}] (\text{catch } (\mathbf{every\ mistake})))) =$
 $\lambda e \lambda x [3 \text{ copy editors}(x) \ \& \ \text{agent}(x)(e) \ \& \ \lambda y [\text{mistake}(y) \ \sqcap \ \lambda e' [e' \leq e \ \& \ \text{catch}(y)(e')]]]$
 From (4), (6) by Functional Application.
8. $\lambda e \lambda x [3 \text{ copy editors}(x) \ \& \ \text{agent}(x)(e) \ \& \ \lambda y [\text{mistake}(y) \ \sqcap \ \lambda e' [e' \leq e \ \& \ \text{catch}(y)(e')]]]$
 From (7), by Existential Closure.

Derivations 1 and 2 yield the same result. They both correctly predict the truth-conditions of sentence (1) = (17). The prettiest aspect of derivation 2 is that English **catch** and logical-conceptual ‘catch’ have matching argument structures. No higher type has to be assumed for the object position of **catch**. Derivation 2, unlike derivation 1, is compatible with a possible constraint that might require a tight match between the syntactic structure projected by lexical items and their logical-conceptual counterparts. Some such constraint would help a child learn the words of her language. However, derivation 2 comes with a price tag as well. It needs to assume that there is something in the grammar that introduces the right kind of external arguments in just the right kind of places. Derivation 2 also needs an additional composition principle, Event Identification⁷. Event Identification is really just a special type of a conjunction operation that makes it possible to chain together

7. The term ‘Event Identification’ is inspired by Higginbotham’s term ‘Theta Identification’ (Higginbotham 1985).

various conditions for the event described by a sentence. The operation is independently needed for adverbial modification, for example. Event Identification, then, is not part of the cost accrued by derivation 2. The main cost for derivation 2 is the assumption about the introduction of the external argument it has to rely on. I conclude that the comparison between derivation 1 and 2 has not yet produced a winner. I will eventually plead for derivation 2. But Schein's examples all by themselves do not settle the case about argument association in the syntax. In fact, the obstacles derivation 2 has to overcome look more serious at this point.

Let me review where we are. I considered two modes of argument association, the neo-Davidsonian method, and the ordered argument method. This distinction opened up a number of possibilities for the association of different kinds of verb arguments at different levels of representation. I critically examined a piece of evidence brought forward by Barry Schein in support of neo-Davidsonian argument association. I explored the consequences of Schein's case for argument association at logical-conceptual structure and in the syntax. Schein's case came out strongest with respect to neo-Davidsonian association of external arguments of verbs at logical-conceptual structure. An equally strong case could not be made for theme arguments, nor for argument association in the syntax.

In the following chapter, I will take up the status of theme arguments. I will argue that if there was a general thematic role predicate 'theme' that introduced theme arguments in a neo-Davidsonian way, this predicate would lack cumulativity, a property that other basic lexical items expressing relations between individuals and events seem to have. Accepting a thematic

role predicate ‘theme’ at logical-conceptual structure, then, would mean loss of a promising candidate for a semantic universal.

Chapter 3

Theme Arguments

3.1 Cumulativity

If arguments of verbs are introduced by secondary predicates denoting general thematic relations like the agent relation or the theme relation, we may wonder whether those relations have any conceptual significance beyond their role in argument association. The answer is a clear ‘yes’ for the agent relation, but an equally clear ‘no’ for the ‘theme’ or ‘object’ relation. The concept of an agent of an action is vague, but is well understood and the subject of numerous papers and books in philosophy and legal reasoning, none of them interested in argument structure. The concept of a ‘theme’ or ‘object’ of an event, on the other hand, has not generated much interest outside of lexical semantics. Parsons 1990 observes that “the use of Theme (“Patient”) is often called the “left-over” case, since so little can be said about it in general...”¹. To say the least, then, the agent relation is a more contentful and more interesting concept than the theme relation. Why should that be so? Is there something wrong with the putative theme relation?

There is a formal property that brings out a conceptual asymmetry between agents and so-called ‘themes’ of events. The property is called ‘cumulativity’ in Krifka 1992 and 1998 and ‘summativity’ in Krifka’s earlier work². Krifka

1. Parsons 1990, 74.

2. See Krifka 1986 and 1987.

assumes cumulativity to be a property shared by all thematic relations. However, cumulativity can only be a property of all thematic relations if the putative theme relation is not taken to be a thematic relation in a technical sense, that is, a relation denoted by a predicate that introduces all theme arguments of verbs in a neo-Davidsonian way.

Before looking at the role cumulativity may play for theories of verb meaning, let me first illustrate what it is. Imagine a young man, Alan, who wants to plant a rose bush in his mother's yard. He picks a sunny spot, removes the sod, digs a hole, and drives off to the garden center. It so happens that unbeknownst to Alan, his brother Brian has the same idea at about the same time. Unlike Alan, however, Brian drives to the garden center first. He buys a rose bush, drives to his mother's house, and discovers the hole in the ground that has just been dug by his brother. He loosens the bottom soil, places the bush in the hole, and walks away. Along comes Campbell, the gardener, who adds manure, leaf mold, and compost. Campbell's back hurts, so he goes home without finishing the job. Neighbor Dunn happens to see what still needs to be done and covers the roots of the rose bush with subsoil and topsoil. The bush is planted. Who did it?

I think it's fair to say that the four men together planted the rose bush. The action of planting is the sum of all the individual actions: Removing the sod, digging a hole, loosening the bottom soil, placing the bush in the hole, adding manure, leaf mold, and compost, and shoveling subsoil and topsoil into the hole. The agent of the rose bush planting is a plural individual consisting of Alan, Brian, Campbell, and Dunn. That plural individual is the sum of the agents of all the constituent actions.

The rose bush example illustrates the cumulativeness of the agent relation. Formally, this kind of cumulativeness is a property of relations between individuals and events, and is defined as follows ('+' stands for the mereological sum operation)³:

- (1) Cumulativeness (relations between individuals and events)

$$\Box [R_{\langle e, \langle s, t \rangle \rangle}(e) \wedge e \sqsubseteq e' \wedge x \sqsubseteq y \wedge [R(x)(e) \wedge R(y)(e')] \rightarrow R(x+y)(e+e')]]$$

The agent relation is a relation between individuals and events, hence is the kind of relation that can fall under definition (1). And it seems to have the property required in (1): whenever x is the agent of event e , and y is the agent of event e' , then the sum of x and y is the agent of the sum of e and e' .

But wait. It looks like there is an important class of exceptions. Suppose the teacher made the student leave. In this case we have an event e of making the student leave, and an event e' of leaving. The agent of e is the teacher and the agent of e' is the student. Assuming cumulativeness of the agent relation, the agent of $e+e'$ should be the sum of the student and the teacher. But isn't e' a part of e , and therefore $e+e' = e$? The sum of the teacher and the student should then also be the agent of e . I think that something rather subtle has gone wrong in the piece of reasoning I just presented. Following Ginet 1990, we should distinguish between events of causing the student to leave and events that cause the student to leave. An event of causing the student to leave is the sum of all the events in some causal chain that has the student's leaving as a final link. An event of causing the student to leave, then, is an event that includes the student's leaving. On the other hand, an event that causes the student to leave is the initial link of some causal chain

³. Krifka 1986, 1987, 1992.

whose final link is the student's leaving. Such an event, then, does not include the student's leaving. If periphrastic causative constructions like **make the student leave** describe events that cause the student to leave, rather than events of causing the student to leave, those causatives do not falsify the claim that the agent relation is cumulative. In our example, the teacher would be the agent of an event e that causes e' , the event of the student's leaving. But now e' is not assumed to be part of e anymore, and consequently, e is no longer identical to $e+e'$. I conclude that periphrastic causative constructions are compatible with the cumulativeness of the agent relation. I will have more to say about causatives in chapter 7. For now, let us move on with the current plot.

Unlike the agent relation, the putative theme relation does not seem cumulative. When Alan removes the sod, the theme of that event is the sod, but when he digs a hole, the theme is the hole. Or is it the soil that is being removed when the hole is dug? Skipping over that last complication, the rose bush is the theme when placed in the hole, and the manure, the leaf mold, the compost, the bottom soil, subsoil and topsoil are the themes of relevant subevents. Assuming that the whole planting event is indeed the sum of all the subevents I mentioned, cumulativeness would require that the sum of the rose bush plus the sod, the hole, the manure, the leaf mold, the compost, and the various portions of soil is the theme of the planting, rather than the rose bush alone. Unlike the agents of actions, then, the themes of actions do not get summed up when the actions are. Themes lack the conceptual independence of agents. Theme arguments seem to be tightly linked to their verbs. Agents are different. Actions seem to have agents independently of how we describe them.

We have seen that the putative theme (or object) relation is not cumulative. On the other hand, the agent relation looks like a good candidate for a cumulative relation. Let us now explore what other relations are cumulative. Take the relation that holds between an event e and an individual x just in case e is an event of planting x . This relation is cumulative if (2) holds:

$$(2) \quad \forall e \forall e' \forall x \forall y [[\text{plant}(x)(e) \ \& \ \text{plant}(y)(e')] \supset \text{plant}(x+y)(e+e')]$$

(2) does seem to be true. If e is an event of planting those roses and e' is an event of planting those lilies, for example, then $e+e'$ is an event of planting those roses and those lilies. We talk about such complex planting events when we say that planting those roses and those lilies must have taken a long time. Or when we complain that the planting of those roses and those lilies was a real strain on the town budget. Other transitive verbs behave in a similar way. If e is an event of constructing those roads, and e' is an event of constructing those bridges, then $e+e'$ is an event of constructing those roads and those bridges, for example. And if e is an event of destroying those barns, and e' is an event of destroying those sheds, then $e+e'$ is an event of destroying those barns and those sheds, and so on.

There is a difficulty with what I just said. I sometimes feel I should have used the plural 'events' instead of the singular noun 'event'. For example: If e is an event of destroying those barns, and e' is an event of destroying those sheds, then $e+e'$ are events of destroying those barns and those sheds. Once we start thinking about this issue, more questions pop up. If more than one barn was destroyed, why can we still talk about 'the destruction of those barns'? Why don't we have to use the plural 'destructions'? In contrast, why does my use of 'the father of those children' imply that there is just one father that is being talked about, hence those children must be siblings? Those

questions are important, but I don't think I have to go into them here. The individuation of events is notoriously vague and context dependent, and it is often not clear whether particular events are singular or plural in any absolute sense. This is bound to lead to insecurities with number marking on event nouns. While verbs are number marked as well, their number marking (if meaningful at all) does not seem to relate to the singularity or plurality of the events they describe. The problem I have been experiencing, then, is a problem created by my English metalanguage. Using the count noun 'event' to describe the events picked out by verbs forced me to make number distinctions that are not imposed by the verbs themselves. To avoid the problem, I could have coined an artificial number-neutral mass term like 'eventure'. I chose not to, but had to add a word of caution.

The generalizations suggested by our discussion so far can be summarized as follows:

- (3) a. The agent relation is cumulative.
- b. The putative theme relation is not cumulative.
- c. Specific thematic relations like the one holding between a
 planting event and what is planted, a construction event and
 what is constructed, a destruction event and what is destroyed
 and so on, are all cumulative.

If agent arguments are, but theme arguments are not true arguments of their verbs, the specific thematic relations mentioned in 3(c) are simply the denotations of the predicates 'plant', 'construct', and 'destroy' respectively, and (3) can be restated as (4):

- (4) a. The agent relation is cumulative.
- b. The putative theme relation is not cumulative.
- c. Verb denotations that are relations between individuals and events are cumulative.

If theme arguments are not neo-Davidsonian even at logical-conceptual structure, there is no thematic role predicate ‘theme’, and the theme relation has no status as a cognitively significant relation. The theme relation may not qualify as a ‘natural’ category at all. What are natural categories? Any theory of lexical acquisition must make some distinction between categories that are natural and those that are not. Very roughly, the natural categories are those that humans take to be candidates for denotations of simple lexical items, spontaneously and without any explicit instruction or definition. The most famous example of a non-natural category is the property ‘grue’ discussed by Nelson Goodman in the fifties⁴. An object is grue if it is green and has been examined before a fixed time, say December 31, 2010, or else it is blue, and has not been examined before December 31, 2010. All emeralds that have been examined so far are grue as well as green. But for some reason - and this is Goodman’s puzzle - grueness, unlike greenness, is not a category that humans come up with naturally when presented with emeralds, grass, or frogs, for example. To be sure, the concept of grueness can be grasped by human minds, but if it is, it’s on the basis of a verbal definition. The theme relation may not be quite as gruesome as grueness, but unlike the agent relation, it may still not qualify as a natural relation. If it doesn’t, it is not a candidate for the denotation of a thematic role predicate at logical-conceptual structure. If there is neo-Davidsonian association in logical-

⁴. Goodman 1954.

conceptual structure, a child acquiring the meaning of verbs must come up with the necessary thematic role predicates without definition or explicit instruction. The denotations of those relations, then, must be natural. Excluding the theme relation, but not the agent relation from the set of natural relations, we can state a constraint that might very well apply to all relations between individuals and events:

$$(5) \quad \begin{aligned} & \Box e \Box e' \Box x \Box y \Box R_{\langle e \langle st \rangle \rangle} [[\text{natural}_{\langle \langle e \langle st \rangle \rangle t \rangle}(R) \ \& \ R(x)(e) \ \& \ R(y)(e')] \\ & \Box R(x+y)(e+e')] \end{aligned}$$

Constraints like (5) are humble contributions to a theory of semantic acquisition. They cut down the candidate set for the denotations of lexical items a language learning child might have to consider.

Suppose what I just said is all wrong, and the theme relation is a natural relation after all⁵. If both agent and theme arguments were neo-Davidsonian at logical-conceptual structure, the counterparts of ordinary transitive verbs would merely denote properties of events. But there would still be the fact that the relation between the events described by verbs and the individuals denoted by their theme arguments is constrained by cumulativity, and this generalization would have to be captured. We would have to do this using the two postulates in (6), or some such set of statements:

$$(6) \quad \begin{aligned} \text{a.} \quad & \Box e \Box e' \Box x \Box y [[\text{agent}(x)(e) \ \& \ \text{agent}(y)(e')] \Box \text{agent}(x+y)(e+e')] \\ \text{b.} \quad & \Box e \Box e' \Box x \Box y \Box P_{\langle st \rangle} [[\text{natural}_{\langle \langle st \rangle t \rangle}(P) \ \& \ P(e) \ \& \ P(e') \ \& \ \text{theme}(x)(e) \\ & \ \& \ \text{theme}(y)(e')] \Box P(e+e') \ \& \ \text{theme}(x+y)(e+e')] \end{aligned}$$

⁵. “The notion of Theme may be one that comes naturally to human language learners, since we have an intuitive understanding of it...”. Parsons 1990, 81.

(6) lacks the simplicity and elegance of (5). (5) captures the cumulativity constraint through a constraint on a whole semantic domain. (6) needs two conditions to account for the same constraint, including particular stipulations for the predicates ‘agent’ and ‘theme’. The clumsiness of (6) may be a pointer to a misguided conceptualization of the domain of investigation. If we don’t make the right theoretical choices, we often end up with odd generalizations.

If the theme relation is not natural, theme arguments of verbs cannot be introduced by a secondary predicate denoting the theme relation, since the meaning of such a predicate would have to be learned without instructions or definitions. Theme arguments could still be introduced by less general secondary predicates, however. Such secondary predicates might denote the more specialized thematic relations discussed in Krifka 1987: gradual effected patient (‘write a letter’), gradual consumed patient (‘eat an apple’), gradual patient (‘read a letter’), affected patient (‘touch a cat’), or stimulus (‘see a horse’)⁶. As soon as there is a choice of thematic relations that can introduce direct objects, the question whether each and every kind of direct object can be matched up with one of those relations pops up. The search for thematic relations suitable for introducing direct objects of all kinds has been unsuccessful so far, in spite of many attempts documented in the literature. Commenting on that lack of success, Levin 1999 lists the direct objects in (7) and (8) as hard to classify in terms of commonly used thematic role inventories:

6. See also Ramchand 1997, who argues for some additional object relations.

- (7)
 - a. **The engineer praised the bridge.**
 - b. **The engineer touched the bridge.**
 - c. **The engineer avoided the bridge.**
 - d. **The engineer owned the bridge.**
 - e. **The engineer imagined the bridge.**
 - f. **The engineer studied the bridge.**

- (8)
 - a. **The engineer ignored the architect.**
 - b. **The engineer praised the architect.**
 - c. **The engineer greeted the architect.**
 - d. **The engineer selected the architect.**
 - e. **The engineer supervised the architect.**
 - f. **The engineer fought the architect.**
 - g. **The engineer met the architect.**
 - h. **The engineer visited the architect.**
 - i. **The engineer followed the architect.**

Even worse, among the object relations listed by Krifka, the affected object relation is not cumulative. For example, when the rose bush was planted in our earlier example, it was presumably an affected patient of the planting event. That event in turn was the sum of events with affected patients of their own: the sod, the different layers of soil that were removed and shoveled back, the manure, the leaf mold, the compost, and then again the rose bush. Summing up the affected patients of those subevents doesn't give us the rose bush. The concept of an 'affected patient', then, fares no better than the general 'theme' concept, as far as cumulativeness is concerned. Events do not seem to have affected patients independently of the way they are described. Unfortunately, the affected patient relation would have to be posited for a

very large class of transitive verbs on Krifka's proposal, presumably including all of the following, which seem to have affected objects:

- (9) **Abduct, absorb, adjust, affect, amputate, arrest, attach, banish, buy, chase, check, clasp, clench, decorate, deport, dump, edit, educate, execute, feed, fix, grab, grasp, greet, harm, hurt, ignite, immerse, jar, jostle, jumble, kidnap, kill, label, loot, mail, maim, neglect, nip, occupy, operate, paralyze, pare, quench, record, recycle, slay, slice, sprain, squash, thrust, usher, veil, vend, wag, wreck,**

There are good reasons, then, for being skeptical about the viability of neo-Davidsonian association for each and every kind of direct object. It is important, however, to keep in mind that so far, we haven't seen any arguments that would rule out the possibility that at least some direct objects are introduced by secondary predicates, either at logical-conceptual structure alone, or both at logical-conceptual structure and in the syntax. The skepticism I expressed concerned the assumption that all direct objects are uniformly introduced by secondary predicates, and was also directed against particular secondary predicates – those expressing non-cumulative relations like the general theme or the affected object relation. I certainly do not want to deny that transitive predicates can be constructed syntactically and that direct objects can be introduced by secondary predicates, including adjectives, adverbs, particles, and other verbs, as in a serial verb construction. I will explore the syntactic construction of transitivity in chapter 7.

3.2 The lexical representation of theme arguments

If a secondary predicate is to be usable for the neo-Davidsonian introduction of theme arguments in the syntax, it must have a denotation of its own and has to participate in the semantic composition process in a predictable way. If it is unpronounced, it has to have a recoverable meaning and satisfy whatever general constraints on unpronounced heads there may be. It's hard to see how all transitive verbs in, say, a language like English could be syntactically decomposed into primary predicates expressing properties of events and secondary predicates responsible for the neo-Davidsonian introduction of direct objects. If there isn't a single all-purpose thematic relation 'theme', how could primary predicates select the thematic relations that are right for them?

Take the English verb **construct** and suppose it is a mere predicate of events. It would then denote the function $\lambda e \text{ construct}(e)$. It would describe construction events, that is. How would we know that **construct** takes direct objects that denote the creations of the construction activity? Construction creates things, hence we expect effected patients. But construction also affects materials like boards or bricks, and the environment like the plot a house is built on. Why, then, don't we get affected patients with **construct**? Finally, construction work consumes time and money, and might destroy good views and open farmland, so we should find consumed patients with **construct**, too. But we don't. Among the three possibilities illustrated in 10(a) to (c), only 10(a) is realized.

- | | | | |
|------|----|-------------------------------------|--------------------|
| (10) | a. | construct this barn | (effected patient) |
| | b. | * construct those plots | (affected patient) |
| | c. | * construct my yearly income | (consumed patient) |

- (13) a. **Nelly drilled a hole.**
b. **Nelly drilled herself deaf.**

In 13(a), we are talking about an event of causing a state that consists in the existence of a hole and is a drilling activity. In 13(b), the drilling is also claimed to be an event of causing a state - Nelly's deafness. I will discuss resultatives in more detail chapter 7. For now, it should be sufficient to see that we do not necessarily have to assume that there are both transitive and intransitive versions of verbs like **build**. We can start out with the intransitive alternant, and construct the transitive alternant in the syntax using whatever composition process is at work in 13(b). The only adjustment needed is that we have to shift the property of being a hole into the property of being a state consisting in the existence of a hole, hopefully an independently motivated move. The prediction is that as far as the syntax is concerned, any intransitive verb should be able to appear in the construction illustrated by 13(a), and the only limits should be set by the semantics. Not every intransitive verb describes an activity that can be readily understood as creating things. But with a little imagination, suitable scenarios can be found even for activities like sneezing holes into a wheel of cheese.

So far so good, but what is now the status of the verb **construct**, a verb that also describes building activities, but is obligatorily transitive, with an object that also picks out the things that are being built? What does that verb look like when it enters a syntactic derivation? What is it that guarantees its transitivity? In recent work (Borer forthcoming), Hagit Borer explores the interesting idea that all transitivity is syntactically constructed. More generally, I understand the claim as implying that in, say, sentences like 13(b), the conceptual system provides nothing but the concept 'Nelly' for the

root **NELLY**, the concept ‘drill’ for the root **DRILL**, and the concept ‘deaf’ for the root **DEAF**. All other components that contribute to the meaning of 13(b) are provided by the functional structure of the sentence. Highly relevant for our current discussion is that according to Borer, the conceptual system does not provide any argument structure for lexical (substantive, as opposed to functional) items. Let’s start to spell out the proposal to see what the consequences are. We could think of the root **DEAF**, for example, as naming a kind, where in this particular case, the kind would have to be a kind of eventuality, rather than a kind of individual. If **DEAF** is a name for a kind, it doesn’t have arguments. Names never do. It would then be the job of functional structure, hence the syntax proper, to map names for kinds into predicates with arguments. A name for an event kind, for example, could be mapped into a property of events in a first step, and then into a relation between individuals and events as the derivation proceeds. This is a very radical proposal that implies that all relational concepts are the product of syntactic derivation. It is syntax that gives us relational concepts. If a relational concept doesn’t directly correspond to a basic functional category, it would have to be syntactically constructed with the help of such a category.

How realistic is Borer’s proposal? It runs into a first class of problems with irreducibly relational concepts. Take spatial concepts like ‘farness’ or ‘closeness’. There just can’t possibly be kinds of states consisting in mere closeness or farness. Closeness is always ‘closeness to’, and farness is always ‘farness from’. True, we use prepositions with **close** and **far**, but those prepositions can’t be the force that makes the concepts expressed by those two roots relational. Other relational concepts that look irreducible include ‘lack’, ‘resemblance’, ‘ancestry’, ‘possession’, ‘content’, ‘inclusion’, ‘ability’, ‘clue’, ‘goal’, ‘difference’, ‘connection’, ‘advantage’, ‘preference’, and countless others. It is hard to see how those concepts could be decomposed into an

argumentless substantive core and a piece contributed by a well-motivated functional structure providing precisely the right kind of arguments in each case. While the program of rethinking the role of functional structure in the creation of what we used to think were basic lexical items is extremely promising and has already been successful in several areas (in particular adverbial modification, telicity, and the mass/count distinction⁷), I still have to see proof that even the toughest cases of relational concepts can be syntactically decomposed in the way proposed by Borer.

Relational concepts play an important role in research on the minds of animals. Hauser 2000 writes that Herrnstein, “who spent close to twenty years working on this problem, concluded that nonhuman animals have concepts, but not abstract relational concepts; important supporting evidence was obtained from studies of pigeons that failed to understand the relational concept of inside versus outside (Herrnstein et al. 1989).”⁸ However, Hauser reports on a study he himself conducted with two colleagues (Hauser, Kralik and Botto) exploring the presence of relational concepts ‘on’, ‘off’, ‘connected’, and ‘disconnected’ in cotton-top tamarins, small New-World monkeys. The study provided evidence for the presence of spatial relational concepts in the absence of human language. Relational categories in the social domain have been identified in primates. According to Tomasello 1999, “primates are selective in choosing their coalition partners, selecting as an ally, for instance, an individual who is dominant to their potential adversary – indicating their understanding of the relative dominance ranks of these two

⁷. Apart from Borer’s book, Marcin Morzycki’s program of Mediated Modification is most relevant here. Morzycki does in fact commit himself to concrete decompositions in the area of adverbial modification, and spells out what the proposed division of labor between substantive core and functional structure is. Morzycki (forthcoming).

⁸. Hauser 2000, 561.

individuals. They also seek retribution for attacks against themselves not just on the attacker, but also in some circumstances on the attacker's kin – in this case evidencing an understanding of third-party kinship relations. And there is even some evidence that primates understand whole categories of third-party social relationships across different individuals, for example, many different instances of the relationship “mother-child”...”⁹ Given that animals without human language are able to form at least certain kinds of relational concepts, it is rather implausible that humans should have to rely on the functional structure provided by their uniquely human language faculty to build such categories in the syntax. What is known about relational categories in human and non-human primates, then, seems to support the view that humans have at least some basic relational concepts that are not syntactically constructed. Unlike their primate cousins, however, humans are also able to build more complex relational concepts by syntactic means. A mixed source for relational concepts in humans fits well with the conclusions in a recent paper by Elizabeth Spelke: “What makes humans smart? According to the first answer, human intelligence depends on a biological endowment of species-specific, core knowledge systems. According to the second answer, human intelligence depends both on core knowledge systems that are shared by other animals and on a uniquely human combinatorial capacity that serves to conjoin these representations to create new systems of knowledge. The latter capacity, I suggest, is made possible by natural language, which provides the medium for combining the representations delivered by core knowledge systems. On the second view, therefore, human intelligence depends both on a set of core knowledge systems and on the

⁹ . Tomasello 1999, 17.

human language faculty. Recent research on human infants, nonhuman primates, and human adults now seems to me to favor this view.”¹⁰

I take it then, that, in all likelihood, there are relational concepts that are carried by syntactic atoms without the help of functional structure. But as soon as you have relational concepts, you already have arguments. Not all argument structure is likely to be syntactically constructed, then, though I am ready to grant that a substantial part of it is. There is such a thing as initial, irreducible, transitivity, and we have to think about how to represent it. My proposal is that the semantics can do the job. If obligatorily transitive verb roots like **construct** denote relations between individuals and events, they have an individual and an event argument¹¹. The verb root’s denotation, then, is what makes it obligatorily transitive. There is no separate representation of argument structure. The root **construct** could then have the logical-conceptual representation given in (14):

(14) $\lambda x \lambda e \text{ construct}(x)(e)$

The format in (14) is a most economical way of representing the information that **construct** has an obligatory object denoting what is being constructed.

If all theme arguments were neo-Davidsonian in the syntax in the way Borer proposed, we would need some piece of functional structure that would be capable of accounting for the difference between **build** and **construct**. We

¹⁰ . Spelke 2003, 305.

¹¹ . This proposal might not hold up in the end. It might be that verbs of creation like **construct** denote relations between properties and events. This complication does not affect the point I want to make here, however. The verb’s denotation is what makes it obligatorily transitive.

would need some device that could formally mark obligatory transitivity.
There would have to be something that rules out (15):

(15) * **Nelly constructed.**

We might invoke a syntactic feature [transitive], to be checked against a matching feature of a direct object. If that object is to be neo-Davidsonian in the syntax, however, we still need a syntactic mechanism, a secondary predicate for example, to introduce it. Most importantly, we have to make sure that that syntactic mechanism introduces the right kind of object relation. As mentioned earlier, we have to exclude 16(b) and (c), for example:

- (16) a. **construct this barn** (effected patient)
 b. * **construct those plots** (affected patient)
 c. * **construct my yearly income** (consumed patient)

We also saw that we run into the same problem with **build**.

- (17) a. **build this barn** (effected patient)
 b. * **build those plots** (affected patient)
 c. * **build my yearly income** (consumed patient)

We have been granting that **build** is intransitive and that 17(a) is syntactically constructed. If both **construct** and **build** are intransitive, whatever excludes 17(b) and (c) should also exclude 16(b) and (c). But what is it that excludes 17(b) and (c)? 17(a) is a causative construction. As we will see in chapter 7, it is a concealed causative construction of a very general kind that is freely available with any intransitive verb. As far as 17(b) and (c) are concerned, we seem to be forced to the conclusion that there can be no

concealed syntactic mechanism introducing affected or consumed patients in a neo-Davidsonian way. There are no semantic reasons to exclude 17(b) or (c). In fact, we find verbs for building activities in German that have precisely that kind of direct objects:

- (18) a. **das Grundstück bebauen**
 the plot be-build
 ‘cover the plot by building’
 $\Box x \Box e$ [build(e) & cover (x)(e)]
- b. **das Geld verbauen**
 the money ver-build
 ‘use up the money by building’
 $\Box x \Box e$ [build(e) & use-up(x)(e)]

It seems, then, that the only way to exclude 17(b) and (c) in a theoretically satisfying way would be to say that the syntactic mechanism for adding the relevant kinds of objects is not available. It doesn’t exist. But this means trouble for the radical neo-Davidsonian. If all direct objects are neo-Davidsonian in the syntax, we would now predict that there couldn’t be any transitive verbs with affected or consumed patient arguments. This is very wrong. We already looked at a large collection of obligatory transitive verbs with affected patient objects, for example:

- (19) **Abduct, absorb, adjust, affect, amputate, arrest, attach, banish, buy, chase, check, clasp, clench, decorate, deport, dump, edit, educate, execute, feed, fix, grab, grasp, greet, harm, hurt, ignite, immerse, jar, jostle, jumble, kidnap, kill, label, loot, mail, maim, neglect, nip, occupy, operate, paralyze, pare, quench, record, recycle, slay, slice, sprain, squash, thrust, usher, veil, vend, wag, wreck,**

The story I am defending readily admits that there are affected or consumed patient arguments. What it denies is that they have to be neo-Davidsonian in the syntax. There doesn't seem to exist a freely available syntactic mechanism whose job is to introduce direct objects in a neo-Davidsonian way relying on general thematic relations like 'affected patient' or 'consumed patient'. Those relations do not seem to play a theoretical role in argument association.

Not surprisingly, the German verbs **bebauen** ('cover by building') and **verbauen** ('use up by building') are obligatorily transitive. The thematic role of their objects depends on the presence of the verb prefixes **be-** and **ver-**, but this dependence is non-compositional. Both prefixes are common transitivizers, but cannot be tied to affected or consumed patients consistently. Moreover, as the glosses for 18(a) and (b) make clear, the semantic contribution of the prefixes **be-** and **ver-** in connection with the verb **bauen** ('build') goes beyond merely contributing a general or only slightly more specialized thematic role. From a semantic point of view, German transitive prefix+verb compounds look more like largely non-compositional serial verb constructions. Two verbal meanings seem to be glued together, one of them already relational. This impression is confirmed when we inspect other compounds with **bauen**.

- (20) a. **die Scheune anbauen**
 the barn an-build
 'add the barn by building'
 $\Box x \Box e$ [build(e) & add(x)(e)]

- b. **das Haus umbauen**
 the house um-build
 ‘change the house by building’
 $\lambda x \lambda e$ [build(e) & change(x)(e)]

- c. **den Speicher ausbauen**
 the attic aus-build
 ‘finish the attic by building’
 $\lambda x \lambda e$ [build(e) & finish(x)(e)]

If the meanings of the verbal compounds in 20(a) to (c) are constructed from two more basic concepts, the component concepts are substantive, and most importantly, one is relational already. At the conceptual level, a property of events might have combined with a relation between individuals and events via Event Identification. The conceptual complexity of those verbs is still mirrored by their compound structure, but the conceptual atoms cannot generally be retrieved in a compositional way. The prefixes are mere pointers to a relational ingredient. They allow us to distinguish among the different object relations that are possible for a particular intransitive root, but they can't be said to denote any one of those relations. German has hundreds - if not thousands - of obligatorily transitive verbs that are built in this way, an utter nightmare for adult learners, who have to struggle with the subtle meaning differences thus conveyed. There are a lot of myths surrounding German verbal compounds: Myths trying to link their prefixes to telicity as in Slavic, myths trying to equip the prefixes with compositional meanings. True, there are productive pockets here and there, and there are some connections with telicity. But even a very superficial look at a dictionary will deliver a pervasive picture of non-compositionality defying any quick generalizations. This is typical of compounds. Interestingly, recent work by Gruber & Collins, Nishiyama, Cummings, and Collins has established a connection between verbal compounds and serial verb constructions in a

variety of languages¹². In the morphological literature, it is usually assumed that compounding does not involve functional structure¹³. Compounding joins roots or stems before they have had the opportunity to put on inflection. If that's so, the proposals of Collins, Gruber, Cummings and Nishiyama imply that even the syntactic addition of object arguments via serialization does not involve functional structure. William Snyder's work on the connection between the availability of resultative constructions and compounding (Snyder 2000) points to a similar conclusion. I will take up this important topic in chapter 7.

Returning to our English example, I conclude that while **build** is intransitive and its transitive uses are likely to be syntactically constructed, **construct** is already transitive when it enters the syntactic derivation. A host of other verbs should be, too - presumably all those listed in (19). To summarize, I have made a case against the assumption that all direct object arguments are neo-Davidsonian in the syntax. I first discredited the idea of a very general thematic role 'theme' by pointing out that it would lack cumulativity, an important property that other basic lexical items have. I then considered the possibility that direct object arguments might be introduced by less specialized, but still fairly general, thematic role predicates like 'effected patient', 'affected patient', and their kin. It turned out that some of those thematic role predicates express non-cumulative relations as well. Setting that problem aside, the assumption of neo-Davidsonian association with a whole repertoire of general direct object roles ran into an impasse when we started wondering how to ensure that verb roots would be connected to just

¹². Gruber & Collins 1997, Nishiyama 1998, Cummings 2001, and Collins 2002. But we would then have to wonder what it is that entitles a particle to adjoin to Voice.

¹³. This is not an entirely uncontroversial assumption. I will discuss (and defend) it in chapter 7. See Selkirk 1982 for what the main issues are.

the right kind of objects. Without any particular demands made by verb roots, we would expect many more transitivity alternations in natural languages than we do in fact observe. All verbs would start out intransitive, and would then be able to take a whole range of different kinds of objects. The verbs in (19), for example would be expected to allow all available object types in the repertoire, the only constraint being that no semantic inconsistency result. The vocabularies of natural languages do not seem to be constructed that way. The survey of English transitivity alternations given in Levin 1993 presents a very different picture, for example. In the following section, I will look at the major transitivity alternations in Levin's book from the point of view developed in this essay.

3.3 Transitivity alternations in Levin 1993: An agenda for theories of verb meanings

Some alternations classified as transitivity alternations by Levin would not come out as transitivity alternations on the present account. Those include the middle alternation illustrated in (21), and the causative/inchoative alternation illustrated in (22).

- (21) a. **The butcher cuts the meat.**
b. **The meat cuts easily.**

Levin 1993, 26.

- (22) a. **Janet broke the cup.**
b. **The cup broke.**

Levin 1993, 29.

The verbs in 22(a) and (b) are both built from a root expressing a relation between individuals and events. Such a root would be transitive from the

present point of view. It has a direct object argument denoting the thing that breaks. Crosslinguistically, roots of this kind are used to build unaccusatives or causatives, and that's what the alternation is about. The argument structure of the root **break** all by itself does not alternate in those constructions, assuming, as we do, that external arguments are not arguments of the verb root itself.

The middle alternation, too, seems to preserve the basic transitivity of the verb roots involved. This fact might be masked in English, but can be observed more clearly in the German translation of 21(b):

- (23) **Das Fleisch schneidet sich leicht.**
 The meat cuts itself easily.
 The meat cuts easily.

In (23), the direct object of the verb **schneiden** ('cut') is obligatorily represented as a reflexive pronoun. The subject of (23), then, does not realize the direct object of the verb, suggesting that the same situation might obtain in English, which would then have to be assumed to have an unpronounced reflexive pronoun in constructions like 21(b). The same difference between English and German shows up with respect to Levin's reflexive and reciprocal object alternations:

- (24) a. **Jill dressed herself hurriedly.**
 b. **Jill dressed hurriedly.**
 Levin 1993, 35.

- (25) a. **Anne met Cathy.**
 b. **Ann and Cathy met.**
 Levin 1993, 37.

German would have to use a reflexive pronoun in both 24(b) and 25(b). I will discuss middles and reflexives in chapter XXX. The phenomenon exemplified in Levin's middle, reflexive, and object alternations will be argued to be related to a voice alternation, hence to an alternation relating to external not internal arguments. The argument structure of verbal roots remains unaffected in those constructions.

More has to be said about roots like **cut**, however. **Cut** does have intransitive uses, as illustrated in 26(b). (26) is an example of the conative alternation.

- (26) a. **Margaret cut the bread.**
b. **Margaret cut at the bread.**

Levin 1993, 41.

The conative alternation is an alternation of the kind that is of great interest here. Isn't the bread an affected patient when you cut it? It is, but the example brings out a recurrent problem with the 'affected patient' role. It is still too general to be of practical use. If the bread is an affected patient of my cutting activity, so is the knife and the cutting board. If we could use the 'affected patient' relation for the neo-Davidsonian association of objects, we would expect the following unattested constructions, among many others:

- (27) a. * **Margaret cut the knife.**
 'Margaret used the knife to cut something.'
b. * **Margaret cut the cutting board.**
 'Mary cut something on the cutting board.'

Within the present framework of assumptions, we could consider the possibility that there are two transitive stems, **cut** and **cut at**, expressing two slightly different cutting relations. Or else there might be an intransitive verb root **cut**, and a corresponding transitive root that sounds the same because it has a zero affix attached to it functioning like a German prefix. Generalizations for related paradigms in English and other languages should help us choose the right option. Whatever the outcome may be, we see one more time that with actual transitivity alternations like the conative one, general thematic relations are of no help. The verb roots themselves ask for very specific object relations. Moreover, as in the case of **build** and **construct**, there are English verbs describing cutting activities that are obligatorily transitive. Among the non-alternating verbs Levin mentions, the following ones seem obligatorily transitive: **chop**, **crop**, **dice**, **mince**, **mow**, **prune**, **slice** and **slit**. Having to distinguish those from the alternating verbs **chip**, **clip**, **cut**, **saw**, **slash**, and **snip** seems to doom any theory advocating neo-Davidsonian association for all direct objects.

A related transitivity alternation is Levin's Locative Preposition Drop alternation illustrated in (28).

- (28) a. **Martha climbed up the mountain.**
b. **Martha climbed the mountain.**

Levin 1993, 43.

Here, German marks the contrast with an overt prefix, suggesting that English, too, might use a zero-prefix to mark the transitive alternant.

- (29) a. **Martha stieg auf den Berg.**
Martha climbed up the mountain.
- b. **Martha bestieg den Berg.**
Martha climbed the mountain.

The problems for neo-Davidsonian association are familiar by now. General thematic roles like ‘affected patient’ are too general to pick out the right kind of objects. When you climb a mountain, your boots and feet are affected, too. Yet we don’t climb our hiking boots or our tired feet. Consumed patients are not possible at all. Even though mountain climbing consumes a lot of energy, we do not climb a lot of energy. And so on. Here, too, alternating verbs contrast with non-alternating verbs that have closely related meanings. Levin mentions the alternating verbs **canter, climb, cross, fly, gallop, hike, jog, jump, leap, prowl, ramble, ride, roam, rove, row, run, shoot** (rapids), **stroll, swim, traipse, tramp, travel, trudge, vault, wade, walk**, and **wander**, which contrast with the following ones that only have intransitive uses: **bounce, drift, drop, float, glide, move, roll, slide, swing, spin, turn, twirl, whirl, wind**. In this case, then, a class of verbs that only has intransitive uses contrasts with one that has both transitive and intransitive uses. This seems to exclude neo-Davidsonian association for the transitive uses. If neo-Davidsonian association was responsible for the transitive uses of the first group, how could we prevent that very same mechanism from applying to the second group as well?

If all direct objects were neo-Davidsonian in the syntax, verbs could not exercise much control over their direct objects. Whatever control there was would have to be mediated by the event argument. As a consequence, the range of possible objects a verb can take would be much bigger than what we in fact observe. There is a crucial asymmetry between external and internal

arguments in this respect, then, that I will return to again and again in this essay.

So far, I have addressed neo-Davidsonian addition of direct objects in the syntax. What about logical-conceptual structure? If there is such a thing as logical-conceptual structure, we expect the cumulativity requirement to apply at that level as well. Consequently, there would be no logical-conceptual predicate ‘affected patient’, for example, that could introduce direct object arguments in a neo-Davidsonian way. Cumulativity would exclude that possibility. But what about logical-conceptual representations like the one in 30(b)? Are there any reasons to exclude 30(b) in favor of 30(a), for example?

- (30) a. $\Box x \Box e \text{ construct}(x)(e)$
 b. $\Box x \Box e [\text{construct}(e) \ \& \ \text{make}(x)(e)]$

In matters of logical-conceptual structure, the situation is rather unclear, and quite controversial. The logical-conceptual representations in 31(a) to (c), for example, capture the meanings of verbs all right, but to be honest, I really do not have the necessary evidence to justify the particular decompositions I posited.

- (31) a. **die Scheune anbauen**
 the barn an-build
 ‘add the barn by building’
 $\Box x \Box e [\text{build}(e) \ \& \ \text{add}(x)(e)]$
- b. **das Haus umbauen**
 the house um-build
 ‘change the house by building’
 $\Box x \Box e [\text{build}(e) \ \& \ \text{change}(x)(e)]$

- c. **den Speicher ausbauen**
 the attic aus-build
 ‘finish the attic by building’
 $\Box x \Box e$ [build(e) & finish(x)(e)]

As mentioned in chapter 1, we may wonder whether there is any genuine lexical decomposition at logical-conceptual structure at all, that is, decomposition that is not matched by parallel decomposition in the syntax. In a series of publications since 1970, Jerry Fodor and Fodor and Lepore have criticized standard arguments that allegedly support lexical decomposition¹⁴. Hale and Keyser’s lexical theory reduces all decomposition to syntactic decomposition¹⁵. In the interest of learnability, we expect the relationship between the syntactic representations of lexical items and their logical-conceptual counterparts to be highly constrained, and in the best of all possible worlds, the two representations would match completely. For my current project, I do not have to try to resolve the issue. Argument association in the syntax is what I am primarily interested in, and here, we have at least some evidence suggesting that direct objects are not introduced by a general thematic role predicate ‘theme’, nor by any of the slightly less general thematic role predicates found in common thematic role inventories.

My strategy for discrediting the theme relation relied on cumulativity, a property that emerged as a possible universal for the denotations of simple transitive verbs and thematic role predicates. Cumulativity as a constraint on basic predicative meanings is an important topic all by itself, even though, in this chapter, I have primarily used it as ammunition against a thematic

14 . Fodor 1970, Fodor and Lepore 1998 and other references cited there.

15 . Hale and Keyser 1993, 1999.

role predicate ‘theme’. In the following chapter, I will supplement the anecdotal evidence for cumulativity that I have given so far with an in-depth theoretical evaluation. This evaluation will take us far into the semantics of plurals and distributivity. If only the status of the theme relation was at stake, such an excursion might very well be too time consuming and costly. Since my case against the theme relation is at the same time a case for a powerful universal restricting the range of possible basic verb meanings, however, the excursion will be well worth its money in the end.

Chapter 4

Cumulativity as a possible universal

In this chapter I will explore the cumulativity properties of verbal projections. My immediate goal is to find solid evidence for Lexical Cumulativity of basic verbs and thematic role predicates. But in the course of our investigation, we will also have the opportunity to gain insights into many other phenomena affected by cumulativity and events: collectivizing adverbs, downward entailing quantifiers, durational adverbials, collective nouns, groupings of pluralities, plural agreement morphology, and neo-Davidsonian argument association. We will find that Lexical Cumulativity interacts with many other phenomena in just the right way, supporting the points made in other places, in this book and in other books, by this author, and by others. In the end, we will have a comprehensive assessment of verbal cumulativity that brings together the efforts and agonies of many scholars in this lively field of semantic research.

4.1 Cumulativity

In this section, a Cumulativity Universal is introduced for basic lexical predicates in natural languages. It is then argued that simple nouns satisfy the Cumulativity Universal.

Over the last 15 years or so ¹, Manfred Krifka has explored cumulativity as an important property of nominal and verbal predicates, and in the course of this work, the possibility emerged that cumulativity might correspond to a significant semantic universal: “simple predicates in natural language typically are cumulative”.² Here is a way of stating the universal:

(1) Cumulativity Universal

The denotations of simple predicates in natural languages
are cumulative.

Let us first see how the notion of cumulativity has to be generalized so as to apply to the full range of cases we are interested in, and then investigate whether the scope of the proposed universal can be extended beyond the cases we looked at in the last chapter.

Within the framework assumed here, the denotations of 1-place predicates are (the characteristic functions of) subsets of the domain of individuals D_e or the domain of events and states (that is, eventualities) D_s . According to Link 1983, the domain of individuals D_e contains singular, or atomic, individuals and plural individuals³. Plural individuals are sums of atomic individuals. The domain D_e is closed under sum formation. Closure under sum formation can mean that whenever x and y are members of D_e , then $x+y$ (the sum of x and y) is a member of D_e . For non-finite D_e , closure under sum formation can

¹. Krifka 1986, 1987, 1992, 1998.

². Krifka 1998, p.200.

³. I am neglecting denotations for mass nouns. Mass nouns do not pose a threat for the Cumulativity Universal in any case.

also mean that any non-empty subset of D_e , whether it is finite or not, has a sum in D_e . To keep things simple, I will mostly neglect the possibility of non-finite domains. Adjustments to the non-finite case are straightforward if needed. If the domains D_e is cumulative, certain subsets of D_e are cumulative as well, and cumulativity can be extended to the characteristic functions of those sets, that is, to the members of $D_{\langle et \rangle}$. We have:

(2) Cumulativity (properties of individuals)

$$\Box P_{\langle et \rangle} \Box x \Box y [[P(x) \ \& \ P(y)] \Box P(x+y)]]$$

Assuming that the domain of events and states D_s is also cumulative, cumulativity for the members of $D_{\langle st \rangle}$ comes out as expected:

(3) Cumulativity (properties of events)

$$\Box P_{\langle st \rangle} \Box e \Box e' [[P(e) \ \& \ P(e')] \Box P(e+e')]]$$

Cumulativity for the denotations of other relevant types of predicates can be defined in a parallel way.

We can now begin to look into the question whether it is true that the denotations of basic lexical predicates in natural languages are cumulative. Blatant counterexamples seem to be singular count nouns like **child**, **chair**, or **chin**. Following Link, the extensions of singular count nouns are usually taken to be sets of singularities, hence are not assumed to be cumulative. However, for count nouns, the singular feature has a completely compositional interpretation. When singular, a count noun always denotes a

set of atomic individuals⁴. It should be possible, then, to derive the denotation of a singular count noun compositionally from the denotation of the number feature, which might be a piece of inflection, and the denotation of a number-neutral noun stem, which is the lexical item itself. Müller 2000 proposes to derive the denotations of number marked count nouns in Brazilian Portuguese in exactly that way. The singular noun **child**, for example, is analyzed as consisting of the number-neutral lexical item **CHILD** and the meaningful number feature **singular**. Consequently, singular count nouns are not lexical items, but lexical items plus inflectional morphology, and can therefore have non-cumulative denotations without violating the Cumulativity Universal. The semantic separation of nouns and their number feature has to be present at some level of representation. One possibility is for number features to correspond to functional heads that are picked up by nouns in the course of a syntactic derivation via head movement⁵. Alternatively, nouns might start out fully inflected, but their inflectional features would be meaningless and would have to be checked against matching meaningful features carried by higher inflectional heads. In either case, we have to posit number neutral denotations for the nouns themselves. I am aware of two options that have been proposed, both of which are compatible with the Cumulativity Universal. On the first option, **CHILD** is predicative and denotes the smallest set that contains all atomic children and is closed under sums. This kind of number-neutral noun denotation is the one considered by Müller. It is a plural, but not a proper plural denotation in the sense of Link 1983. It is not a proper plural denotation since it contains

⁴. For the status of collective nouns like **committee** see below.

⁵. Halle and Marantz 1996, section 6, has some discussion of syntactic processes of ‘picking up’ meaningful morphological features.

atomic individuals along with their sums. The inflectional feature **singular** now has to be a function that picks out the largest subset of atomic individuals from any set. Alternatively, we might assume that all inflectionless nouns are referential, and that they refer to kinds. Proposals of this kind are made in Longobardi 1994 and Krifka 1995⁶. The feature **singular** now denotes a function that maps kinds into the set of its atomic realizations. If nouns (all by themselves) denote sets of individuals that are closed under sums, they have cumulative denotations. If they denote kinds, they satisfy the Cumulativity Universal trivially. I conclude that we do not have to worry about nouns. In one way or other, they comply.

4.2 Cumulativity for verbs and thematic role predicates

This section illustrates the view that the denotations of all basic verbs and thematic role predicates are cumulative from the very start. Crucially, verbal cumulativity is not specifically linked to atelicity, as sometimes assumed erroneously.

As for verbs and the thematic role predicate ‘agent’, I already presented evidence in the previous chapter that motivated the hypothesis that there might be such a thing as a Cumulativity Universal in the first place. Cumulativity is also at the heart of Schein’s argument we looked at in chapter 2. This chapter is about the repercussions of cumulativity within verbal projections. Is the Cumulativity Universal compatible with a mereological theory of plurality? Is lexical cumulativity enough? If there is

6. Longobardi’s and Krifka’s proposals were inspired by Carlson 1977. However, Carlson’s actual proposal is different. On Carlson’s account, singular and plural count nouns start out with predicative denotations. The denotation of a plural count noun can then be lifted to a generalized quantifier that corresponds to a set of properties of kinds. Chierchia 1998 argues that nouns might denote kinds or properties as a matter of parametric variation between languages.

non-lexical cumulativity, where does it come from? What is the role of verbal number agreement?

In the nominal domain, we were able to maintain the Cumulativity Universal by distinguishing between the denotations of nouns themselves and the semantic impact of number features contributed by meaningful nominal inflection. The recognition that meaningful number marking is not necessarily a property of the nouns *per se* led to our positing number-neutral denotations for inflectionless nouns. In English, number marking is also part of the inflection of verbs, and verbs and their subjects agree in number. If there is an unpronounced agreement marker for accusative objects, there might also be non-overt number agreement between a verb and its object. It is not clear whether there is any number inflection related to the event argument. Event plurality might be indicated by pluractional markers, however. Crosslinguistically, the morphological status of pluractional markers is still an open question. Most known pluractional markers seem to be affixes, “frequently reduplicative, most often derivational rather than inflectional...”⁷.

“The usual view of pluractional morphemes is that they function as a kind of plural marker for verbs. Of course plural marking on verbs is familiar from the phenomenon of number agreement with a plural argument, exhibited by a wide variety of languages; but here we mean something different. Pluractional markers do not reflect the plurality of a verb’s arguments so much as plurality of the verb itself: the verb is understood to represent the occurrence of multiple events.” Lasersohn, 1995, p. 241.

⁷. Lasersohn, 1995, p. 238.

Is overt number agreement on English verbs meaningful? Roberts 1987, 1990 has argued that verbal number morphology is a purely syntactic agreement phenomenon. One of her examples is (4):

(4) **John bought a house, and Bill and Mary did, too.**

Roberts' point is that assuming that in VP-ellipsis the elided VP and its antecedent have to have the same denotation, (4) seems to show that singular and plural VPs cannot have different denotations. From the present perspective, it is significant that in the VP ellipsis construction (4), the second conjunct has its own inflection, and therefore its own number marking (carried by the auxiliary **do**). The elided constituent in (4) is an inflectionless VP that is anaphoric to another inflectionless VP in the first conjunct. VP ellipsis cases like (4), then, do not establish that number marking on verbs is a mere syntactic agreement phenomenon.

When we ask about the meanings of verbs and VPs within the current framework of assumptions, we are talking about the meanings of 'bare' verbs and VPs, which are verbal projections that do not yet include any functional structure. Suppose the functional projections of verbs are built step by step in the course of a syntactic derivation by introducing ('merging') functional heads with possibly meaningful features. This is compatible with the view that verbs enter a syntactic derivation fully inflected⁸, as long as the features of those initial pieces of inflection are not meaningful themselves. Possible carriers of inflectional meaning would be matching features carried by functional heads. The question is now whether the number features of verbal

⁸. Chomsky 1995.

functional heads are ever meaningful. Suppose they are. Verbs and VPs, - and in fact all verbal projections below the point where functional heads with number features come in, - should then have denotations that allow us to construct singular and plural denotations with the help of number operators. Alternatively, suppose that verbal number features are not meaningful. In that case, verbs and verbal projections should have denotations that, without any further modification by number operators, directly make the right contributions to the truth-conditions of the sentences they occur in. The important point is that in either case, number-neutral denotations are needed for bare verbs and VPs. It is those number neutral denotations that I will be concerned with. The chunks of a verb's extended projection that we will be examining in this book are mostly located below the point where number features might leave their mark. We will mostly have to consider number-neutral denotations, then, that is, denotations that have not yet been affected by number operators, - if indeed they ever will be.

The prime candidates for number-neutral verb denotations are cumulative denotations, of course. If the denotations of verbs and thematic role predicates are cumulative from the start, the availability of cumulative⁹ readings for sentences like 5(a) is expected, as pointed out in Krifka 1992. Assuming initial cumulativity, we can represent those readings as in 5(b)¹⁰:

⁹. The term is due to Scha 1981,1984. Kroch 1974 coined the name 'serially distributive reading' for the salient reading of **The men in the room are married to the girls across the hall** (p. 204 f.). Sauerland 1998 uses the term 'co-distributive reading'.

¹⁰. Both Krifka 1992 and Landman 1996, 2000 work within a neo-Davidsonian event semantics and rely on initial cumulativity of verbs and thematic role predicates for the logical representation of cumulative readings of sentences analogous to 5(a).

- (5) a. **Twenty children ate ten pizzas.**
 b. $\exists e \exists x \exists y [\text{children}(x) \ \& \ /x/ = 20 \ \& \ \exists \text{agent}(x)(e) \ \& \ \text{pizzas}(y) \ \& \ /y/ = 10 \ \& \ \exists \text{eat}(y)(e)]$ ¹¹

5(b) improves on early analyses of cumulative readings. Take Remko Scha's¹². Scha's paraphrase for the cumulative reading of 5(a) would presumably be: The total number of children who ate a pizza was 20, and the total number of pizzas that were eaten by a child was 10. In contrast, 5(b) allows sharing of pizzas, and it doesn't impose 'exactly'-readings for the two numerals. Sharing of pizzas is automatically taken care of by neo-Davidsonian separation of the agent argument, as proposed by Schein. Landman uses a similar example as an objection to Schwarzschild's theory of plurality¹³, which is not based on events. While 5(b) is a problem for Schwarzschild's actual theory, it is still not an argument for the highly structured nominal domains that Landman has proposed.

All predicates in 5(b) have cumulative denotations. As in Landman (1996, 2000), the basic verb and thematic role predicates of the logical representations are singular predicates that are pluralized with a $\exists$ -operator that maps properties and relations into their smallest cumulative extensions. If every basic verb and thematic role predicate has a cumulative denotation

¹¹. For any individual x , $/x/$ is only defined if there is a set of atomic individuals that x is the sum of. If defined, $/x/$ is the number of atomic individuals that x is the sum of, that is, $/x/ = / \{y: y \leq x \ \& \ \text{atom}(y) \} /$.

¹². Scha1981, 1984.

¹³. Landman 1996, p. 455.

from the start, there is no need to repeat that information for every lexical item, of course. However, using the $\sqcup$ -operator even for those predictable cases is still pedagogically useful as a reminder that we are dealing with cumulative denotations. I will follow this practice for clarity.

To see the impact of cumulativity on verb denotations within an event semantics in more technical detail, let us examine a very simple example, the small clause in (6):

(6) **(We made) two children lift two boxes.**

On the intended cumulative reading, the small clause in (6) is compatible with a wide range of situations, as long as two children did the lifting and two boxes were lifted in all. The children might have acted individually or jointly. The boxes might have been lifted one at a time or both together. And either box or the two boxes together might have been lifted once or several times. How does (6) manage to cover so many different kinds of situations? Suppose the two children are Casey and Stacey, and the two boxes are Red and Green. Casey lifted Red on her own once, and Stacey did so twice. In addition, Casey and Stacey jointly lifted Green. We have four events, e_1 , e_2 , e_3 , and e_4 , then, which can be characterized as follows:

	Box lifted	Box lifter
e_1	Red	Casey
e_2	Red	Stacey
e_3	Red	Stacey
e_4	Green	Casey+Stacey

In truth-conditional semantics, the extensions of predicates depend on relations that hold in the actual world. A customary, non-cumulative, extension for ‘lift’ would pair actual lifting events with the objects actually lifted, for example. Disregarding Schönfinkelization, the relation would include the pairs listed in (7) (assuming our scenario is true):

(7) Extension of ‘lift’

{<e₁, Red>, <e₂, Red>, <e₃, Red>, <e₄, Green>,.....}

The customary extension of ‘agent’ would include the pairs in (8):

(8) Extension of ‘agent’

{<e₁, Casey>, <e₂, Stacey>, <e₃, Stacey>, <e₄, Casey +Stacey>,.....}

So far, we have a close match between what might be ‘basic’ relations in the actual world and the relations in the extensions of the predicates. Intuitively, there are four box lifting events and three different agents. One of the agents is a plural individual, and that means that there is collective action. These kinds of denotations reflect nicely what is going on in the world as we see it. At this stage, extensions satisfy what Fred Landman has called the ‘Collectivity Criterion’¹⁴. All plural individuals paired with an event are collectively involved in that event. All plural agents are collective agents, then. Enters Cumulativity, and our extensions turn to mush:

¹⁴ . Landman 1996, 2000.

- (9) a. Extension of 'lift'
- {<e₁, Red>, <e₂, Red>, <e₃, Red>, <e₄, Green>, <e₁+e₂, Red>, <e₁+e₃, Red>, <e₁+e₄, Red+Green>, <e₂+e₃, Red>, <e₂+e₄, Red+Green>, <e₃+e₄, Red+Green>, <e₁+e₂+e₃, Red >, <e₁+e₂+e₄, Red+Green>, <e₁+e₃+e₄, Red+Green>, <e₂+e₃+e₄, Red+Green>, <e₁+e₂+e₃+e₄, Red+Green>, }
- b. Extension of 'agent'
- {<e₁, Casey>, <e₂, Stacey>, <e₃, Stacey>, <e₄, Casey+Stacey>, <e₁+e₂, Casey+Stacey>, <e₁+e₃, Casey+Stacey>, <e₁+e₄, Casey+Stacey>, <e₂+e₃, Stacey>, <e₂+e₄, Casey+Stacey >, <e₃+e₄, Casey+Stacey>, <e₁+e₂+e₃, Casey+Stacey>, <e₁+e₂+e₄, Casey+Stacey>, <e₁+e₃+e₄, Casey+Stacey>, <e₂+e₃+e₄, Casey+Stacey >, <e₁+e₂+e₃+e₄, Casey+Stacey >, }

The cumulative extensions in 9(a) and (b) include more than just the basic relations between individuals and events we might be prepared to recognize. There are more lifting events than we ever dreamed of, and, strangely, the sum of Casey and Stacey is the plural agent of most of them. There is nothing intrinsically bad about this state of affairs, however, as long as the truth conditions we predict are right. Are they?

Assuming the cumulative denotations partially listed in (9), the logical representation 10(a) correctly comes out true. The open sentence 10(b) is satisfied by several variable assignments, including the one in 10(c):

- (10) a. $\exists e \exists x \exists y [\text{children}(x) \ \& \ /x/ = 2 \ \& \ \exists \text{agent}(x)(e) \ \& \ \text{boxes}(y) \ \& \ /y/ = 2$

- & λ lift(y)(e)]
- b. children(x) & /x/ = 2 & λ agent(x)(e) & boxes(y) & /y/ = 2
 & λ lift(y)(e)
- c. 'e' $\sqcup$ e₁+ e₂+ e₃+ e₄
 'x' $\sqcup$ Casey+Stacey
 'y' $\sqcup$ Red + Green

Having cumulative, number-neutral, denotations yields correct results not only for plural VPs, as in 11(a) and (b) below, but also for singular VPs, as in 11(c) and (d). Assume the same scenario as before and look at the following sentences:

- (11) a. **Casey and Stacey lifted Red.**
 b. **Casey and Stacey lifted Green.**
 c. **Casey lifted Red (at least) once.**
 d. **Stacey lifted Red (at least) twice.**

11(a) to (d) should all come out true, and they do. 11(a) is verified by e₁+e₂, e₁+e₃, and e₁+e₂+e₃. 11(b) is verified by e₄. The fact that Stacey, but not Casey, lifted Red twice is in principle retrievable from 9(a) and (b) as well. There is an event (namely e₂+e₃) that has Stacey as its agent, and also has two proper subevents, each of which is a lifting of Red by Stacey. As for Casey's liftings of Red, there is only one such event, e₁.

4.3 The challenge: Together

Assuming cumulative denotations for verbs and thematic role predicates seems to blur the distinction between collective and distributive predication. However, in the presence of adverbs like **together**, that distinction still needs to be made. In his dissertation and later

work, Peter Lasnik has shown that within an event semantics, the distinction between collective and distributive predication can be retrieved, even if the predicates involved have cumulative denotations. “The basic idea was to base the semantics of *together* and related expressions ... around the part/whole structure of the events themselves.”¹⁵ Crucially, Lasnik’s method makes it possible to maintain the Cumulativity Universal without forcing us to posit more complicated denotations for plural definite DPs.

Within mereological frameworks, the potentially most serious problem with cumulative verb denotations is that the distinction between distributive and collective involvement of plural individuals in events might get lost. For those who have singular and plural predicates at their disposition, collective predication is the result of applying a singular property to a plural individual, and distributive predication comes about by applying a plural property to a plural individual. If we nevertheless insist on cumulativity for verbs and thematic role predicates, doesn’t that mean that we have to put up with a more populated universe of pluralities in the nominal domain? In fact, Krifka’s 1992 pitch for cumulative verb denotations concludes with the remark that

“to cover collective readings, as e.g., *John and Mary (jointly) own three houses*, we need of course a different representation, which will not be developed here.”¹⁶

Following the spirit of Schwarzschild¹⁷, but in a mereological framework, I have been relying on a simple ‘sums theory’ for pluralities. In view of the strong arguments Schwarzschild has given for this approach to pluralities, I

¹⁵. Lasnik 1995, 189.

¹⁶. Krifka 1992, 44.

¹⁷. Schwarzschild 1991, 1992, 1994, 1996.

would be reluctant to abandon it. However, we now have to take very seriously the question whether the assumption that verb denotations are cumulative from the very start still allows us to distinguish collective and distributive predication without overpopulating our universe of pluralities. Simplicity for verb denotations should not come at the cost of inflation in the nominal domain.

To get a feeling for the problem that Krifka was alluding to, look at the following sentences:

- (12) a. **Casey and Stacey lifted Green.**
b. **Casey and Stacey lifted Red.**
- (13) a. **Casey and Stacey lifted Green together.**
b. **Casey and Stacey lifted Red together.**
- (14) a. **Casey and Stacey lifted Green individually.**
b. **Casey and Stacey lifted Red individually.**

Given our earlier scenario, 12 (a) is true when understood as collective predication, and false when understood as distributive predication. 12(b), on the other hand, is false as collective predication, and true as distributive predication. Even if we were reluctant to say that 12(a) and (b) are truly ambiguous, we would still have to be able to retrieve the distinction between collective and distributive predication from the extensions of verbs, since adverbs like **individually** and **together** depend on it. 13(a) and 14(b) are true, and 13(b) and 14(a) are false. The denotations we posit for verbs and

thematic role predicates, then, must contain enough information to provide adverbs with sufficiently fine-grained extensions to operate on.

In his dissertation and later work, Peter Lasnik¹⁸ has developed an analysis for collectivizing adverbs within an event semantics that is compatible with cumulative verb denotations without automatically triggering inflation in the nominal domain. To see an example of the kind of analysis he proposed, let us go back to the extensions of ‘ λ lift’ and ‘ λ agent’ we looked at earlier:

(9) a. Extension of ‘ λ lift’

{<e₁, Red>, <e₂, Red>, <e₃, Red>, <e₄, Green>, <e₁+e₂, Red>, <e₁+e₃, Red>, <e₁+e₄, Red+Green>, <e₂+e₃, Red>, <e₂+e₄, Red+Green>, <e₃+e₄, Red+Green>, <e₁+e₂+e₃, Red>, <e₁+e₂+e₄, Red+Green>, <e₁+e₃+e₄, Red+Green>, <e₂+e₃+e₄, Red+Green>, <e₁+e₂+e₃+e₄, Red+Green>, }

b. Extension of ‘ λ agent’

{<e₁, Casey>, <e₂, Stacey>, <e₃, Stacey>, <e₄, Casey+Stacey>, <e₁+e₂, Casey+Stacey>, <e₁+e₃, Casey+Stacey>, <e₁+e₄, Casey+Stacey>, <e₂+e₃, Stacey>, <e₂+e₄, Casey+Stacey>, <e₃+e₄, Casey+Stacey>, <e₁+e₂+e₃, Casey+Stacey>, <e₁+e₂+e₄, Casey+Stacey>, <e₁+e₃+e₄, Casey+Stacey>, <e₂+e₃+e₄, Casey+Stacey>, <e₁+e₂+e₃+e₄, Casey+Stacey>, }

¹⁸ The 1988 dissertation was published in 1991. The later works include Lasnik 1990, 1995 and 1998.

Take $e_1 + e_2$. The event $e_1 + e_2$ is an event of lifting Red that has Casey+Stacey as agent. This is sufficient to make 12(b) true. We now want to find a condition that tells us why 13(b) is false. Here is a possibility: 13(b) is true if there is some event of lifting Red that has Casey+Stacey as agent, but doesn't have any subevent that is a lifting of Red by anybody but Casey+Stacey. Our scenario doesn't provide such an event. The only events that are events of lifting Red that have Casey+Stacey as agent are $e_1 + e_2$, $e_1 + e_3$, and $e_1 + e_2 + e_3$, but each of those events has subevents that are liftings of Red and have either Casey or Stacey alone as agents. 13(b), then, is correctly ruled false. Next, look at 13(a). Applying the same condition, 13(a) is true if there is some event that is a lifting of Green that has Casey+Stacey as agent, but lacks any subevent that is a lifting of Green by anybody but Casey+Stacey. Since we have a suitable event in our scenario, namely e_4 , 13(a) winds up true, as it should. The condition we have just looked at can be stated as in (15)¹⁹:

(15) $T(\text{together}) =$

$$\exists R_{\langle e \langle st \rangle \rangle} [\exists y \exists e [R(y)(e) \ \& \ \text{plural}(y) \ \& \ \exists e' \exists z [[e' \leq e \ \& \ R(z)(e')]] \ \& \ z = y]]$$

¹⁹. (15) is basically Lasnik's condition for collectivizing **together** (Lasnik 1988, 1990), adapted to the semantic framework I am using. Lasnik's semantics is a situation semantics in the sense of Barwise and Perry 1983, whereas I am working within a Davidsonian event semantics. Rather than talking about events that are liftings of this box by Stacey, for example, Lasnik would talk about events in which Stacey lifted this box. Having a Davidsonian event semantics makes neo-Davidsonian association of external arguments possible, an option Lasnik does not have. This in turn has consequences for the analysis of Schein sentences. Lasnik 1995 has a revised analysis of **together**, which makes assumptions about team credit that are not compatible with the present work.

Using (15), the denotation of sentences like 13(a) or (b) can be derived as follows:

1. $T(\text{lift Green}) = \lambda e [\lambda \lambda \text{lift}(\text{Green})(e)]$
2. $T([\text{active}] (\text{lift Green})) = \lambda y \lambda e [\lambda \text{agent}(y)(e) \ \& \ \lambda \text{lift}(\text{Green})(e)]$
3. $T(\text{together}) =$
 $\lambda R_{\langle e, \langle st \rangle \rangle} \lambda y \lambda e [R(y)(e) \ \& \ \text{plural}(y) \ \& \ \lambda e' \lambda z [[e' \leq e \ \& \ R(z)(e')]] \ \& \ z = y]$
5. $T([\text{active}] (\text{lift Green})) \text{ together} =$
 $\lambda y \lambda e [\lambda \text{agent}(y)(e) \ \& \ \lambda \text{lift}(\text{Green})(e) \ \& \ \text{plural}(y) \ \& \ \lambda e' \lambda z [[e' \leq e \ \& \ \lambda \text{agent}(z)(e') \ \& \ \lambda \text{lift}(\text{Green})(e')]] \ \& \ z = y]$
6. $T(\text{Casey and Stacey}) = \text{Casey+Stacey}$
7. $T(\text{Casey and Stacey} ([\text{active}] (\text{lift Green})) \text{ together}) =$
 $\lambda e [\lambda \text{agent}(\text{Casey+Stacey})(e) \ \& \ \lambda \text{lift}(\text{Green})(e) \ \& \ \text{plural}(\text{Casey+Stacey})$
 $\ \& \ \lambda e' \lambda z [[e' \leq e \ \& \ \lambda \text{agent}(z)(e') \ \& \ \lambda \text{lift}(\text{Green})(e')]] \ \& \ z = \text{Casey+Stacey}]$

As is, (15) might be too strong. (16) below, for example, seems compatible with a situation where one of the copy editors was looking for mistakes, but didn't find any, or where two copy editors found the very same mistakes:

(16) **The 10 copy editors together caught those 20 mistakes.**

In such situations, the events verifying (16) have subevents in which a proper subgroup of the 10 copy editors caught the 20 mistakes. Maybe what **together** does is convey that none of the copy editors alone caught those

mistakes. It is a marker of non-distributivity²⁰. We might then have (17) instead of (15):

(17) $T(\text{together}) =$

$$\lambda R_{\langle e, \langle st \rangle \rangle} \lambda y \lambda e [R(y)(e) \ \& \ \lambda z [[z \leq y \ \& \ \text{atom}(z)] \rightarrow \lambda e' [e' \leq e \ \& \ R(z)(e')]]]$$

(17) assumes wrongly that the relevant parts of a plurality are always the atoms it is composed of. This is not necessarily so, however, a point Roger Schwarzschild has emphasized in his works on plurality. Suppose the ten copy editors work in teams of two, and each of the five teams found all 20 mistakes. (16) would not be true in such a situation, even though none of the individual copy editors alone found the 20 mistakes. Quantification over plural parts of pluralities is context dependent. (17) might reflect this context sensitivity with the help of a free variable ‘C’:

(17') $T(\text{together}) =$

$$\lambda R_{\langle e, \langle st \rangle \rangle} \lambda y \lambda e [R(y)(e) \ \& \ \lambda z [[z \leq y \ \& \ C(z)] \rightarrow \lambda e' [e' \leq e \ \& \ R(z)(e')]]]$$

Following Schwarzschild, possible values for ‘C’ might be restricted to properties picking out cells of a contextually salient partition or cover of y ²¹. I will not try to choose between (15) and (17). The following discussion will target the kind of account given to collectivizing adverbs by Lasersohn,

²⁰ . Schwarzschild 1993-94 captures this intuition in a non-event-based framework.

Schwarzschild 1993-94 also has provisions for the context dependency affecting **together** and discusses possible restrictions for what the relevant parts of a plurality are.

²¹ . A cover of a plurality y is any set of parts of y whose sum is y . A partition is a cover, but without overlap between its members.

rather than the exact conditions for togetherness imposed. To stay close to the Lasersohn-Schwarzschild debate in this area, I will continue with (15) while keeping its competitor (17') in the background as a possible alternative.

We are now ready to deal with a complication that I have completely neglected so far: Different positions of **together** can produce different readings. Look at the following pair of sentences, for example:

- (18) a. **Casey and Stacey washed every single car together.**
 b. **Casey and Stacey together washed every single car.**

18(a) and (b) differ in meaning possibilities. 18(a) implies that Casey and Stacey collaborated on every single car. In addition to true collaborative action cases, 18(b) can also be used to describe events where Casey and Stacey didn't work together at all, but between them, they happened to wash all the cars. 18(b) covers all the varied scenarios familiar from cumulative readings: We sum up Casey and Stacey's collective and individual car washing actions and the cars they washed.

What is it that accounts for the difference between 18(a) and (b)? A possible answer is "quantifier scope". In 18(a), the quantifier phrase could be scoped out, and the input for semantic interpretation would then be a structure of the following kind, whose interpretation is straightforward:

- (19) **(Every single car)₁ (Casey and Stacey washed t₁ together).**

18(b) is a Schein sentence of the sort we discussed in chapter 2. As we saw then, the reading we are interested in can be obtained by combining the

denotation of the verb and the quantifier phrase directly. Assuming neo-Davidsonian association of the agent argument in the syntax, the next step would be to add the agent argument. The adverb **together** could then operate over the resulting constituent:

1. $T(\text{every single car}) = \lambda R_{\langle e, \langle st \rangle \rangle} \lambda e \lambda x [\text{car}(x) \wedge \lambda e' [e' \leq e \ \& \ R(x)(e')]]$
2. $T(\text{wash}) = \lambda x \lambda e \text{wash}(x)(e)$
3. $T(\text{wash}(\text{every single car})) = \lambda e \lambda x [\text{car}(x) \wedge \lambda e' [e' \leq e \ \& \ \text{wash}(x)(e')]]$
4. $T([\text{active}]) = \lambda y \lambda e \text{agent}(y)(e)$
5. $T([\text{active}] (\text{wash}(\text{every single car}))) = \lambda y \lambda e [\text{agent}(y)(e) \ \& \ \lambda x [\text{car}(x) \wedge \lambda e' [e' \leq e \ \& \ \text{wash}(x)(e')]]]$
6. $T(\text{together}) = \lambda R_{\langle e, \langle st \rangle \rangle} \lambda y \lambda e [R(y)(e) \ \& \ \text{plural}(y) \ \& \ \lambda e' \lambda z [[e' \leq e \ \& \ R(z)(e')] \wedge z = y]]$
7. $T(\text{together} ([\text{active}] (\text{wash}(\text{every single car})))) = \lambda y \lambda e [R(y)(e) \ \& \ \text{plural}(y) \ \& \ \lambda e' \lambda z [[e' \leq e \ \& \ R(z)(e')] \wedge z = y]], \text{ where}$
 $R = \lambda y \lambda e [\text{agent}(y)(e) \ \& \ \lambda x [\text{car}(x) \wedge \lambda e' [e' \leq e \ \& \ \text{wash}(x)(e')]]]$.
 Etc.

According to this derivation (once you finish it), 18(b) says that there is an event e , Casey+Stacey is the plural agent of e , e is an event in which every single car is washed, and e has no subevents in which every single car is washed by anybody but Casey or Stacey. These are the correct truth-conditions for 18(b), granting some simplifications²².

²² . We really want to talk about completed events of washing every single car, rather than about events in which every single car is washed. This can be accomplished by amending the interpretation of **every single car** as follows, using Link's $\sqsubset$ -operator:

If quantifier scope alone accounted for the ambiguity between 18(a) and (b), there should be no ambiguity if the direct object is not a quantifier phrase. This is not so, however. Look at (20), due to Roger Schwarzschild²³:

- (20) a. **Leakey and Livingston together excavated the cave.**
 b. **Leakey and Livingston excavated the cave together.**

For (20), Schwarzschild invokes a situation where Livingston excavated the western part of the cave, and many years later, Leakey arrived on the scene and excavated the eastern part. In this situation, 20(a) is true, and 20(b) is false. Unlike 20(a), 20(b) requires true collaboration between Leakey and Livingston. In 20(a), **together** is structurally higher than in 20(b). It might even be part of the subject DP, as Schwarzschild has suggested. Be this as it may, how come a merely cumulative interpretation is only available if **together** is in the higher position? With 18(b), we saw that we can obtain the intended reading if we leave the object *in situ* and introduce **together** just after **[active]**. If we do the same in 20(a), we also get the right result. 20(a) describes events *e* that are excavations of the cave, Leakey and Livingston are the agents of *e*, and there are no parts of *e* that are excavations of the cave with anybody but Leakey and Livingston as agents. This is right. If the higher position of **together** is just above **[active]**, where is the lower position? Given the semantic type of **together**, we would expect a lower **together** to operate over the verb. It could do so at a stage of the

$T(\text{every single car}) = \lambda R_{\langle e, st \rangle} \lambda e [\lambda x [\text{car}(x) \wedge \lambda e' [e' \leq e \wedge R(x)(e')]] \wedge e = \lambda e' \lambda x [\text{car}(x) \wedge R(x)(e') \wedge e' \leq e]]$. It is now required that *e* be identical to those of its subevents that are washings of a car.

derivation where the verb hasn't moved up to a higher inflectional head yet. However, using definition (15) wouldn't give us the meaning of 20(b):

1. $T(\mathbf{excavate}) = \lambda x \lambda e [\lambda \text{excavate}(x)(e)]$
2. $T(\mathbf{together}) =$
 $\lambda R_{\langle e, \text{st} \rangle} \lambda y \lambda e [R(y)(e) \ \& \ \text{plural}(y) \ \& \ \lambda e' \lambda z [[e' \leq e \ \& \ R(z)(e')]] \ \& \ z = y]$
3. $T(\mathbf{together \ excavate}) =$
 $\lambda y \lambda e [\lambda \text{excavate}(y)(e) \ \& \ \text{plural}(y) \ \& \ \lambda e' \lambda z [[e' \leq e \ \& \ \lambda \text{excavate}(z)(e')]]$
 $\& \ z = y]$
4. $T(\mathbf{(together \ excavate) (the \ cave)}) =$
 $\lambda e [\lambda \text{excavate}(\mathbf{the \ cave})(e) \ \& \ \text{plural}(\mathbf{the \ cave}) \ \& \ \lambda e' \lambda z [[e' \leq e \ \& \$
 $\lambda \text{excavate}(z)(e')]] \ \& \ z = \mathbf{the \ cave}]$

Since the cave is an atom, and not a plural individual, the property derived in step 4 is not true of any event. Having the plural **the caves**, rather than the singular **the cave**, is a little better, even though it is not all too clear how to excavate the caves as collectives, rather than as individual caves. Change the verb to **sell**, and things become very plausible. I can sell the caves together in a single transaction, or individually. Combining **sell** directly with one or the other adverb gives us that difference. This is all to the good, but we still don't know how to derive the meaning of 20(b). In 20(b), **together** enforces a team action reading. This is the only reading, given that **together** can't successfully relate to the singular direct object in 20(b).

Lasersohn has pointed out that postverbal **together** has a whole variety of closely related uses. In addition to collaborative or team action, it may

²³. Schwarzschild 1993-1994, p. 246.

indicate temporal or spatial proximity, or social accompaniment, for example. Lasersohn has also emphasized that one of the main attractions of an analysis of preverbal **together** along the lines of (15) is that it generalizes to its other uses:

- (21) a. **We sat together.**
 b. **We stood up together.**
 c. **We worked together.**
 d. **We went to Brazil together.**
 e. **We put the bike together.**
 f. **I can only sell you the hat and the gloves together.**

Following the spirit of Lasersohn, but exploiting neo-Davidsonian argument association, the team action reading of postverbal **together** might come via (22), where f_{agents} is a partial function that maps actions with plural agents to their agents, and is undefined for any other kind of eventuality.

$$(22) \quad T(\text{together}) = \\ \lambda R \lambda y \lambda e [R(y)(e) \ \& \ \lambda e' \lambda z [[e' \leq e \ \& \ R(z)(e')] \ \& \ f_{\text{agents}}(e) = f_{\text{agents}}(e')]]$$

Using (22), we derive the meaning for the relevant part of 20(b) as follows:

1. $T(\text{excavate}) = \lambda x \lambda e [\lambda \text{excavate}(x)(e)]$
2. $T(\text{together}) = \\ \lambda R \lambda y \lambda e [R(y)(e) \ \& \ \lambda e' \lambda z [[e' \leq e \ \& \ R(z)(e')] \ \& \ f_{\text{agents}}(e) = f_{\text{agents}}(e')]]$
3. $T(\text{together excavate}) = \\ \lambda y \lambda e [\lambda \text{excavate}(y)(e) \ \& \ \lambda e' \lambda z [[e' \leq e \ \& \ \text{excavate}(z)(e')]]$

- $\Box f_{\text{agents}}(e) = f_{\text{agents}}(e')]]$
4. $T(\textbf{together excavate}(\textbf{the cave})) =$
 $\Box e[\Box \text{excavate}(\textbf{the cave})(e) \ \& \ \Box e' \Box z [[e' \leq e \ \& \ \Box \text{excavate}(z)(e')]] \Box$
 $f_{\text{agents}}(e') = f_{\text{agents}}(e)]]$

According to this derivation, 20(b) is expected to imply that Leakey and Livingston collaborated on every relevant stage of the excavation of the cave. This is what team action is all about. If Leakey and Livingston have any claim to having excavated the cave together, then in some way or other, they were jointly in charge of all parts of the excavation. The individuation of events is notoriously underdetermined and context dependent, of course, a fact emphasized and exploited by Lasersohn and discussed extensively in Moltmann 1997. (22) allows for realistic cases of collaboration, then. It does not commit us to the view that Leakey and Livingston literally dug out each and every part of the cave together.

As is, (22) closely follows the format of (15), but might contain a redundant part. Whenever e is an excavation of something, then any subaction of e is an excavation of something, too. Likewise, whenever e is a reading, eating, building, pushing, or petting of something, and e' is a subaction of e , then e' is a reading, eating, building, pushing, or petting of something as well. If this is a general principle for verb meanings, we can replace (22) by (23) without any losses for postverbal **together**:

- (23) $T(\textbf{together}) =$
 $\Box R \Box y \Box e [R(y)(e) \ \& \ \Box e' [[e' \leq e \ \& \ \text{action}(e')]] \Box f_{\text{agents}}(e) = f_{\text{agents}}(e')]]$

If there are a variety of meanings **together** can have, we expect each of them to be available, unless excluded for some principled reason. We wouldn't want any one of those instances of **together** to come with a stipulation for where it can appear in a syntactic derivation. We have already seen that (15) is available for postverbal **together**, but will then automatically relate to the object. There is nothing to prevent (23) from being used for preverbal **together**, so long as the predicate it operates over is an action predicate. It will then impose a team action reading, hence require collaboration. Since team action is also covered by using (15) instead of (23), no real ambiguity is perceived for preverbal **together**. There are collectivizing adverbs, however, which only have a team action reading, - even in pre-VP positions. German **gemeinschaftlich** is an example:

- (24) a. **Newton und Einstein haben zusammen die moderne**
 Newton and Einstein have together the modern

Physik begründet.

physics founded.

Newton and Einstein together created modern physics.

- b. **Newton und Einstein haben die moderne Physik**
 Newton and Einstein have the modern physics

zusammen begründet.

together founded.

Newton and Einstein created modern physics together.

- (25) a. **Newton und Einstein haben gemeinschaftlich die**
 Newton and Einstein have jointly the

moderne Physik begründet.

modern physics founded.

Newton and Einstein jointly created modern physics.

- b. **Newton und Einstein haben die moderne Physik**
 Newton and Einstein have the modern physics
gemeinschaftlich begründet.
 jointly founded.
 Newton and Einstein created modern physics jointly.

Since Newton and Einstein lived too far apart in time to collaborate as a team, 24(b) and 25(a) and (b) are all false. Only 24(a) is true: The cumulated achievements of the two men created the field of modern physics. In 25(a) and (b), I used the English adverb **jointly** to translate German **gemeinschaftlich**. With action verbs, **jointly** also imposes a team action reading in preverbal and postverbal position. Unlike **gemeinschaftlich**, however, **jointly** is also acceptable with verbs like **imply**, for example, hence doesn't necessarily relate to collaborative action or collective ownership. The existence of collectivizing adverbs that require collaboration even in pre-VP positions supports (23) against (22). (22) would not generally produce a team action interpretation in those positions. Consequently, (23) has to be available as a meaning assignment, even if my rationale for obtaining it from (22) might eventually turn out to be flawed.

Here is a summary of the account of **together** and **gemeinschaftlich** I have proposed. To start with the conclusion, - we need both (23) and our old (15):

(15) T(**together**) =

$$\neg R_{\langle e \rightarrow st \rangle} \neg \exists y \neg e[R(y)(e) \ \& \ plural(y) \ \& \ \neg e' \neg z [[e' \leq e \ \& \ R(z)(e')] \ \neg \ z = y]]$$

In pre-VP position, adverbs with the denotation in (15) are expected to produce interpretations that are compatible with both collaborative and merely cumulative action. Pre-VP **together** has that range of interpretations, hence should denote (15). On the other hand, pre-VP **gemeinschaftlich** is not compatible with merely cumulative action, hence needs to denote (23). With that denotation, it is also expected to generate a collaborative action reading even if it directly combines with the verb. This is so. In postverbal position, English **together** can relate to the direct object, as in 21(f) from above:

(21) f. **I can only sell you the hat and the gloves together.**

This interpretation is automatically generated if post-verbal **together** has the denotation in (15) and combines directly with the verb. Since postverbal **together** can also have a collaborative action interpretation, we have to assume that **together** is ambiguous and can optionally have the meaning specified in (23). Between them, (15) and (23) cover quite a bit of ground, then.

The notion of collaboration formalized in (23) is based on a crucial assumption about team action that needs more discussion. A first intuition about what teams or groups are is that they are pluralities ‘seen as ones’ - as singularities, that is. Since pluralities can’t literally be singularities²⁴, it is likely that they are seen as ‘ones’ because they behave as ‘ones’. We should now look into the behavior of singular agents. What might very well be the

²⁴ . There is, of course, the trickery made possible by set theory. I am siding with Lewis 1991 here. For a different opinion, see Landman 1989, 1996, 2000.

most distinctive property of singular agents is that they satisfy what I want to call the ‘Single Agent Constraint’: even complex actions by singularities do not have subactions by anyone but those very same singularities. Actions with singular agents obey (26), then:

(26) Single Agent Constraint

If e is an action, and x is the agent of e , then x is the agent of any subaction of e .

I am just one person, and all of my actions are in line with (26). If e is my cooking dinner tonight, for example, then I am the agent of all subactions of e . Plural agents do not generally satisfy (26). If you and I cooked dinner yesterday independently of each other, and e is that event, then the sum of you and me are the agents of e . But that sum is not the agent of all subactions of e .

That (26) holds for actions with singular agents follows from what we have been assuming: Singular individuals are atoms; any action has a unique agent, and the agent relation is cumulative. Suppose e is an action with a singular agent x , and e' is a subaction of e whose agent is y . Cumulativity of the agent relation implies that $x+y$ is the agent of $e+e'$. Since e' is part of e , $e+e' = e$. Consequently, $x+y$ is the agent of e . But agents are unique, hence $x+y = x$, and therefore $y \leq x$. Since x is atomic, $y = x$.

What I want to propose is that (26) also holds for group actions. It’s not that group agents have to be singularities. They might as well be regular pluralities, - as long as they behave like singularities. For group agents to behave like singularities means to satisfy (26) with respect to their actions.

If we watch the Red Sox play, for example, we might see that game or a part of it as group action. Consequently, that action would be represented as conforming to (26) in our domain of eventualities. There might have to be ‘team credits’ for home runs, then:

“It is common practice to attribute to a group an action which is properly performed only by some (or even just one) of the group’s members. This is especially true in a context where the group acts as a team; consider sentences such as *The Islanders scored a goal*, for example. We accord “team credit” to the Islanders, even though the goal might have been scored through the efforts of just one player.”²⁵

Lasersohn extends the notion ‘team credit’ beyond the cases of true collaborative action, however, which is going too far, I think. (27) might very well be true, for example, even though Newton wasn’t part of a team that developed the Theory of Relativity, and the Laws of Motion weren’t discovered by a crew that included Einstein.

(27) (Between them,) Newton and Einstein are responsible for every single influential physical theory of modern times.

Since Lasersohn doesn’t have neo-Davidsonian association of agent arguments, he invokes team credit even for the merely cumulative interpretations of Schein sentences. According to Lasersohn, (27) could only be true, for example, if the plural individual consisting of Newton and Einstein was responsible for each and every influential physical theory of modern times. For this analysis not to be obviously wrong, Lasersohn relies on team credit. “Team credit is more-or- less automatic whenever the

²⁵ . Lasersohn 1995, 197.

combined effects of a group's actions are pragmatically relevant"²⁶. The combined efforts of Newton and Einstein are pragmatically relevant for (27), yet we can't grant team credit in the absence of any actual team grouping. Since Newton and Einstein weren't a team on anybody's account, they should not be entitled to team credit. Teams aren't created by the mere powers of imagination. They correspond to substantive groupings of pluralities in the actual world²⁷. The Red Sox are a team, but Newton and Einstein were not. On the analysis I am pursuing, team credit may be given in cases of true collaborative or team action, but not when we are merely summing up the actions of singular individuals. Most importantly, this hard-line approach to team credit gives us an account of the differences between collectivizing adverbs that we observed in (20), (24), and (25). Some collectivizing adverbs tolerate mere cumulativity, but there are others that require true collaboration.

If there are group actions, there should also be group states, states of being a team, for example. Theoretically, a group state would have to be a state of a plurality that has no substate whose possessor is anything but that very same plurality. It is very important to keep in mind that mereology alone gives us nothing in the way of substantive pluralities. If our domains D_e and D_s are closed under sums, they have countless members that are a far cry from anything we might be willing to recognize as actually existing groupings. To see one of those abnormal specimens, take David Lewis' trout-

²⁶ . Lasersohn 1995, p. 198.

²⁷ . Substantive groupings of pluralities are 'integrated wholes' in the terminology of Moltmann 1997.

turkey²⁸:

“I accept a principle of *Unrestricted Composition*: whenever there are some things, no matter how many or how unrelated or how disparate in character they may be, they have a mereological fusion. ... That means that if I accept individuals and classes, I have to accept mereological fusion of individuals and classes. Like the mereological fusion of the front half of a trout plus the back half of a turkey, which is neither fish nor fowl, these things can be mostly ignored. They can be left out of the domains of all but our most unrestricted quantifying. They resist concise classification: all we can say is that the salt beef sandwich is part animal, part vegetable, part mineral; the trout-turkey is part fish and part fowl; and the mereological fusion of Possum plus the class of all cat-whiskers is part individual and part class.”

I suggest that group states and actions are states and actions of substantive actual pluralities. This gives the notions ‘group state’ and ‘group action’ their empirical bite. You can’t just stipulate the existence of substantive groups. They are there in reality. Actual teams, piles, bunches, flower arrangements, clubs, committees, congregations, and what have you - all correspond to substantive groupings of pluralities in the actual world. My proposal is a particular hypothesis about the connection between substantive groupings of pluralities and the part structure of events: Actions by substantive groups satisfy the Single Agent Constraint, and states of substantive groups satisfy an analogous Single Possessor Constraint.

It’s easy to mix up mere mereological sums and substantive pluralities. Let’s try not to.

“I myself take mereology to be perfectly understood, unproblematic, and certain. This is a

²⁸ . Lewis 1991, p. 7f.

minority opinion. Many philosophers view mereology with the gravest suspicion.

Sometimes they suspect that originally the notion of part and whole was understood not as topic-neutral, but rather as a spatiotemporal - or merely spatial - notion. They conclude that any application of it to things not known to be in space and time is illicit. The original idea, supposedly, was that x is part of y iff y is wherever x is.

That is wrong thrice over...”.²⁹

Spatial and temporal proximity plays an important role in the creation of substantive groupings of pluralities, however. This is illustrated by an example of Barry Schein. Schein asks us to imagine a tree whose leaves are allergenic when they come in bunches larger than three. We are now asked to consider

“ a context for (107) in which the fallen leaves of the allergenic tree have been raked up into several bunches scattered on the lawn:

(107) All the bunches of leaves (on the lawn) are allergenic.

...(107) is true if each of the bunches one sees on the lawn contains more than three leaves. The sentence is not falsified by the one-, two-, or three-leafed bunches that can be created from the actual bunches lying there. The domain of the quantifier *all the bunches of leaves (on the lawn)* is thus not closed under combinations of the atoms. Note that the domain of actual bunches is essentially a partition of the atoms on the lawn, since no leaf is in two places at once. A domain of actual individuals observed at a given moment will never attribute to its atoms more than one location. This property always holds even of stuff in constant flux, like the bits of glass, the atoms, constantly regrouping in a kaleidoscope to form new clusters, the individuals.”

Schein 1993, p. 104

²⁹ . Lewis 1991, p. 75.

Given the connection between substantive actual groupings and spatial and temporal proximity, fewer different meanings for English **together** might have to be posited than Lasersohn suspected³⁰. Look at the following examples from above:

- (21) a. **We sat together.**
b. **We stood up together.**
d. **We went to Brazil together.**

With activities like sitting, standing up, or going to Brazil, spatial proximity of the agents and temporal closeness and coordination of their actions contributes essentially to establishing them as substantive groups, and their actions as collective actions. It might very well be, then, that we do not need separate spatial and temporal uses of **together**. When we sat together, we sat as a group. When we stood up together, we stood up as a group. And when we went to Brazil together, we traveled as a group. The groups were all substantive groups, and this means that there were group actions and states that came with them, satisfying the Single Agent or Possessor Constraint.

It may be helpful to compare the present account of group states and actions to the account of groups in Link 1984 and 1991³¹. Link has group individuals that are ‘impure atoms’. They are denoted by singular collective DPs such as **the committee**, **the choir**, or **the army**. In addition to the soldiers that constitute the army and are a plurality, then, there is also the army, which is a group, and hence a singularity. Link 1991 posits a group forming operation

³⁰ . A unified analysis based on flexible part structures is advocated in Moltmann 1997.

³¹ . A related account is given in Landman 1989, 1996, 2000.

that maps all sums of ‘pure atoms’ like the soldiers or the children into corresponding groups. The outputs are entities that have no longer any pure atoms as parts. In contrast, our group actions might very well have regular pluralities as agents. Their distinctive property is that they are not sums of actions by singularities. Likewise, group states might very well be states of ordinary sums of individuals. What’s special about them is that they are not sums of states of singularities. Most importantly, I have suggested that group actions and states are actions and states of substantive groupings of pluralities. As a consequence, the inventory of group states and group actions of our world is a matter of contingent fact.

Having group actions and states opens up new possibilities for the semantics of collective nouns. Rather than describing special breeds of individuals - Link’s groups - collective nouns like **committee**, **choir**, or **army** might in truth be relational. They might denote relations between regular pluralities (sums, that is) and particular groupings of those pluralities, which are collective states. **Choir**, for instance, would express a relation that holds between a plurality *x* and a state *s* just in case *s* is a choir grouping of *x*. An immediate consequence of this proposal is that it accounts for the fact that one and the same plurality might be grouped in more than one way at the same time, a possibility that Link’s proposal does not cover. The same children might be the school choir and the school lunch committee, for example. On the present account, this would mean that there are two different groupings of the children, hence two different collective states for

them, a school choir grouping and a school lunch committee grouping³². Both states might happily coexist over a stretch of time.

If **choir** is relational, and its external argument is a state argument, we expect that the number marking of **choir** does not reflect the singularity of the individuals that constitute choirs, but the singularity of the grouping of those individuals. Likewise, when we count choirs, we count groupings of plural individuals, not plural individuals themselves. Two different choirs may be made up by the very same singers. The way we count choirs is reminiscent of the way we count passing ships³³:

(28) **Four thousand ships passed through the lock last year.**

(28) could be true, for example, if a total of 200 hundred ships passed through the lock again and again, adding up to 4000 different passages. Each of those 4000 events contains a different stage of a ship. Rather than counting passages, then, we might as well count passing stages of ships. If the noun **ship** has a state argument that can refer to temporal stages of individual ships, we understand why we are allowed to count ship stages when evaluating the truth of (28). Here is an actually attested case of quantification over stages. The Boston Globe of August 6, 2001 quotes an unidentified Republican as saying “In most White Houses, the chief of staff is first among equals.” As far as buildings go, the quote does not assume there to be any other White Houses apart from the one in our nation’s capital.

³² . Those relatively permanent groupings correspond to the guises of Landman 1989. See also Moltmann 1997 for discussion of related cases.

³³ . Krifka 1990.

What is being quantified over are temporal stages of the White House that correspond to different presidencies. The current president's chief of staff, who "leaves no footprints in the sand", is compared in the article to the high visibility of the chiefs of staff of former presidencies. It seems, then, that common nouns - and even proper names - can have eventuality arguments, an assumption that fits well into the program of Higginbotham 1989, 2000. James Higginbotham has argued over the years that predicates of all kinds can have eventuality arguments.

To have at least a concrete proposal on the table for further thought, (29) is a first idea of what an analysis of collective nouns might eventually look like³⁴:

- (29) a. **A boys' choir sang.**
 b. $\exists x \exists e [\text{choir}(x)(s) \ \& \ \text{boys}(x) \ \& \ \exists \text{agent}(x)(e) \ \& \ \exists \text{sing}(e) \ \& \ \exists e' \exists y [[e' \leq e \ \& \ \exists \text{agent}(y)(e')] \ \& \ [y = x \ \& \ s < e']]]$
 c. 'There is a choir of some boys x and, as a choir, x is the collective agent of a singing event'.

29(b) accounts for the fact that 29(a) requires group action by a plurality of boys, and implies that a choir grouping of them be present throughout the event³⁵. In the end, then, we might not need Link-style groups for the

³⁴. To flesh out that proposal, we would have to derive all the hybrid properties of collective nouns in an insightful compositional way. I will not do this here, since the issue is not that central to my plot.

³⁵. If the analysis is right, group actions by some plurality x can include group states. If so, the possessors of those states should have to be identical to x .

semantics of collective nouns any longer³⁶. Collective actions and states might be able to take their place. Those, we need anyway.

Abolishing Link-style groups in favor of collective states and actions could make life easier in other ways, too. Take (30):

(30) **John destroyed that pile of plates.**

On the relational view of collective nouns, we might take the denotation of **that pile of plates** to be a pair consisting of the plates and that particular pile state they were in. The direct object of **destroy** in (30) would then denote such a pair, and we could easily account for the fact that destroying that pile of plates might involve destruction of the state or the plates. This looks good. (30) is true in situations where John destroyed the actual pile arrangement without destroying the plates. Or he might have destroyed the state along with the plates. He couldn't have destroyed the plates (completely) without destroying the pile, though. How come? If an individual is gone, its particular states are gone, too. Perfect.

³⁶. Link himself does not subscribe to his group theory any longer. Referring to Schwarzschild's work he writes: "Recently, intriguing arguments have been advanced to the effect that there is no need to introduce a new kind of entities over and above the mereological i-sums." Link 1998, p. 174. Link does not discuss the status of collective nouns in this connection, though. Schwarzschild 1996, chapter 9: "In the end we concluded that in a purely extensional theory, collectives and plural individual noun phrases could not be coreferent, however we raised the possibility that this conclusion could be avoided by adopting a non-extensional theory to explain the data." (p. 192.) My proposal above is to avoid that conclusion through a relational theory of collective predicates.

What would a theory based on Link-style groups say about (30)? Here is what I imagine. Our domain of individuals would contain two different objects: those plates, which are the sum of singular plates, and the pile of plates, which is a group, hence an impure atom. **That pile of plates** might be ambiguous. It might denote the sum or the group. Suppose John destroyed the group. What would happen to the sum? And what if he destroyed the sum? What would happen to the group? Could he destroy the mere arrangement by destroying the group? Not if groups are essentially like the corresponding sums, except for their being atoms.³⁷ Then the groups should be gone whenever the sums are, and the other way round. Suppose John could destroy the arrangement by destroying the group and leaving the sum alone. Then we would have to say something special about why it is that he couldn't destroy the sum and leaving the group alone. Or could he? I have no clue. Things just can't seem to fall in place in any obvious way.

I will have more discussion of group states and substantive pluralities in the following section. For now, let us retain that we need mereology, because we need to be able to merely sum up individuals, actions, states, and other kinds of events. Otherwise, we couldn't account for the merely cumulative readings we have encountered on our way. But we also need to be able to talk about substantive actual groupings of pluralities. Otherwise, we couldn't do justice to collective nouns and the full range of collectivizing adverbs. What we do not seem to need, however, is groups of the kind originally proposed by Link and developed more fully in the work of Landman:

³⁷ . Link 1984, 1998 (p. 81).

“I am arguing for a theory that reduces distributivity to semantic plurality. Without groups, cases of distribution to collections are an insurmountable problem for such a theory. These cases require a certain amount of ‘grid’ that sums alone are not able to provide: i.e. we want to distribute in these cases, but not all the way down to the individuals.”³⁸

It looks like the part structure of states and events might very well be able to provide the ‘grid’ that sums of individuals alone are not able to provide.

The connection between substantive pluralities and part structures is a central topic in Moltmann 1997, which surveys a large range of phenomena where substantive pluralities (her ‘integrated wholes’) play a role, including plural and mass quantification, collective nouns, and collectivizing adverbs. Moltmann abandons standard mereology and develops a new theory of part structures, however. Moltmann’s work documents the role of substantive pluralities in the semantics of natural languages and presents novel analyses based on this notion. While following Moltmann in exploiting the notion ‘substantive plurality’ for the analysis of phenomena involving collectivity, I have raised the possibility that an account of substantive pluralities can be given while maintaining a standard mereological account of pluralities³⁹. The only special assumption needed was that there are such things as truly collective eventualities. Among those, I have singled out collective states and actions for further discussion. Collective actions are actions by pluralities that have no subactions by anybody but those very same pluralities. Likewise, collective states have plural possessors who are also the possessors

³⁸ . Landman 2000, p. 160.

³⁹ . I do not mean to suggest that non-mereological part relations have no role in natural language semantics. The situation semantics introduced in Kratzer 1989, for example, has a part relation for situations that cannot be identified with the mereological part relation, nor can the spatio-temporal part relation.

of all of their substates. Pluralities are perceived as integrated wholes, then, because they are the participants in perceptually salient collective states and events.

The individuation of events is known to be a thorny matter. When attorney John Lord O'Brian delivered his closing arguments, was that a speech by him alone or by the defense team that also included his colleague Hollingsworth? The answer depends in part on the facts of the case. Was Hollingsworth present, standing close to O'Brian? Did he help prepare the statement? But even with satisfactory answers to those last two questions, we can't always expect to come up with an obvious answer to the question at the beginning of this paragraph. Was there a single collective speech? A single individual speech? Or were there two speeches, one by an individual, and another one by a team? The last possibility sounds absurd, but remember that when it comes to quantification over events in everyday English, we couldn't have both of those speeches in our quantification domain, even if there were in fact two of them. If O'Brian's speech and the one by the defense team can't both be in one and the same quantification domain (in non-philosophical discourse), it would simply be false for us to claim: **There were two speeches**. There weren't. You can only have one of the two. Which one? Context may settle the question. Or not. There is only so much a context can do for you.

What is it that could prevent O'Brian's speech and the defense team's speech from entering the same quantification domain? One contributing factor is likely to be complete or almost complete overlap in spatio-temporal location. Avoidance of too much spatio-temporal overlap is a known constraint for quantification over locations. Suppose someone claims that there is only one

place in the whole world where you can eat well, and that's the 'Student Prince' (a restaurant in Springfield, Massachusetts). I personally think that claim is true. Yet, strictly speaking, it couldn't possibly be. If you can eat well in the 'Student Prince', you might also eat well in its Heidelberg Room. And you will certainly be able to eat well in Springfield, in Massachusetts, in the United States, and so forth. There is nothing wrong with acknowledging the existence of all of those places. We just can't have all of them in one and the same quantification domain. Likewise, I suggest, there is nothing wrong with assuming the existence of two speeches in our story, as long as we keep them out of the same quantification domain.

Let me summarize where we are in our plot. We have seen that within an event semantics, cumulative verb denotations do not eliminate the difference between distributive and collective predication. They are compatible with a promising semantics for collectivizing adverbs that preserves the spirit of Lasersohn. Most importantly, we can have this without giving up standard mereology. We do not need a new theory of part structures as advocated by Moltmann, nor special breeds of pluralities as advocated by Landman. Standard mereology allows us to make all the necessary distinctions. This is a good result. However, we are not over the hill yet. There are some nasty looking counterexamples for Lasersohn's analysis of collectivizing adverbs that have to be cleared before we can be confident that 'cumulativity from the start' is even a viable option for basic lexical items in natural languages.

4.4 Collective states and covers

This section starts out with a potential counterexample to Lasersohn's semantics of **together** that seems to threaten the whole enterprise of retrieving collective predications that I have been pursuing. I will argue that the counterexample goes away if we assume - as

we did in the previous section - that there are group states satisfying the Single Possessor Constraint. To further boost the case for the group states we are committed to, I will argue that that very commitment also yields an account of cover effects, that is, for the context-dependent groupings of pluralities.

Let us begin with what looks like a very threatening example by Roger Schwarzschild⁴⁰:

(31) **The ax and the box together are light enough to carry.**

Schwarzschild's point is that whenever an ax and a box together are light enough to carry, it follows that the ax and the box alone are light enough to carry, too. Doesn't this go right against (15)? Wouldn't we predict that (31) should be necessarily false?

(15) **T(together) =**

$$\Box R_{\langle e \langle st \rangle \rangle} \Box y \Box e [R(y)(e) \ \& \ plural(y) \ \& \ \Box e' \Box z [[e' \leq e \ \& \ R(z)(e')] \ \Box \ z = y]]$$

Instead of discussing Schwarzschild's example directly, I will first examine (32), which prepares for Laserson's reaction to Schwarzschild's threat:

(32) **Casey and Stacey together fit this ad.**

Suppose the ad is by a theater company that is looking for either a child with red hair and freckles or a pair of children who are twins. Casey and Stacey fit the ad both individually and collectively. They are children with red hair and freckles, and they are twins. We have the following situation:

⁴⁰. Schwarzschild 1993-1994, 213.

s ₁	State of Casey fitting the ad (red hair & freckles)
s ₂	State of Stacey fitting the ad (red hair & freckles)
s ₃	State of Casey+Stacey fitting the ad (twins)

Assuming cumulativity for the denotation of **fit** gives us (33):

(33) Extension of 'fit'

{<s₁, Casey, the ad>, <s₂, Stacey, the ad >,
 <s₃, Casey+Stacey, the ad >, <s₁+s₂, Casey+Stacey, the ad >,
 <s₁+s₃, Casey+Stacey, the ad>, <s₂+s₃, Casey+Stacey, the ad >,
 <s₁+s₂+s₃, Casey+Stacey, the ad >, }

Now consider the following sentences:

- (34) a. **Casey and Stacey together fit this ad.**
 b. **Casey and Stacey each fit this ad.**

Both sentences are intuitively true in the situation I set up. But they are also predicted to be true by our analysis of **together**. Since we are dealing with statives, only (15) can be used as a meaning assignment for **together**. The verifying state for 34(a) is Casey and Stacey's fitting the ad in virtue of the group state s₃, then, while the verifying state for 34(b) is Casey and Stacey's fitting the ad in virtue of s₁+s₂. The important observation is that there is no reason to assume that the two ways of fitting the descriptions should correspond to one and the same state. Intuitively, the state of Casey's fitting the ad in virtue of her red hair and freckles, for example, is not part of the group state of Casey+Stacey's fitting the ad in virtue of being twins. (32),

then, is no problem for (15). Lasersohn reacts to Schwarzschild's counterexample (31) in exactly this way in his 1995 book:

“Although it will normally be the case that if there is an eventuality of the axe and the box together being light enough to carry, there will also be an eventuality of the axe being light enough to carry and an eventuality of the box being light enough to carry, I see no reason to assume that these latter eventualities must be parts of the first.”⁴¹

In the conceptual framework of the last section, the assessment is that (31) talks about the existence of a group state - the collective weight of the axe and the box. That state should not be confused with the mereological sum of the two individual weights. The possessors of collective weights are substantive pluralities, most likely occupying adjacent or nearly adjacent spatial regions. The analysis of **together** I have been endorsing predicts that (31) can only come out true if the axe and the box are a substantive plurality. This prediction is confirmed by the following example from Lasersohn 1995:⁴²

“.... the ease with which something can be carried depends not just on its weight, but on the ratio of its weight to that of a comparable volume of air (or whatever the surrounding medium may be). For example, an anvil might be too heavy to carry, but the anvil and a helium balloon together could still turn out to be light enough to carry.”

With Lasersohn's scenario in mind, look at the following sentences:

(35) a. **The anvil was too heavy to carry.**

41. Lasersohn 1995, 225.

42. Lasersohn 1995, p. 217, footnote 14.

- b. **The anvil and the helium balloon together were too heavy to carry.**

The truth of 35(a) in no way guarantees the truth of 35(b). Given the weight of the anvil, whether or not 35(b) is true depends on the actual arrangement of the anvil and the helium balloon, hence on a substantive grouping of the two.

Here is another example confirming that the likes of (31) are about substantive pluralities. Suppose I am corresponding with some far away stranger on the Internet and take a liking to him. After a while, I ask him what he looks like. Here are two versions of a reply:

(36) I have no idea what YOU look like. But I can tell you for sure that

- a. **You and I together are heavy.**
b. **You and I together would be heavy.**

There is something slightly inappropriate about 36(a). The fact that that inappropriateness can be repaired by 36(b) suggests that 36(a) presupposes some substantive actually existing grouping of the two of us, more true togetherness than there was at that point.

Why is it that 36(a) requires a substantive grouping of me and the stranger to be true? Here is an answer. For 36(a) to be true, there must be some state *s* such that the pair consisting of *s* and the sum of the stranger and me is in the extension of '□heavy'. There are only two possibilities for this to be so. The pair <*s*, the stranger+I > might have gotten into the extension of '□heavy' via

cumulation. This is precisely the case excluded by **together**. The pair $\langle s, \text{the stranger+I} \rangle$ must therefore already be in the extension of the uncumulated predicate 'heavy'. It then has to satisfy Landman's Collectivity Criterion, and s must be a collective state, hence a state of a substantive plurality.

Interestingly, a slight change of example yields different judgements - and is expected to. Suppose I revealed to that stranger how much I weigh. He might have replied:

- (37) I can now tell you that
- a. **You and I together weigh 250 pounds.**
 - b. **You and I together would weigh 250 pounds.**

This time round, 37(a) is quite appropriate. Unlike 37(b), it doesn't even evoke the possibility of physical togetherness. Why? Look at the logical form of 37(a), where f_{pound} is a measure function defined for states of weight:

- (38) $\exists s [\Box \text{weigh}(\text{the stranger+I})(s) \ \& \ f_{\text{pound}}(s) = 250 \ \& \ \Box s' \Box x [[s' \leq s \ \& \ \Box \text{weigh}(x)(s') \ \& \ f_{\text{pound}}(s') = 250] \rightarrow x = \text{the stranger+I}]]$

Suppose the verifying state for (38) is s . The pair $\langle s, \text{the stranger+I} \rangle$ could now have entered the extension of ' $\Box \text{weigh}$ ' via cumulation of our individual weights without running against the demands of **together**. 37(a) doesn't suggest physical togetherness, then, because the participating pluralities could in principle be the result of mechanical cumulation.

We have seen that collective weights must be distinguished from sums of individual weights. In our domain D_s states must be represented so as to let us draw that distinction. We need group states that are not mere sums of individual states. The distinction is a very important one for the semantics of adjectives. Take the reasoning in 39(a):

- (39) a. **Each of those 100 plates are light. Therefore, those
 100 plates are light.**
 b. **Those 100 plates are light.**

Is 39(a) intuitively valid? It depends. In a sense it is, in another sense it is not. Is 39(a) predicted to be valid on our analysis? Yes, so it seems.

Whenever the first sentence of 39(a) is true, there exists a verifying state for the second sentence. Given the cumulativity of **light**, the sum of the 100 individual states of being light in weight is a state characterized by the AP of 39(b). What about the ‘it depends’, then? Wouldn’t we want to predict that judgement? Shouldn’t there be an ambiguity? How can we account for the reading of 39(b) that’s behind the intuition that 39(a) is not necessarily valid?

39(b) does not have to be understood as a claim about the existence of a state. We can also use it to refer to a particular state. The 100 plates’ collective weight would be a plausible state to refer to. If this is a group state, it might not be in the extension of the AP of 39(b), even if the first part of 39(a) is true and **light** is cumulative.

If adjectives have eventuality arguments that are represented as variables in logical-conceptual representations, those variables do not have to be quantified. They may be left free and would then have to be supplied with a

value by the utterance context. We could have the following kind of logical-conceptual representations, for example:

(40) $\Box\text{light}(\text{the 100 plates})(s)$

The predicate ‘ $\Box\text{light}$ ’ in (40) expresses a cumulative relation between individuals and states that can be compared to a standard weight. Those states should then themselves be weights. The ‘ $\Box\text{light}$ ’ relation holds between you and a state s , for example, if s is your weight and s is low on the weight scale⁴³. If the variable ‘ s ’ is left free, contextually plausible values for ‘ s ’ would be the collective weight of the 100 plates, or else the mereological sum of their individual weights. If the two states are different, the truth of (40) can vary from one context to the next, even if the 100 plates and their individual weights don’t change. The ‘it depends’ judgement for 39(a) is thus explained while sticking to the cumulativity of **light**.

⁴³. The proposal suggests that with adjectives, the eventuality argument can take over the role of what is customarily assumed to be a degree argument, - an extra bonus of having a state argument for adjectives. When a rope is 20 yards long, for example, it is in a state of length that measures 20 yards. That adjectives have eventuality arguments is again in line with James Higginbotham’s program that provides eventuality arguments for lexical predicates of all kinds. Higginbotham 1989, 2000. If **light** has an eventuality argument, it does not automatically follow that phrases like **be light** have eventuality arguments, too, an issue that is crucial for making Higginbotham’s proposal compatible with Kratzer 1989, 1995. **Be** might be an aspectual operator that quantifies off the eventuality argument of **light**. **Be light**, then, could denote a relation between individuals and times, rather than a relation between individuals and states. This account of the difference between **light** and **be light** would have to be evaluated by looking at the many pertinent facts discussed in Rothstein 1999, 2001. (40) neglects the contribution of **be** so as to avoid complications that are not relevant just now.

That natural languages have demonstrative devices that can refer to events and states in the world and are described by sentences was advocated by John Austin:

“.....there must be two sets of conventions:

Descriptive conventions correlating words (=sentences) with the *types* of situation, thing, event, &c., to be found in the world.

Demonstrative conventions correlating the words (=statements) with the historic situations, &c., to be found in the world.

A statement is said to be true when the historic state of affairs to which it is correlated by the demonstrative conventions (the one to which it ‘refers’) is of the type with which the sentence used in making it is correlated by the descriptive conventions.”

J. L. Austin, 1979, 121 f.

Quoting Peter Strawson as saying

“What ‘makes the statement’ that the cat has mange ‘true’ is not the cat, but the *condition* of the cat, i.e. the fact that the cat has mange. The only plausible candidate for the position of what (in the world) makes the statement true is the fact it states; but the fact it states is not something in the world.”

Austin replies:

“I cannot swallow this because it seems to me quite plain:

- (1) that the condition of the cat is a fact;
- (2) that the condition of the cat is something-in-the-world - if I understand that expression at all.

How can Strawson have come to say that the condition of the cat is *not* something in the world?”

J. L. Austin, 1979, 156.

Austin's views were taken up in Barwise and Perry's *Situation Semantics*, and also by Lasersohn.

"The crucial insight needed goes back to Austin. ... As Austin put it, a statement is true when the actual situation to which it refers is of the type described by the statement. That is, just as there are conventions about what situations can be used as resource situations, there are also conventions about what situation a person is describing. One can make a false statement by violating these conventions just as surely as one can by using a different statement." Barwise and Perry 1983, 160.

Within an event semantics, eventuality arguments can be the demonstrative devices used by natural languages to make claims about particular states and events in the actual world⁴⁴. Within our present framework, eventuality variables may remain free in logical-conceptual representations and then receive a value from the context of use. Alternatively, we might allow contextual restrictions for the state argument, which could then be existentially quantified at a later point in the derivation. In 39(b), the values of the state variable would be restricted to states corresponding to salient groupings of plates and their sums, and the referential interpretation would come out as a special case.

Roger Schwarzschild has defended the context dependency of talk about the groupings of pluralities in his works, which are not framed within an event semantics. For Schwarzschild, the utterance context partitions the domain of plural individuals into possibly overlapping groups, and plural predication

⁴⁴. I am very open to the possibility that eventuality arguments are also represented in the syntax. In fact, I am pretty convinced that they are. I am just not exploring this issue in this book.

depends on such contextually provided groupings. On the account I am pursuing, substantive groups of individuals are the possessors of a host of group states. If the plates are in a pile, for example, their sum is the possessor of many collective states, including a pile state, a collective weight, a collective height, and what have you. An adjective's state argument is a variable that can refer to those states. (38) is context dependent because there is a state variable whose values can be provided or constrained by context. Salient states of weight are possible values in that particular case. In the absence of other salient acceptable values, the collective weight of the plates or the mereological sum of their individual weights are the two default candidates. Consequently, (40) can have a distributive or a collective interpretation.

Schwarzschild posits a free cover variable as part of logical representations. (41) illustrates one way of executing his proposal:

(41) The plates Distr_C light.

Adapted to a mereological framework, possible values for the cover variable 'C' in (41) would be those subsets of the domain of individuals D_e whose sum is identical to the sum of D_e itself. Those subsets 'cover' D_e , hence are covers of D_e . I am assuming that D_e is closed under sums, and I take that to mean here that any non-empty subset of D_e has a sum in D_e . The generalized distributivity operator Distr in (41) introduces universal quantification over relevant parts of the plates. (41) is then true in a context just in case all parts of the plates that are members of the cover that the context assigns to 'C' are in the extension of **light**. For the purposes of illustrating Schwarzschild's proposal, I take **light** to denote a mere set of individuals, rather than a

relation between individuals and states. Moreover, I am (temporarily) not assuming that the extension of **light** is cumulative. It contains all the things that are light individually or collectively, but excludes all those pluralities that are merely light through cumulation. If the cover assigned to ‘C’ has only atomic plates in it, we get the distributive reading of (41). If it has the sum of all the plates, but no atomic plates or other sums of plates, the collective reading results.

If the denotation of **light** is cumulative from the start, no distributivity operator is needed to account for the distributive reading of 39(b). That reading is already taken care of by lexical cumulativity, and so is the collective interpretation. Yet there is ambiguity. And there is evidence for the relevance of Schwarzschild’s covers for plural predication. If there is no distributivity operator, what is it that could produce cover sensitivity in sentences like 39(b)? Contextual restrictions for eventuality variables are the obvious candidates. Within an event-based framework, the context dependency of Schwarzschild’s groupings of plural individuals would boil down to the context dependency brought in by eventuality variables. No special devices would be employed by natural languages to produce grouping effects in plural predications. The possibility that Schwarzschild’s cover effects might ultimately come from the vagueness and context dependency of the individuation of states and events was raised by Barry Schein in correspondence with Roger Schwarzschild. Schwarzschild 1996, p. 96 reports the following:

“The following example, based on one from Barry Schein (p.c.), is a particularly surprising case in which a mentioned cover is nevertheless unlikely to produce the relevant reading:

(227) The vegetables, which are the beets and the carrots, weigh 5 lbs.

Even though the partition into beets and carrots is mentioned, Schein would find the intermediate distributive interpretation impossible.”

In footnote 27 on the same page, Schwarzschild notes:

“If I understood him correctly, the source of the problem in Schein’s view is that explicit mention of a covering is insufficient and what is needed are individuating events.”

There are no covers in Schein’s own work on plurality, which precedes Schwarzschild’s. In their stead we find contextual restrictions for event variables⁴⁵. The impact of event individuation on plural quantification is a topic that Schein has consistently pursued since his 1986 dissertation, a time when events were not yet commonly accepted in semantic theorizing.

To have an illustration for how cover effects reduce to constraints on event quantification, let us look at a slightly more complicated case, one of many very convincing examples that Schwarzschild uses to illustrate the importance of his covers⁴⁶:

“Imagine a situation in which two merchants are attempting to price some vegetables. The vegetables are sitting before the merchants, piled up in several baskets. To determine their price, the vegetables need to be weighed. Unfortunately, our merchants do not have an appropriate scale. Their grey scale is very fine and is meant to weigh only a few vegetables at a time. Their black wholesale scale is coarse, meant to weigh small truckloads. Realizing this, one of the merchants truthfully says: [sentence 42(a) below, A.K.]”.

⁴⁵ . Schein 1986, 1993.

⁴⁶ . Schwarzschild 1996, 67.

- (42) a. **The vegetables are too heavy for the grey scale and
 too light for the black scale.**
 b. **The vegetables are heavy for the grey scale.**
 c. **The vegetables are light for the black scale.**

To make things easier, let us work with 42(b) and (c). On Schwarzschild's scenario, there is a contextually salient cover that groups the vegetables in the same way as they are in fact grouped by the baskets. All vegetables that share a basket are lumped together. On Schwarzschild's account, 42(b) and (c) both quantify over those parts of the vegetables that are in the contextually supplied cover, and that's all basketfuls of vegetables.

Let's play with Schwarzschild's story. Imagine that the merchants have a range of options, as far as baskets go. There are various sizes available, including small crates. Suppose further that the grey scale can handle a bit more than just a few vegetables at a time. Upon discovering that the vegetables were delivered in biggish baskets, one of Schwarzschild's merchants complains:

- (43) **I wished the vegetables were light. (Then I could weigh them
 with the grey scale.)**

That merchant isn't necessarily longing for new breeds of vegetables that are lighter in weight. More likely, he would have preferred the vegetables to be arranged differently, in smaller crates, for example. The salient cover in the utterance context is the same we had before. It groups the vegetables in the way they are arranged in the actual world. But by uttering (43), the

merchant in fact requests a grouping of the vegetables that is different from the one in the actual world.

The example shows that in intensional contexts, the grouping involved in plural predication does not have to be the same as any grouping in the actual world. The covers relied on, then, must be capable of changing from one possible world to the next. This kind of phenomenon is familiar from the semantics of modals⁴⁷. Modals depend for their interpretation on a contextually provided set of accessible worlds. But when we look at modals in intensional contexts, we find that the accessible worlds they depend on must be able to change from one world to the next. Consequently, the context must supply an accessibility function, not a mere set of possible worlds. Accessibility functions are functions from possible worlds to sets of accessible worlds. Amending Schwarzschild's account of plural predication, we would want to say that plural predication depends on contextually provided cover functions, not just on contextually provided covers. What are cover functions, then, and how can utterance contexts provide them?

A cover function is a partial function that assigns covers to eligible possible worlds or situations. Just any old covers? No way. We cannot evaluate (42), for example, in a world where the vegetables are actually arranged in baskets, using a cover where the grouping of those very same vegetables mirrors the quantities that would fill a whole silo.

(44) The vegetables Distr_C light.

⁴⁷. Kratzer 1978 has much discussion of this kind of context dependency, and Kratzer 1977 has some. See also Schwarzschild 2000.

The values of cover variables cannot be chosen independently of what is the case in the evaluation world. We are only allowed to consider covers that correspond to existing arrangements in the worlds we are looking at. That constraint also has a parallel in the semantics of modals. An accessibility function for an epistemic modal, say, must assign to any possible world a set of possible worlds that are compatible with what is known in that world. Returning to cover functions, they should be functions that assign to any possible world w in their domain a cover that - as far as our vegetables are concerned - lumps them in a way that matches one of their existing arrangements in w . That means that what a cover function does for our vegetables is pick a salient actual arrangement of them, and link it to existing comparable arrangements in other possible worlds. In our example, a basket arrangement in the actual world might be linked to crate arrangements in other worlds.

The insight we have just gained is that covers (and cover functions) have to respect substantive pluralities in the evaluation worlds. But that means that covers correspond directly to sums of group states in the evaluation worlds, with individuals being special cases of groups. Take the basket arrangement of the vegetables. Each basketful of vegetables is a substantive plurality of vegetables. The vegetables in each basket are the possessors of a wide range of group states. Among those are collective weights. The sum of those collective weights is a state itself, and is moreover a state of the sum of all the vegetables, assuming cumulativity for the possessor relation. Such a sum of group states has all the information of a Schwarzschild cover. Let's look at an illustration. We are still dealing with vegetables and Schwarzschild's

merchant. Suppose we have just three baskets of vegetables, one with eggplants, one with zucchini, and one with carrots:

Basket 1	x = the eggplants in 1	s_1 : collective weight of x
Basket 2	y = the zucchini in 2	s_2 : collective weight of y
Basket 3	z = the carrots in 3	s_3 : collective weight of z

$$\text{Cover} = \{a: \exists s [s < s_1 + s_2 + s_3 \ \& \ \text{Possessor}(a)(s)]\} = \{x,y,z\}.$$

The state $s_1 + s_2 + s_3$ is the sum of the three group states s_1 , s_2 , and s_3 . Being group states, they satisfy the Single Possessor Constraint, the analogue of the Single Agent Constraint. As a consequence, any substate of s_1 is a state of x , any substate of s_2 is a state of y , and any substate of s_3 is a state of z . The cover determined by $s_1 + s_2 + s_3$ is thus $\{x,y,z\}$, which is what we want. Let us now return to 42(b) and (c) and see what our analysis would say about those examples. Not fussing too much about the exact semantic representation of adjectives (cheating, in fact - are those adjectives really basic predicates?), the logical representations for 42(b) and (c) could be 45(a) and (b) respectively:

- (45) a. $\neg \text{heavy}_{\text{grey scale}}(\text{the vegetables})(s)$
 b. $\neg \text{light}_{\text{black scale}}(\text{the vegetables})(s)$

45(a) and (b) have free state variables. Given our story, the state $s_1 + s_2 + s_3$ would be a contextually salient value for the free variable 's' in 45(a) and (b). The resulting statements would both be true, assuming cumulativity for both

adjectives, and given that each basketful of vegetables is heavy for the grey scale, and light for the black scale. The facts come out right, then.

Let us now look at the intensional case (43). On the account I am pursuing, the merchant's complaint (43) might have a free state variable that would refer to $s_1 + s_2 + s_3$ in the context of our story. The merchant would then say that he would prefer to live in a world where the pair consisting of the counterpart of the vegetables and the counterpart of $s_1 + s_2 + s_3$ is in the (cumulative) extension of **light**. The intensional context introduces talk about the counterparts of the actual state $s_1 + s_2 + s_3$. However, what would the counterparts of $s_1 + s_2 + s_3$ be in the worlds that correspond to the merchant's wishes? In those worlds, the counterparts of the vegetables are arranged differently, in small crates, for example. The vegetables in each crate would be a substantive plurality of vegetables, and we would have corresponding group states, including collective weights. For our account to go through, the sums of those collective weights would have to be the counterparts of the actual state $s_1 + s_2 + s_3$. This might be stretching the notion of 'counterpart' beyond tolerable boundaries. We would have to accept, for example, that $s_1 + s_2 + s_3$ can have counterparts in worlds in which s_1 , s_2 , and s_3 themselves do not have any counterparts at all. More plausibly, then, (43) involves radically restricted existential quantification over plural states⁴⁸. Quantification could then be restricted in each possible world to

⁴⁸. See Schwarzschild 2000 for the use of such radical (singleton) quantifier restrictions for indefinites. If radically restricted existential quantification rather than reference is at work here, we might wonder whether radically restricted existential quantification isn't generally at work when apparently referential expressions occur in the scope of intensional operators. Maybe that's how individuals are identified across possible worlds. Another possibility to be considered is that events might not be particulars at all, but functions that

‘analogues’ of $s_1 + s_2 + s_3$, that is, sums of collective weights of vegetables in baskets, crates, and other kinds of containers. Neglecting details, we could have the following logical representations⁴⁹:

(46) $\text{Should}_w \Box w' \Box s [C(w')(s) \ \& \ \Box \text{light}_{w'} (\text{the vegetables}_{w'})(s)]$

(46) has an existential quantifier over states, which is restricted by the free cover variable ‘C’. Plausible values for ‘C’ in the context of our story are highly specific properties of states that hold of just $s_1 + s_2 + s_3$ in the actual world and of analogous states corresponding to possibly different arrangements of the vegetables in other possible worlds. As expected, overt event quantifiers over states can be restricted in precisely this way, too. The vegetable merchant might have said: “Whenever the vegetables are light, I can weigh them with the grey scale,” for example. Or: “Only twice have the vegetables been light so far. Usually, they are too heavy for the grey scale.” In all those cases, the states quantified over are sums of collective weights of substantive pluralities of vegetables in the evaluation worlds.

Not everything is left to the vagaries of context when it comes to filling in a value for the free variable ‘C’ in (46). When we talked about Schwarzschild’s covers, we encountered in fact a generalization: The cells of a cover have to correspond to salient substantive pluralities. Here is a rather dramatic example to underline this point. Suppose you and I each have a donkey and a

map possible worlds to one of their parts. A related notion of events is argued for in David Lewis’ paper “Events”, in his *Philosophical Papers*, volume II, chapter 23, p. 241-269.

⁴⁹ . For the purposes of this discussion, we are temporarily switching to an intensional language that has overt quantification over possible worlds. The variables w and w' range over possible worlds.

cat, and these are all the animals we have. It so happens that my donkey looks just like your donkey, and my cat looks just like your cat. In this situation, 47(a) is true, and 47(b) is false. How come?

- (47) a. **My animals look just like your animals.**
 b. **My animals look very different from your animals.**

Given the facts of the case and cumulativity of the relations 'look like' and 'look different', both 47(a) and (b) should be true. But 47(b) seems false, and the intuition that there is a difference between 47(a) and (b) is very strong. We have:

Look like

My donkey $\longleftrightarrow$ Your donkey s₁

My cat $\longleftrightarrow$ Your cat s₂

Look different

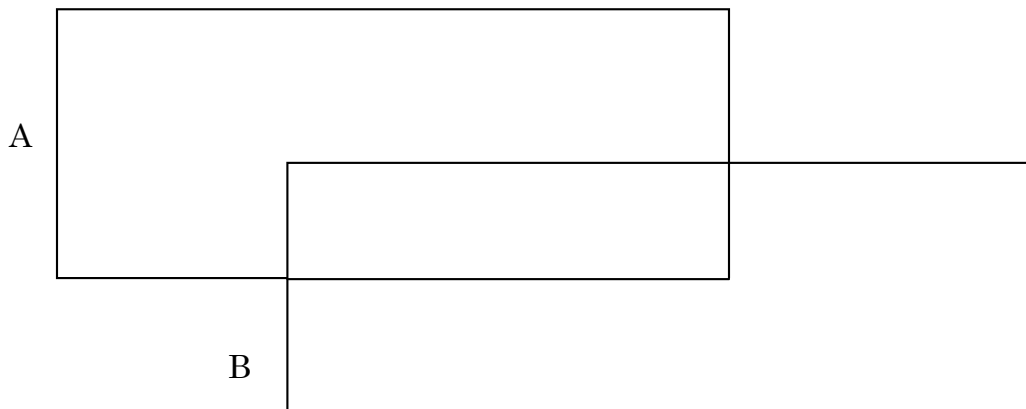
My donkey $\longleftrightarrow$ Your cat s₃

My cat $\longleftrightarrow$ Your donkey s₄

It's not that 'look alike' behaves very differently from 'look different' as far as cumulativity is concerned. If my donkey looked different from your donkey, and my cat looked different from your cat, my animals would look different from your animals. It seems, then, that the difference between 47(a) and (b) in the scenario I set up is due to a difference in substantive pluralities. 47(a)

and (b) have radically restricted existential quantification over (or reference to) plural states, and the domain of eligible plural states should not include states like $s_3 + s_4$. The participants in s_3 and s_4 are not substantive pluralities. If we are comparing looks, we have to compare donkeys with donkeys and cats with cats. The substantive pluralities in the situation we are looking at are my donkey + your donkey, and my cat + your cat, but not my donkey + your cat, and your donkey + my cat. Substantive pluralities are parts of natural kinds in this case.

Not surprisingly, grouping effects are well documented with spatial examples. Take Scha's rectangles⁵⁰ and 48(a) and (b):



- (48) a. **The sides of rectangle A are parallel to the sides of rectangle B.**
- b. **The sides of rectangle A are perpendicular to the sides of rectangle B.**

⁵⁰ . Scha 1981, 1984.

Assuming cumulativity of ‘parallel to’ and ‘perpendicular to’ and no further constraints, 48(a) and (b) should both be true. Yet only 48(a) is commonly judged true in Scha’s scenario. Grouping seems to account for the difference. You are comparing each side of triangle A with the corresponding side of triangle B. Each side of triangle A, then, forms a substantive plurality with the corresponding side of triangle B. No side of triangle A forms a substantive plurality with a non-corresponding side of triangle B.

Relations group individuals. In our earlier example, the ‘look alike’ relation groups our donkeys together, and our cats. Both of those pluralities are substantive and are parts of natural kinds. In contrast, the ‘look different’ relation groups my donkey and your cat together, and your donkey and my cat. None of those pluralities is substantive in the absence of context.

Quantification over states, then, respects substantive pluralities:

Constraint for quantification over states

Quantification over states is restricted to sums of states whose participants are substantive pluralities.

The constraint covers quantification over singular states or states of singular individuals as well. This is so because singular states are also sums, and singular individuals are always substantive groups. The constraint is likely to be a special case of a more general constraint, but for our present concerns

it will do as is⁵¹. It amounts to a substantial constraint for quantification over states and does justice to the strength of the intuitions that we have regarding examples like (47) and (48).

Examples (47) and (48) have relational predicates and are thus cases where Schwarzschild would have to posit cover variables whose values are ‘paired covers’. Beck 1999, 2001, and 2002 observes that all convincing examples of relational cover effects present us with situations that are made up of salient subsituations⁵². Schwarzschild’s own comments on Scha and Stallard’s frigates and carriers example (his example (208)) are revealing in this respect⁵³:

(208) The frigates are faster than the carriers.

“..... We just need to think partitionally. Imagine, for example, that (208) is uttered in a context in which it is clear that these ships are sent out in teams to different areas of the globe with each team consisting of frigates and carriers. It may be that one area calls for very fast action while another will tolerate a sluggish response. If that were the case, I would judge (208) true just in case the frigates in a given area were faster than the carriers of that area, regardless of what speed relations obtained between ships of different areas.”

⁵¹ . For a more general discussion of plural quantification, see Schein 1986, 1993, Roberts 1987, 1990 (3.2.4), as well as Schwarzschild’s work. Schein very explicitly links quantification over plural individuals to event quantification, an important move.

⁵² . Beck does not go so far as to eliminate all statements about cover dependence in her logical representations. Her argument only targets the use of Schwarzschild’s paired covers. She still uses cover dependent restrictions for the argument variables of verbs that are affected by a cumulation operator.

⁵³ . Schwarzschild 1996, p. 87. Scha and Stallard 1988.

That apparent cover effects are triggered by salient states or events is entirely compatible with a cover approach, of course. Those covers just have to be restricted so as to respect salient states or events. This is one way how context can affect covers. The cover approach is only threatened if we have independent motivation to posit devices that take care of quantification over or reference to salient states and events. Davidsonian event and state variables are such devices. We do not need separate quantification over covers, then.

To summarize this section, we started out with Schwarzschild's apparent counterexample (31) to Lasersohn's analysis of **together**:

(31) **The ax and the box together are light enough to carry.**

I supported Lasersohn and argued that (31) can only be true if the ax and the box are a substantive plurality. (31) must then describe a group state, which has to satisfy the Single Possessor Constraint. To muster additional support for the theoretical importance of the notion of a group state, I ended this section by showing that, given that notion, cover effects in plural quantification can be reduced to constraints on event quantification, as Schein suggested.

4.5 More Schwarzschild vs. Lasersohn

Schwarzschild has also argued that Lasersohn's method of retrieving collective predication within an event semantics fails in cases that have objects with downward entailing quantifiers. While Schwarzschild's arguments do not literally affect Lasersohn's actual analysis, they fully apply to the adaptation of it I have been endorsing. I will argue that

there are independent reasons to assume that the problematic quantifier phrases are not interpreted in the domain of **together**. They are scoped.

All would be well for **together** and cumulative verb denotations, if Schwarzschild 1993-94 hadn't set up what looks like yet another fatal trap for definition (15) and its kin.

(15) $T(\text{together}) =$

$$\lambda R_{\langle e, \langle st \rangle \rangle} \lambda y \lambda e [R(y)(e) \ \& \ plural(y) \ \& \ \lambda e' \lambda z [[e' \leq e \ \& \ R(z)(e')] \ \& \ z = y]]$$

As before, if we lose (15), we lose our method for distinguishing distributive and non-distributive predication, and that puts the whole enterprise of starting out with cumulative verb denotations in jeopardy. Sentence (49) is a sentence of the kind Schwarzschild has drawn attention to:

(49) **The two copy editors together caught fewer than three mistakes.**

In this case, Schwarzschild's objection would go as follows: According to (15) (plus routine interpretation procedures including existential closure of the event argument), (49) says that there was an event e that had the two copy editors as agents, fewer than three mistakes were caught in e , and there was no subevent of e where fewer than 3 mistakes were caught, and whose agent was anybody but the two copy editors. But this is not right. (49) might very well describe an event e that has a subevent in which just one of the copy editors found just one mistake, for example. Such an event e , then, has a

subevent whose agent found less than three mistakes, but who is not identical to the two copy editors. This looks like a deadly blow for (15) ⁵⁴.

For Schwarzschild's argument to go through, we have to assume that (49) does not necessarily describe collaborative action. If it did, we could dismiss the case right away. (49) would not be a problem for (15), because we are assuming that any true collective action by the two copy editors could only have subactions that are themselves actions by the two copy editors. The following examples confirm that sentences following the model of (49) do not necessarily talk about collective states, actions, or mishaps:

- (50) a. **The two real estate agents together own fewer than 10% of all houses in this area.**

⁵⁴. Schwarzschild's objection does not apply to the original analysis of Lasersohn 1988, 1990, as pointed out in Lasersohn 1995, 1998. Here is what Lasersohn's analysis says about (49), simplifying slightly. (49) is true iff there is an event *e*, the two copy editors caught fewer than three mistakes in *e*, and whenever there is a subevent *e'* of *e* in which fewer than three mistakes were caught, the ones who caught fewer than three mistakes in *e'* are the same as the ones who caught fewer than three mistakes in *e*. Suppose the copy editors were Casey and Stacey, and they caught exactly one mistake each. Casey caught Addition, and Stacey caught Omission. Let *e* be the minimal event in which Casey and Stacey caught the mistakes they did. (49) is intuitively true in *e*. Now let *e'* be the subevent of *e* that includes Casey's catching Addition, but excludes Stacey's catching Omission. It's true in *e'*, then, that Casey caught fewer than three mistakes. According to Lasersohn, it would also be trivially true that Stacey caught fewer than three mistakes in *e'*, and the same is true for all other individuals in the universe. Consequently those who caught fewer than three mistakes in *e'* are the same as those who caught fewer than three mistakes in *e*. This argument assumes a semantics for the quantifier phrase, where the mere fact that Stacey does not exist in *e'* is

- b. **The three cows together produced less than 100 gallons of milk last month.**
- c. **Those 50 patients together suffered less than 5 heart attacks last year.**

In 50(a) to (c), all we do is sum up houses owned, milk produced, and heart attacks suffered by the relevant individuals, and count or measure the result⁵⁵.

In order to address Schwarzschild's objection, we have to get clear about the interpretation of downward entailing quantifiers. Downward entailing quantifiers are known to require special care in semantic frameworks based on events or situations⁵⁶. Take (51):

(51) **Casey caught fewer than three mistakes (yesterday).**

What are the conditions for (51) to be true? The first important observation is that the existence of an event of Casey catching one, two, three, or no mistakes at all isn't sufficient to make (51) true. Even if Casey caught a hundred mistakes, there are still events that are events of Casey catching no

sufficient for it to be true that she caught fewer than three mistakes in e', an undesirable consequence.

⁵⁵. 50(c) shows quite clearly that we want the meaning assignment (17') for **together**. We do not want to require that each of the patients contribute a heart attack.

⁵⁶. Krifka 1989, Kratzer 1989, Bonomi and Casalegno 1993, Schein 1993, Lasersohn 1995, and Landman 1998, 2000.

mistake, one mistake, two mistakes, or three mistakes. To find out whether (51) is true, we have to find out about the total number of mistakes Casey caught. (51) is true just in case that number is less than three. The interpretation of **fewer than three mistakes** in (51), then, depends on the total number of mistakes caught by the agent during a contextually specified reference time. Within the current framework of assumptions, this means that we can't obtain the correct interpretation for (51) if we leave the direct object *in situ*. It would then be interpreted at a stage where the agent argument has not yet been added, hence no information about the total number of mistakes caught by the agent of the action described by the verb is available. We can conclude that **fewer than three mistakes** has to scope beyond the point where the agent argument is introduced. Since we are interested in the number of mistakes caught during a reference time, **fewer than three mistakes** has to end up in a place where the events being described are related to a reference time. This gives us a lowest possible landing site: The specifier position of **Aspect**.

Kratzer 1998 argues that at some point in the hierarchy of inflectional heads, properties of events⁵⁷ are mapped into properties of times, and I conjectured that aspectual heads related to viewpoint aspect in the sense of Smith 1991 have the function to carry out this mapping. **Aspect** heads mark a switch of perspective from events to reference times. Perfective aspect, for example, locates events within a reference time. More technically, it maps a property of events P into a property of times that is true of a time t just in case t

⁵⁷. If the event argument is to receive an indexical interpretation, we could 'pseudo-saturate' the event argument position by applying the operator $\lambda e [P(e) \ \& \ e = e_0]$ to the relevant predicate of events, rather than applying that predicate to the variable 'e₀'.

includes the running time of a P-event. If i is the type for intervals of time, and $\Box(e)$ is the running time of e , we have:

1. $T([\textbf{perfective}]) = \Box P_{\langle st \rangle} \Box t_i \Box e [P(e) \ \& \ \Box(e) \Box t]$
2. $T(1 ([\textbf{perfective}] (\textbf{Casey} ([\textbf{active}] (\textbf{caught } t_1))))) =$
 $\Box x \Box t \Box e [\Box \text{catch}(x)(e) \ \& \ \Box \text{agent}(\text{Casey})(e) \ \& \ \Box(e) \Box t]$

If downward entailing quantifier phrases are to move to a position beyond **Aspect**, they must be given appropriate denotations. Instead of operating over relations between individuals and events, they should operate over relations between individuals and times. The denotation of **fewer than three mistakes** might be as in (52), then⁵⁸:

- (52) $T(\textbf{fewer than 3 mistakes}) =$
 $\Box R_{\langle e \langle it \rangle \rangle} \Box t \Box x [[\text{mistakes}(x) \ \& \ R(x)(t)] \Box \ /x/ < 3].$

If a quantifier phrase must operate over relations between individuals and times, scoping beyond **Aspect** is forced by its semantic type. Continuing our most recent derivation we end up with the following:

⁵⁸. Verbs with intensional object positions show that quantifier phrases like **fewer than 3 mistakes** or **less than 10 new winter coats** might require decomposition. See Hackl 2000. **The children need less than 10 new winter coats** is true on the relevant reading, for example, iff for all n such that the children need n -many new winter coats, $n < 10$. A decomposition analysis is unlikely to affect the essence of the point made in this section.

3. $T((\text{fewer than 3 mistakes})(1([\text{perfective}](\text{Casey}([\text{active}]$
 $(\text{caught } t_1)))) =$
 $\lambda t \lambda x \lambda e [\text{mistakes}(x) \ \& \ \text{catch}(x)(e) \ \& \ \text{agent}(\text{Casey})(e) \ \& \ (e) \sqsubseteq t]$
 $\lambda [x/ < 3]$.

Applying the property of times we just derived to a reference time (e.g. yesterday), we obtain the correct denotation for sentence (51) from above:

- (51) a. **Casey caught fewer than 3 mistakes (yesterday).**
 b. $\lambda x \lambda e [\text{mistakes}(x) \ \& \ \text{catch}(x)(e) \ \& \ \text{agent}(\text{Casey})(e) \ \& \ (e) \sqsubseteq \text{yesterday}] \lambda [x/ < 3]$

(51) says that the total number of mistakes Casey caught yesterday is less than 3, and (52) brings about this result, provided that the object is quantified in right above **Aspect**. Downward entailing quantifier phrases, then, would be forced by their very meaning to a position above **Aspect**. But this means that downward entailing quantifier phrases do not stay within the scope of **together**, - and that's how we can safely get around Schwarzschild's objection. It is crucial to that objection that the downward entailing quantifier phrase be interpreted within the scope of **together**.

The explanation I offered for why downward entailing quantifier phrases have to scope beyond **Aspect** implies that they should have to scope out of any constituent that denotes a property of events. That this might be at least a preference is suggested by the following examples:

- (53) a. **The construction of fewer than 10 barns was captured on video.**
 Marginal: Fewer than 10 barns were constructed and that was captured on video. Preferred: There were fewer than 10 barns whose construction was captured on video.
- b. **We saw fewer than 10 guests leave.**
 Marginal: Fewer than 10 guests left and we saw that. Preferred: There were fewer than 10 guests whose departure we saw.
- c. **The teacher made fewer than 10 students stand up.**
 Marginal: Fewer than 10 students stood up, and the teacher made that happen. Preferred: There were fewer than 10 students that the teacher made stand up.

The underlined expressions in 53(a) to (c) denote properties of events and include a downward-entailing quantifier phrase. In every case, the quantifier phrase prefers to scope out. Narrow scope interpretations are not readily available. Apparently, those quantifier phrases do not like to be interpreted *in situ*.

That downward entailing quantifiers prefer to scope to a higher position can be detected directly in languages where surface order reflects scope relations more directly than is the case in English. The German sentence 54(a) is awkward, for example, since a downward entailing quantifier phrase is trapped within the scope of a manner adverb that selects a property of events.

- (54) a. ? **Ich hab' genüsslich weniger als 3 Äpfel gegessen.**
 I have with enjoyment less than 3 apples eaten
 I ate less than 3 apples with enjoyment.

- b. **Ich hab' weniger als 3 Äpfel genüsslich gegessen.**
 I have less than 3 apples with enjoyment eaten.
 I ate less than 3 apples with enjoyment.
- c. **Ich hab' genüsslich 3 Äpfel gegessen.**
 I have with enjoyment 3 apples eaten
 I ate 3 apples with enjoyment.

If uttered out of the blue, there is clear contrast in acceptability between 54(a) on the one hand, and 54(b) and (c) on the other. In 54(b), the quantifier phrase has scope over the adverb, rather than the other way round. In 54(c), the quantifier phrase is not downward entailing, and is comfortable within the scope of the adverb.

In English, a deviance like the one in 54(a) appears when we prevent a downward entailing quantifier phrase from scoping out of an event nominalization:

- (55) ? **John's construction of fewer than 10 barns was captured on video.**

In (55), the genitive within the event nominalization acts as a barrier for movement, and forces the quantifier phrase to stay within the DP. The result sounds awkward. Good, - but why only awkward? Why aren't 54(a) and (55) outright ungrammatical? Why is it that in 53(a) to (c) the quantifier phrase only prefers to scope out? Don't we need a more solid result?

(55) and its kin become acceptable if the right kind of context is created. Suppose John is a participant in a contest where a series of 10 Lego barns have to be constructed within two hours, following specifications that become

increasingly harder with every barn. Few participants manage to build all 10 barns within the two hours allotted to them. John's time was up after barn 8. Talking about that particular event, my uttering (55) would not be awkward. Similar contextual restrictions improve the German sentence 54(a). What is going on?

Since (55) is acceptable in the context I just described, it must be possible for downward entailing quantifier phrases to be interpreted within predicates denoting properties of events after all. In addition to (52), we might have a the meaning assignment in (56).

- (56) a. $T(\text{fewer than 10 barns}) =$
 $\lambda R_{\langle e, st \rangle} \lambda e \lambda x [\text{barns}(x) \ \& \ R(x)(e) \ \& \ /x/ < 10]$
- b. $T((\text{construction of fewer than 10 barns})) =$
 $\lambda e \lambda x [[\text{barns}(x) \ \& \ \lambda e [\text{construction}(x)(e)] \ \& \ /x/ < 10]$

56(a) is the kind of definition that I didn't want to consider earlier. For good reasons. It yields properties of events that are too easy to satisfy. Even if a thousand barns were constructed in all, there are still events in which less than 10 barns were constructed. However, the acceptability of (55) in certain contexts suggests that 56(a) might be available if the context provides the right quantifier restrictions. The scenario I set up for (55) characterized a particular event that was presupposed to be a construction of barns. I used (55) to make a claim about that very event, and informed you that it was one where fewer than 10 barns were constructed in all. Properties of events like 56(b) are only dangerous if the event argument is quantified without further contextual constraints. With strong contextual restrictions, those properties can do no harm. In those cases, then, downward entailing quantifier phrases

might not have to scope out. They, too, might be interpreted within predicates that denote properties of events.

Quantifier interpretations following the model of 56(a) are also needed for the cumulative reading of (57):

(57) (In this department), at most 5 professors are supervising at most 10 dissertations.

As documented in Schein 1993, the likes of (57) pose considerable challenges to theories of logical form. They are often taken to require the resources of branching or binary quantification⁵⁹. I will not be able to even begin to face the empirical coverage of relevant cases in Schein's work. I have to stick to a few simple examples so as to not lose track of what I am after. What does (57) mean on its cumulative interpretation? On the intended reading, (57) is most natural in a context where ongoing thesis supervisions in a particular department are being discussed. It implies that at most 10 dissertations are being supervised, and that at most 5 professors are supervising dissertations. The two quantifier phrases do not scopally interact. This suggests strongly that they have existential interpretations as in (58):

⁵⁹. Sher 1990, and references cited there. For a different perspective within an event-based semantics, see Landman 2000. Landman aims at a unified account of maximalization claims for both asserted and implicated meaning components, a project that is much more ambitious than the one I am engaged in here. Landman proposes an interpretation system with several new design features affecting the grammar as a whole, which makes it difficult to assess the cost of his analysis in comparison with other approaches.

- (58) a. $T(\text{at most 5 professors}) =$
 $\Box R_{\langle e \langle st \rangle \rangle} \Box e \Box y [\text{professors}(y) \ \& \ R(y)(e) \ \& \ /y/ \leq 5]$
- b. $T(\text{at most 10 dissertations}) =$
 $\Box R_{\langle e \langle st \rangle \rangle} \Box e \Box x [\text{dissertations}(x) \ \& \ R(x)(e) \ \& \ /x/ \leq 10]$
- c. $T(\text{at most 5 professors supervising at most 10 dissertations}) =$
 $\Box e \Box y [\text{professors}(y) \ \& \ \Box \text{agent}(y)(e) \ \& \ /y/ \leq 5 \ \& \\ \Box x [\text{dissertations}(x) \ \& \ \Box \text{supervise}(x)(e) \ \& \ /x/ \leq 10]]]$

What is still needed for an appropriate logical form of (57) on the intended reading is event quantification without commitment to any actual dissertation supervisions, and the usual provision for contextual restrictions. We should understand (57) as a contextually restricted universal quantification over possibly plural events, then:

- (59) $\Box e [[C(e) \ \& \ \text{now} \ \Box \Box(e) \ \& \ \text{in}(\text{this department})(e)] \ \Box \Box y [\text{professors}(y) \ \& \ /y/ \leq 5 \ \& \ \Box \text{agent}(y)(e) \ \& \ \Box x [\text{dissertations}(x) \ \& \ /x/ \leq 10 \ \& \ \Box \text{supervise}(x)(e)]]]$

Context would be expected to restrict the event quantifier in (59) to sums of dissertation supervisions, and quantification is then over sums of current dissertation supervisions in this department. Any such sum you may pick (including singular thesis supervisions, of course) has at most 5 supervising professors and at most 10 supervised dissertations. Assuming that those who use (59) intend to say something non-trivial and true, it is easy to fill in the

contextual restriction. The consequent of the conditional has all the information needed.

One way of thinking about (59) is to take it as the formalization of a statement that could be more explicitly expressed by 60(a) or 60(b):

- (60) a. **What is going on in this department is that at most 5 professors are supervising at most 10 dissertations.**
- b. **All that's happening in this department is that at most 5 professors are supervising at most 10 dissertations.**

If downward entailing quantifier phrases can in principle be interpreted at a stage where the event argument is still available, where does this leave us with respect to Schwarzschild's threat? Doesn't it come to bite us after all? It would, if we found cases where a downward entailing quantifier phrase must stay within the scope of **together**. (61) looks like a relevant example:

- (61) **The 5 professors together taught at most 50 students in at most 10 classes.**

(61) only has a cumulative interpretation, which means that there are no scope interactions between **at most 50 students** and **at most 10 classes**. We know that to get that interpretation, we need to interpret those two quantifier phrases before the event argument is quantified off. But then they can't be scoped beyond **Aspect**. Could they still be scoped beyond **together**, though? Before we explore this possibility, it is time to be serious about the syntactic status of **together**. Roger Schwarzschild has shown that preverbal

together is in fact in a DP internal position⁶⁰. Here are some of his examples:

- (62) a. **The credit risk and the interest rate together can
 affect the value of the bond in complicated ways.**
- b. **At least ten bullets hit John and Mary together.**
- c. **To John and Mary together, I bequeath my sterling
 silverware.**

If preverbal **together** is always in a DP-internal position, it has to modify DPs or NPs. Not just any NP or DP. DP internal **together** can only modify referential DPs:

- (63) a. * **At least 2 professors together are currently supervising
 10 dissertations.**
- b. * **Professors together are currently supervising
 10 dissertations.**
- c. * **Many professors together are currently supervising 10
 dissertations.**

DP-internal **together** could now have the denotation 64(a) or (b), corresponding to (15) and (17') respectively:

⁶⁰ . Schwarzschild 1993-94, p. 244

- (64) a. $T(\text{together}) =$
 $\lambda y \lambda R_{\langle e, \langle st \rangle \rangle} \lambda e [R(y)(e) \ \& \ plural(y) \ \& \ \lambda e' \lambda z [[e' \leq e \ \& \ R(z)(e')]] \ \& \ z = y]]$
- b. $T(\text{together}) =$
 $\lambda y \lambda R_{\langle e, \langle st \rangle \rangle} \lambda e [R(y)(e) \ \& \ \lambda z [[z \leq y \ \& \ C(z)]] \ \& \ \lambda e' [e' \leq e \ \& \ R(z)(e')]]]$

What would a derivation of (61) look like? Let's answer this question for the simpler sentence (65), with the understanding that **at most five students** in (65) is to be given an existential interpretation following the model of (56) or (58):

(65) **The 5 professors together taught at most 50 students.**

The derivation of the Logical Form for (65) is likely to include the following stage:

(66) **(at most 50 students) (1 ((the 5 professors) together) ([agent] teach t_1)))**.

In (66), **at most 50 students** is scoped, but its landing site could still be below **Aspect**, hence the event argument could still be available. Even for (65), then, scoping could in principle allow us to derive the intended interpretation:

(65') $\lambda e [[C(e) \ \& \ \lambda(e) \ \text{then}] \ \lambda \lambda x [\text{students}(x) \ \& \ /x/ \leq 50 \ \& \ \lambda \text{agent}(\text{the } 5 \text{ professors})(e) \ \& \ \lambda \text{supervise}(x)(e) \ \& \ \lambda e' \lambda z [[e' \leq e \ \& \ \lambda \text{agent}(z)(e') \ \& \ \lambda \text{supervise}(x)(e')] \ \lambda z = \text{the } 5 \text{ professors}]]]]$

Context is expected to restrict quantification in (65') to thesis supervisions by those 5 professors.

So far, so good, but is there a stage corresponding to (66) in the derivation of (65)? In (66), the subject is still in the position where it entered the derivation as the argument of **[agent]**. In contrast, the object is caught at a stage where it has moved over the base position of the subject. In much recent syntactic work, objects are assumed to undergo this kind of movement to reach a position where they can check accusative case. The position is an outer specifier position within the **[agent]** projection for some, and in the specifier position of a separate agreement projection for others. As long as the landing site is below **Aspect**, either possibility is compatible with the facts we have seen. Movement of objects is seen overtly in languages with object shift. Research on overt object shift has shown that object shift is only possible for certain types of DPs⁶¹. In particular, it is never possible for bare plurals or mass nouns. If the interpretation of (65) we are interested in comes about through object shift, that kind of interpretation should not be available for sentences with objects that are known not to undergo that shift⁶². We now have a testable prediction. It is borne out:

⁶¹ . Diesing & Jelinek 1995, Diesing 1997.

⁶² . See Kratzer 1988, 1995 for evidence showing the connection between overt object shift in German and covert object shift in English.

- (67) a. **The two real estate agents together own houses in this area.**
- b. **? The three cows together produced milk last month.**
- c. **? Those 50 patients together suffered heart attacks last year.**

67(a) is about collective ownership. 67(b) and (c) sound funny out of the blue. 67(b) might be about a dairy cooperative run by three cows. Or we might think of scenarios where milk production is enhanced by keeping several cows together in the same stall. 67(c) evokes collective suffering of heart attacks. In each case, the plurality denoted by the subject has to be substantive. A mere ‘summing up’ interpretation of **together** is not available. Why? It is not expected to be available if the direct objects are interpreted *in situ*. The resulting claims would then be contradictions. If we end up with milk when we sum up the milk production of the three cows, it can’t be that none of the individual cows produced any at all. The only available ways for 67(a) to (c) to wind up true, then, is if the real estate agents, the cows, and the patients are substantive pluralities engaged in truly collective states and events.

The contrast between 68(a) and (b) makes the same point:

- (68) a. **? Die 3 Kühe zusammen haben keine Milch produziert.**
 The 3 cows together have no milk produced
 ‘The 3 cows together produced no milk.’

- b. **Die 3 Kühe zusammen haben keinen Tropfen Milch**
 The 3 cows together have no drop milk
produziert.
 produced.
 ‘The 3 cows together didn’t produce a drop of milk.’

The object in 68(a) is a weak negative existential. 68(a) sounds funny, as does its English counterpart. As before, we imagine a cow cooperative, or 3 cows put in the same stall to increase milk production. 68(b) has a strong negative existential. It is fully compatible with a mere ‘summing up’ interpretation saying that there isn’t a single drop of milk that the three cows together produced. That is, the milk production of the three cows didn’t add up to a single drop of milk. What we see in 65(a), then, is exactly what we would expect to see in (50) from above if there wasn’t the possibility of object shift:

- (50) a. **The two real estate agents together own fewer than 10% of all houses in this area.**
- b. **The three cows together produced less than 100 gallons of milk last month.**
- c. **Those 50 patients together suffered less than 5 heart attacks last year.**

We have found evidence that scoping of the direct object makes it possible for **together** in 50(a) to (c) to have a mere ‘summing up’ interpretation. If we

block scoping, we see the expected effects, a forced switch to substantive groupings and collective events. In the case of direct objects, the scoping is regular object shift, a movement that all but the weakest objects undergo, most likely for reasons of case. In the case of prepositional phrases, scope shifts are known to be sensitive to the weak-strong distinction as well, but the driving force for their movement is still a matter of debate.

One more time, we have been able to avert a serious threat for Lasersohn's method of retrieving the difference between collective and distributive predication. This time round, I appealed to scoping. We saw interesting new data showing that the scoping I relied on does indeed exist.

4.6 A single source for lexical and phrasal cumulativity?

The preceding sections looked at a series of apparent obstacles to the 'cumulativity from the start' agenda for verbs and thematic role predicates, and got those obstacles out of the way. Not all instances of cumulative readings can be reduced to lexical cumulativity, however. There are irreducibly phrasal cases of cumulativity. Isn't it likely, then, that lexical and phrasal cumulativity have a single source? Whatever that source may be, it seems to

undermine the motivation for having a Cumulativity Universal for basic verbs and thematic role predicates.

That there is no obstacle to assuming cumulative denotations for bare verbs and thematic role predicates doesn't mean that there is strong motivation in favor of such an assumption, of course. Let us briefly remind ourselves why we wanted cumulative denotations in the verbal domain to begin with. We started out with the observation that cumulative denotations for basic verbs and thematic role predicates explain why sentences like (69) can have cumulative readings:

(69) **Ten movers carried 500 boxes.**

On its cumulative interpretation, sentence (69) allows a whole range of relationships between a plurality of 10 movers and a plurality of 500 boxes. Some movers might have jointly carried some of the heavier boxes. Individual movers might have carried more than one of the lighter boxes at a time. Some of the boxes might have been carried more than once by the same movers or by different movers. (69) covers all those scenarios, and the Cumulativity Universal tells us why.

While the existence of cumulative readings for sentences like (69) indicates that verbal projections are pluralized at some point, we can't quite conclude yet that data of this sort show that basic verbs and thematic role predicates are obligatorily pluralized from the very start. Pluralization of those predicates could still come about in a number of ways. Several authors, most

prominently Wolfgang Sternefeld, Uli Sauerland, and Sigrid Beck⁶³, have argued that the denotations of verbs do not have to start out cumulative, but can be rendered cumulative through the optional presence of syntactically represented *-operators which map properties and relations into their smallest cumulative extensions and can be inserted freely. Those operators seem to be independently needed. The vast literature on plurals and distributivity has documented beyond any doubt that lexical cumulativity alone cannot account for all cases of cumulativity there are. Let us look at some representative examples of non-lexical cumulativity:

- (70) a. **The women from Boxborough brought a salad.**
 (Roberts, 1990, p. 102, 146)
- b. **The boys gave the girls a flower.**
 (Winter 2000, p. 39)
- c. **John and Mary made less than \$10,000 last year.**
 (Lasersohn 1990, p. 32)
- d. **John and Mary made more than \$10,000 last year.**
 (Lasersohn 1990, p. 32)

70(a) has an interpretation where each of the women from Boxborough brought a salad. For 70(b), Yoad Winter designed the following scenario: There were two boys, John and Bill, and four girls, Mary, Sue, Ann, and Ruth. John met Mary and Sue and gave them a flower. Bill met Ann and

⁶³. Sternefeld 1998, Sauerland 1998, Beck 2000, Beck and Sauerland 2000, Beck 2001.

Ruth and gave them a flower. 70(b) is true in this case. As for 70(c) and (d), Peter Lasersohn observes that they could both be simultaneously true in a situation where John and Mary each made \$6,000 last year, pointing to an ambiguity. 70(c) would be true, since John and Mary each made less than \$10,000. And 70(d) would be true, since the combined income of John and Mary was more than \$10,000. Those facts cannot be accommodated in a theory that only has lexical cumulativity. Anybody who subscribes to the Cumulativity Universal, then, has to come up with a convincing source for the many instances of non-lexical cumulativity. Optional *-operators would automatically cover both lexical and phrasal cumulativity. They are very serious threats to our Cumulativity Universal, then. They knock the wind right out of it. There would no longer be any need for such a universal.

“A final issue we would like to address is that of lexical predicates. Once we have the $\square$ -operator [a $\square$ -operator mapping binary relations into their smallest cumulative extension, A.K.] at our disposal, this operator can be held responsible for cumulative readings with lexical predicates, too. We have throughout the paper kept open the possibility that there is an independent mechanism that cumulates lexical predicates (the meaning postulates approach from Scha); this was done for the purpose of establishing the need for $\square$ independently of assumptions about lexical predicates. Given our theoretical conclusions, however, we see no need to keep an independent lexical mechanism, and we suggest to use $\square$ on lexical relations instead.” Beck and Sauerland 2000, p. 370.

Beck and Sauerland are not working within an event semantics, hence cannot rely on the event-based characterization of collective action that Lasersohn’s work has made possible. The difference between the collective and the distributive reading of 70(a), for example, is now attributed to an option for the VP denotation, which could be 71(b) or (c):

- (71) a. **Casey and Stacey lifted the piano.**
 b. $\Box x[\text{lifted}(\text{the piano})(x)]$
 c. $\Box \Box x[\text{lifted}(\text{the piano})(x)]$

71(b) has the uncumulated VP denotation, 71(c) the cumulated one. The underlying assumption is that the uncumulated predicate 71(b) can be true of singularities and pluralities, but if it is true of pluralities, those pluralities have to be collective agents⁶⁴. Using 71(b) for the interpretation of the VP in 71(a), then, gives us the collective action reading. If we use 71(c), the resulting reading is compatible with collective or distributive action. Given that the two readings posited are not logically independent, we might wonder why we need the uncumulated V and VP denotations to begin with. We have learned from Lasersohn's work that the real test cases for accounts of the distinction between collective and distributive action come from the family of closely related collectivizing adverbs. But here, event-based accounts are called for independently⁶⁵. It's at least not necessary, then, to account for the ambiguity of 71(a) via absence versus presence of cumulation.

A serious problem for optional $\Box$ -operators is that we miss generalizations about cumulative inference. The following inference is clearly valid, for example:

⁶⁴. The Singularity Constraint of Landman 1996, 2000 has to be satisfied.

⁶⁵. Schwarzschild 1993-94 has an ingenious, non-event-based, account of at least the 'non-distributivity' reading of **together**. See Lasersohn 1995, chapter 11.5 and Schwarzschild 1996, chapter 10.3.1 for challenges and commentary.

Cumulativity Inference

Casey is sleeping happily.

Stacey is sleeping happily.

Casey and Stacey are sleeping happily.

There is no reading of the conclusion of the Cumulativity Inference where the inference does not go through. Yet if **are sleeping happily** could optionally have a non-cumulative denotation, there should be a reading where the Cumulativity Inference is not valid. Generalizations like these were a crucial force behind Landman's reduction of distributivity to pluralization, where pluralization was now uniformly accounted for by cumulation⁶⁶. The same generalization also led to Schwarzschild's assumption that all plural VPs are obligatorily translated with the $\square$ -operator, hence always have cumulative denotations⁶⁷.

"My star, by contrast, appears on the translations of *all* plural verb phrases. I also differ here from Landman (1989), who optionally translates plural verb phrases with a star. My star is obligatory: it is there whenever the verb is plural."

Schwarzschild 1993-94, p. 206

⁶⁶ . Landman 1989. For a current perspective, see Landman 2000.

⁶⁷ . Schwarzschild 1993-94

Linking cumulativity to verbal plural agreement morphology is not an option, of course, for those who maintain that the collective readings of sentences like 71(a) must come from uncumulated VP denotations. We would then be pushed to deny any connection between verbal plural agreement morphology and cumulation. Within an event-based semantics, linking verbal plural agreement morphology and cumulation becomes an option again. Verbal plural agreement morphology could then reemerge as a source of cumulativity. It could in principle be the source of the phrasal cumulativity we see in 70(a) to (d).

Before attempting to derive the cumulative readings of 70(a) to (d) from cumulation operators linked to verbal plural agreement morphology, we have to remind ourselves that up to now, we haven't yet seen any direct argument in favor of lexical cumulativity of basic verbs and thematic role predicates. We have eliminated obstacle after obstacle, but so far, direct evidence for initial cumulativity of basic verbs and thematic role predicates is still missing. Given our concerns in this chapter, we urgently need proof that basic verbs and thematic role predicates are indeed cumulative all by themselves, that is, independently of overt or non-overt verbal plural agreement morphology. The following section will give us that piece.

4.7 Direct evidence for initial cumulativity

This section presents direct evidence for Lexical Cumulativity of verbs. Lexical cumulativity makes it possible for verbs to have iterative interpretations without the help of operators or plural agreement morphology linked to plural verb arguments. If we combine such iterative verbs with a singular indefinite, we should observe a ‘failure of distributivity’ effect. Interestingly, this is exactly what we find. We will also see that the possibility of initial event iterativity solves an old scope puzzle in connection with durational adverbials that was recently discussed in van Geenhoven 2000 and Zucchi and White 2001.

If verbs are cumulative from the very start, they can describe iterated events without the help of operators bringing about that iterativity or plural agreement morphology linked to plural verb arguments. Look at the following examples, which all have singular indefinite objects and describe iterated events:

- (72) What does this intern do?
- a. **She guards a parking lot.**
 - b. **He cooks for an elderly lady.**
 - c. **She waters a garden.**
 - d. **He watches a baby.**
 - e. **She cleans an office building.**

- (73) a. **I dialed a wrong phone number for 5 minutes**⁶⁸.

⁶⁸. Examples like these are discussed in van Geenhoven 2000 and in Zucchi and White 2001. Van Geenhoven proposes lexical decomposition of **dial** into a frequentativity operator and an abstract remnant predicate ‘dial’. The singular direct object **a wrong phone number** would then be given automatic wide scope over the frequentativity operator unless the verb is semantically incorporating in the sense of van Geenhoven 1998, hence has an object position of a property type. Van Geenhoven makes a difference between ‘non-stop’ and

- b. **She bounced a ball for 20 minutes.**
- c. **He kicked a wall for a couple of hours.**
- d. **She opened and closed a drawer for half an hour.**
- e. **I ran up and down a hill for half a day.**

What is remarkable about those sentences is that the singular indefinite objects invariably fail ‘to distribute’⁶⁹. They look as if they were taking wide scope over an event quantifier: A single parking lot is guarded habitually, a single elderly lady is cooked for repeatedly, a single ball is bounced again and again, and so on. What is it that produces this apparent wide-scope effect? The phenomenon is an automatic consequence of the Lexical Cumulativity hypothesis. It shouldn’t exist if we allowed free optional insertion of unpronounced star operators. Nor if plural verbal agreement morphology was the only source of cumulativity. That last possibility is immediately ruled out by the fact that all subjects and objects in (72) and (73) are singular. As for freely inserted star operators, they could immediately

‘again and again’ frequentativity and posits three different realizations of frequentativity in the decomposition of verbs. I suspect that within an event semantics, initial cumulativity might cover all of those different cases. I also conjecture that van Geenhoven’s incorporating verbs might be related to the kind denoting direct objects discussed in the next footnote.

⁶⁹. Apparent exceptions are **she drives a Ford** or **he wears a suit**. The verbs **drive** and **wear** can readily take kind denoting objects, however. Compare **would she drive this car?** or **would he wear this suit?** with **would he travel with this lady?** or **would she take care of this garden?**. In the first two cases, the demonstratives can refer to a kind, in the other two cases they can’t. If **a Ford** can be used to introduce existential quantification over kinds, multiple events of driving a Ford can involve different cars, while still involving the same kind.

produce 74(b) from 74(a), for example, hence derive unattested readings for the sentences in (72) and (73)⁷⁰:

- (74) a. $\Box e \Box x [\text{ball}(x) \ \& \ \Box \text{bounce}(x)(e)]$
 b. $\Box \Box e \Box x [\text{ball}(x) \ \& \ \Box \text{bounce}(x)(e)]$

74(b) describes possibly repeated events in which more than a single ball might be bounced. In contrast, given the singularity of ‘ball’, each event described by 74(a) can only have a single ball in it. The facts in (72) and (73), then, do indeed fall out from Lexical Cumulativity. No scoping has to be stipulated. To see this more clearly look at the computation of the denotation of the VP in (75):

(75) [**bounce a ball**]_{VP}.

- (76) a. $\Box x \Box e \Box \text{bounce}(x)(e)$
 b. $\Box R_{\langle e \langle st \rangle \rangle} \Box e \Box x [\text{ball}(x) \ \& \ R(x)(e)]$
 c. $\Box e \Box x [\text{ball}(x) \ \& \ \Box \text{bounce}(x)(e)]$

‘being a possibly plural event e such that there is a ball x and e is an event of bouncing x ’.

In a Davidsonian event semantics, events are ‘minimal’ in the sense that an event of bouncing this ball, for example, is an event in which this ball is being bounced and which contains nothing above and beyond that ball and whatever it takes to bounce it. Crucially, it can’t have a second ball in it. Assuming lexical cumulativity, the relation 76(a) is cumulative. As a result,

⁷⁰ . To emphasize lexical cumulativity for ‘bounce’, I used the $\Box$ -operator here, which is in principle superfluous, since we are assuming that all basic lexical items are cumulative.

the property 76(c) can describe an iterated event made up of events which themselves have the property 76(c). However, whenever 76(c) is true of an event *e* and a subevent *e'* of *e*, the ball in *e'* is bound to be the same as the ball in *e*. Otherwise, *e* would have two balls, rather than one. Each event in the iteration, then, has the same ball in it.

To summarize, we have found a non-trivial consequence of the Lexical Cumulativity hypothesis. Assuming Lexical Cumulativity, iterativity is possible from the very start, and iterativity without concurrent ‘object distributivity’ is the automatic result of introducing an ordinary singular indefinite in the early stages of a syntactic derivation. That the phenomenon illustrated in (72) and (73) does indeed affect ordinary, non-specific, indefinites in low positions can be shown by looking at comparable constructions in German:

- (77) a. **Ich hab’ 10 Minuten lang einen Hasen gestreichelt.**
 I have 10 minutes long a rabbit petted.
 I petted a rabbit for 10 minutes.
- b. **Ich hab’ einen Hasen 10 Minuten (lang) gestreichelt.**
 I have some / one rabbit 10 minutes (long) petted.
 I petted some / one rabbit for 10 minutes.
- (78) a. * **Ich hab’ 10 Minuten lang manche Hasen gestreichelt.**
 I have 10 minutes long SOME rabbits petted.
 I petted SOME rabbits for 10 minutes.
- b. **Ich hab’ manche Hasen 10 Minuten (lang) gestreichelt.**
 I have SOME rabbits 10 minutes (long) petted.
 I petted SOME rabbits for 10 minutes.

In 77(a) the indefinite DP is within the scope of the durational modifier, hence low. It is then typically pronounced with a pitch accent on **Hasen**, and there is no presupposition that rabbits have been talked about in the previous discourse. However, 77(a) still implies that there was a single rabbit that I petted for 10 minutes, just like its English translation and 77(b). In 77(b), the indefinite DP is outside the scope of the durational adverbial. It is typically pronounced with a pitch accent on the determiner, and then presupposes that rabbits have been under discussion. 78(a) and (b) have the specific indefinite determiner **manche**. DPs headed by **manche** are plain ungrammatical in the scope of a durational adverbial. The apparent ‘wide scope’ behavior in iterative constructions, then, affects non-specific indefinites sitting in low positions within the VP⁷¹. This gives further support to the hypothesis that the apparent ‘wide scope’ effects in (72) and (73) are an automatic consequence of a singular existential combining with a cumulative relation between individuals and events.

⁷¹ . All complex quantifier phrases are awkward in the scope of durational adverbials, e.g. **höchstens zwei Hasen** (‘at most two rabbits’), **weniger als 5 Hasen** (‘less than 5 rabbits’). This is expected, given the discussion in 4.5. Interestingly, the preferred reading of the sentence **The construction of some barns took longer than 5 months** has wide scope for **some barns**, in particular when **some** is emphasized, suggesting that **some barns** prefers not to be interpreted within a predicate denoting properties of events. To see the difference between **a barn** and **some barns**, compare **we vetoed the construction of a barn** and **we vetoed the construction of some barns**. Given those facts, the apparent wide scope effects with singular indefinites in (72) and (73) should not be attributed to the same source as the true wide scope effects with DPs headed by **some**. See Zucchi and White 2001 for extensive discussion of those issues, but a different conclusion.

Starting out with cumulative verb denotations also opens up exciting possibilities for the interpretation of adverbs of duration. We can now posit the following denotation of **for 20 minutes**, for example⁷²:

$$(79) \quad T(\text{for twenty minutes}) = \\ \lambda P \lambda e [P(e) \ \& \ f_{\text{minute}}(e) = 20 \ \& \ e = \sqcup e' [e' < e \ \& \ P(e')]]$$

To see what this meaning assignment does, take the VP **bounce a ball for 20 minutes**. Using (79), that VP is true of e just in case e is an event of bouncing a ball, lasts 20 minutes, and is made up of proper subevents that are themselves events of bouncing a ball. Since e has just one ball, all its subevents must have the same ball. Ergo, one and the same ball is being bounced for twenty minutes. Compare this to the result for the VP **bounce balls for 20 minutes**. We now have events of bouncing balls that last 20 minutes and are made up of proper subevents of bouncing balls. Different balls can be bounced in such events. Finally, look at **eat a bagel for 20 minutes**. This VP is true of an event e just in case e is a completed event of eating a bagel, lasts 20 minutes, and is made up of proper subevents that are themselves completed events of eating a bagel. Since e has just one bagel, it's a completed event of eating that bagel, and our definition now requires that it be made up of proper subevents that are themselves completed events of eating that very same bagel. Looks pretty impossible to me. Similar

⁷². The definition uses Link's $\sqcup$ -operator. In our case, the operator maps the events in the set $\{e': e' < e \ \& \ P(e')\}$ to their supremum. We are talking about the sum of all events e' that are proper parts of e and have the property P . The requirement is that that sum be identical to e .

impossibilities are derived for **eat 2, 3, 4,bagels for 20 minutes, eat this bagel for 20 minutes, and eat the bagel for 20 minutes.**

I have used a measure function in (79), as does Manfred Krifka in his work. What exactly does this measure function measure? What should it measure? Here are some examples:

- (80) a. **I worked in your garden for 7 hours.**
 b. **We climbed Mount Monadnock for 10 years.**
 c. **I have been sick for the last two days.**

Suppose you pay me by the hour for working in your garden. I could then utter 80(a) when I demand my pay. In this case, the measure function measures the times of my working in your garden in hours, and adds up the numbers. If I worked in your garden for one hour for seven days, the result is 7. That's not the kind of measuring that should be done in 80(b). Here we are looking at the times of our climbs of Mount Monadnock, and we measure the smallest interval that contains all of them. 80(b) would be true, for example, if we started a habit of climbing Mount Monadnock in 1990 and continued it until the year 2000. Both 80(a) and 80(b) have both of those readings, of course, it's just that the reading I illustrated for 80(a) is rather implausible for 80(b). (80) can be used to generate both readings depending on what measure function you fill in. In 80(c), the temporal expression **the last two days** denotes a particular interval of time, and in this case, the requirement is that the time of my sickness include that interval. A slight variation of (80) will accommodate that case, too.

Assuming Lexical Cumulativity for verbs, then, we gain an insightful account of durative adverbials, without giving any thought to principles of aspectual composition or the algebraic properties of VPs. The facts fall out on their own. The well-known constraints on combining durative adverbials with different kinds of VPs follow compositionally from the meanings of the participating parties. In that sense, (80) preserves the spirit of the pioneering analysis of Dowty 1979 and the related account of Moltmann 1991, but without the scope problems brought in by letting durational adverbials introduce universal quantification over events or times. If durational adverbials introduced a universal quantifier over events or times, the indefinites in (73) should be able to take narrow scope with respect to that quantifier. But we have seen that that narrow scope reading is absent. If we have cumulativity from the start, we can have iterated events from the start, and we do not need any operator to give us that iteration.

We can now be fairly confident that there is Lexical Cumulativity in the verbal domain. Basic verbs and thematic role predicates are obligatorily cumulative from the very start. We were able to detect that cumulativity in the absence of any plural subjects or objects. Our next task is to identify a plausible source for the many cases of non-lexical cumulativity. What we are looking for, then, is a ‘Lexical Cumulativity Plus’ theory. The overall plausibility of our approach to cumulativity will depend on how well the ‘Plus’ part can be motivated.

4.8 In search of a Lexical Cumulativity Plus theory

After having presented direct evidence for lexical cumulativity of basic verbs, we have to identify possible sources for phrasal cumulativity. We need a Lexical Cumulativity Plus theory, then, where lexical and phrasal cumulativity have different sources. In this section,

an event-based modification of Winter's highly constrained Lexical Cumulativity Plus theory (Winter 2000) is defended against apparent counterexamples.

A promising Lexical Cumulativity Plus proposal is defended in Winter 1998, 2000. Winter argues for two sources of real or apparent non-lexical cumulativity. The first one is linked to the phenomenon of dependent definites. Winter asks us to consider sentence (82) in the context of (81)⁷³:

(81) At a shooting range, each soldier was assigned a different target and had to shoot at it. At the end of the shooting we discovered that

(82) **Every soldier hit the target.**

In the context of (81), (82) is most likely to be understood as reporting that every soldier hit the target assigned to him. To assure this interpretation, we might posit a contextually salient function f that maps each soldier to the target he was assigned. (82) can then be given the interpretation in (83):

(83) $\forall x[\text{soldier}(x) \supset \text{hit}(f(x))(x)]$.

Within an event semantics, **every soldier** could be combined with a verbal projection denoting a relation between individuals and events, and would then most naturally be understood as saying that for every soldier x , there was an event e and x hit the unique target in e . This would allow the targets to vary for the different soldiers in cases like (82), but would all by itself not necessarily deliver the right result. Suppose Winter's soldiers were a rather

⁷³ . Winter 2000, p. 36.

unskilled bunch. They each hit a target, but it wasn't the one assigned to them. In this situation, (82) is intuitively false. Yet it's true that for every soldier x there was an event e and x hit the unique target in e . I conclude that there are such things as dependent definites, and that we need some device that takes care of them.

The second component of Winter's Lexical Cumulativity Plus account is a unary, atomic, D(istributivity)- operator of the kind found in the work of Link, Roberts, and Dowty⁷⁴. This operator has the following denotation:

$$(84) \quad \llbracket P_{\langle et \rangle} \rrbracket x \llbracket y \rrbracket [[\text{atom}(y) \ \& \ y \leq x] \rightarrow P(y)].$$

Within our event semantics, we have to think about where D-operators could be introduced in the course of a syntactic derivation. If they are optional and there are no particular constraints, we have to be prepared for the possibility that they might appear at a point where the event argument is not yet quantified or saturated. As a consequence, there should be a binary D-operator that operates over relations between individuals and events. A first approximation is (85):

$$(85) \quad \llbracket R_{\langle e \langle et \rangle \rangle} \rrbracket x \llbracket e \rrbracket y [[\text{atom}(y) \ \& \ y \leq x] \rightarrow \llbracket e' \rrbracket [e' \leq e \ \& \ R(y)(e')]].$$

The operator in (85) is still faithful to the spirit of Winter, since it 'affects' a single non-event argument at a time. It is furthermore an atomic D-operator, that is, it quantifies over the atomic parts of pluralities, hence contrasts with

⁷⁴ . Link 1991. Roberts 1987, 1990. Dowty 1987.

Schwarzschild's generalized distributivity operator, which can quantify over subpluralities if they are members of a salient cover.

Together, the D-operator and whatever device is needed for dependent definites account for 70(a) to (d), repeated from above:

- (70) a. **The women from Boxborough brought a salad.**
 (Roberts, 1990, p. 102, 146)
- b. **The boys gave the girls a flower.**
 (Winter 2000, p. 39)
- c. **John and Mary made less than \$10,000 last year.**
 (Lasersohn 1990, p. 32)
- d. **John and Mary made more than \$10,000 last year.**
 (Lasersohn 1990, p. 32)

On their distributive readings, 70(a), (c) and (d) have silent D-operators. The fact that 70(c) and (d) can both be true on Lasersohn's scenario is explained by assuming that D-operators are optional, hence create ambiguities. If John and Mary each made \$6,000 last year, then 70(c) comes out true if there is a D-operator, and 70(d) comes out false if there isn't. 70(b) illustrates the need to pay attention to dependent definites. With Lexical Cumulativity alone, we only get a single reading for 70(b), the one where there is a single flower that the sum of the boys gave to the sum of the girls. Adding the possibility of silent D-operators produces three more readings, assuming the possibility of

movement of DPs of all kinds in the Logical Form branch of the derivation⁷⁵: Each of the boys might have given the sum of the girls a flower, each of the girls might have received a flower from the sum of the boys, and each of the boys might have given each of the girls a flower. Winter's scenario is not yet covered by any of those readings. Treating **the girls** as a dependent definite gets that case, too. 70(b) could then be read as saying that each of the boys gave the sum of the girls he met a flower, or that each of the boys gave each of the girls he met a flower.

A theory like Winter's is an attractive, interestingly constrained, version of a Lexical Cumulativity Plus theory. Unfortunately, it has not remained unchallenged. While there seems to be a consensus that there is a phenomenon of 'dependent definites', it is far less clear how much that phenomenon buys us in the way of explaining away apparent cumulative readings. Beck and Sauerland 2000 mention the following example⁷⁶:

(86) **These five teachers gave a bad mark to those 20 protesting students.**

(86) has a cumulative reading. It can be true in a situation where each of the students got a bad mark from only one of the teachers. The cumulative reading of (86) cannot be reduced to a 'dependent definite' effect. Both DPs in (86) are indexicals with numerals, hence tolerate no further restrictions.

⁷⁵ . Heim and Kratzer 1997. Winter 2000 considers the possibility of a D-operator that can directly operate over plural individuals: $\exists x_e \exists P_{\langle et \rangle} \forall y_e [[\text{atom}(y) \ \& \ y \leq x] \supset P(y)]$. See Schwarzschild 1993-94 for discussion of this option.

⁷⁶ . Beck and Sauerland 2000, p. 356.

Beck and Sauerland's point is that if a dependent definite analysis is not available for (86), its intended reading can only be derived by using a polyadic distributivity operator. In this case, we would need an operator that takes the relation $\Box x \Box y \Box z [\text{bad-mark}(z) \ \& \ \text{gave-to}(y)(z)(x)]$ as its argument and simultaneously 'distributes over' both the subject and the indirect object argument as in (87), for example:

$$(87) \quad D_2 = \Box R_{\langle e \langle et \rangle \rangle} \Box x \Box y [\Box x' [[\text{atom}(x') \ \& \ x' \leq x] \Box \Box y' [y' \leq y \ \& \ R(x')(y')]]] \ \& \\ \Box y' [[\text{atom}(y') \ \& \ y' \leq y] \Box \Box x' [x' \leq x \ \& \ R(x')(y')]]] .$$

In our event-based semantics, the intended reading can still be derived within the limits of Winter's theory. (87) can be analyzed as a Schein sentence with neo-Davidsonian association of the agent argument. The indirect object would be moved out of its VP, and the resulting predicate could be pluralized with just the simple D-operator. We have:

1. $T(\text{gave a bad mark to}) = \Box y \Box e \Box z [\text{bad-mark}(z) \ \& \ \Box \text{gave-to}(y)(z)(e)]$
2. $T(D(\text{gave a bad mark to})) =$
 $\Box x \Box e \Box y [[\text{atom}(y) \ \& \ y \leq x] \Box \Box e' [e' \leq e \ \& \ \Box z [\text{bad-mark}(z) \ \& \ \Box \text{gave-to}(y)(z)(e')]]]]$
3. $T((D(\text{gave a bad mark to})) \text{ those 20 students }) =$
 $\Box e \Box y [[\text{atom}(y) \ \& \ y \leq \text{those 20 students}] \Box \Box e' [e' \leq e \ \& \ \Box z [\text{bad-mark}(z) \ \& \ \Box \text{gave-to}(y)(z)(e')]]]]$

The property of events we just computed is not quite the one we want, but all that is needed is to 'size it down' to one that can't be true of events that do not contain parts that have nothing to do with giving a bad mark to those 20

students. We can do so by amending the definition for the D-operator as follows, using Link's $\sqcup$ -operator⁷⁷:

$$(88) \quad \begin{aligned} & \sqcup R \sqcup x \sqcup e \sqcup \sqcup y \sqcup [\sqcup \text{atom}(y) \ \& \ y \leq x] \sqcup \sqcup e' \sqcup [e' \leq e \ \& \ R(y)(e')] \sqcup \sqcup \& \\ & e = \sqcup e' \sqcup \sqcup y \sqcup [\text{atom}(y) \ \& \ y \leq x \ \& \ e' \leq e \ \& \ R(y)(e')] \sqcup \sqcup . \end{aligned}$$

If used in the derivation of the meaning of (86), (88) requires that the event under discussion be made up of subevents of giving a bad mark to one of those students. After making the necessary adjustments in step 3 of the computation above, we add the agent argument, as usual. The predicted interpretation says that these five teachers are the agents of a minimal event in which those 20 students were each given a bad mark. This allows for the possibility that each of the students received a bad mark from only one of the teachers.

As another challenge for Winter, Beck and Sauerland 2000 present (82), which also doesn't submit to a dependent definite analysis:

(89) **The two women wanted to marry the two men.**

The reading of (89) that we are interested in is true in a situation where each of the two women wants to marry one of the two men. According to Beck and

⁷⁷ . In our case, the $\sqcup$ -operator maps the events in the set $\{e': \sqcup y \sqcup [\text{atom}(y) \ \& \ y \leq x \ \& \ e' \leq e \ \& \ R(y)(e')] \}$ to their supremum. We are talking about the sum of all subevents of e that are events of giving a bad mark to one of those students. This sum is required to be identical to e itself. But then e is made up of subevents that are all events of giving a bad mark to one of those students.

Sauerland, the intended reading of (89), too, has to be derived by applying a polyadic distributivity operator to a non-lexical relation, in this case the relation $\Box x \Box y [y \text{ wants to marry } x]$. However, in an event semantics, the cumulative reading of (89) can again be accounted for within Winter's constraints, making certain natural assumptions.

Suppose the infinitival complement in (89) expresses a property of individuals, as many have argued⁷⁸. Within the current framework, **want** should then denote a relation between properties of individuals and states. The relation holds between a property *P* and a state *s* just in case *s* is a wish with content *P*. That a property *P* is the content of the wish *s* means that *s* is only fulfilled in worlds in which the possessor of *s* has *P*.

Assuming lexical cumulativity, the 'want'-relation has to be cumulative:

$$(90) \quad \Box P_{\langle et \rangle} \Box Q_{\langle et \rangle} \Box s \Box s' [[\Box \text{want}(P)(s) \ \& \ \Box \text{want}(Q)(s)] \Box [\Box \text{want}(P+Q)(s+s')]]$$

We now have to think about the sum operation for properties. Most plausibly, it should amount to non-Boolean predicate conjunction, an operation argued for in Link 1984 and Lasnik 1992⁷⁹.

⁷⁸. This doesn't necessarily mean that there is no **PRO** subject for **marry**. Heim & Kratzer 1998 have proposed that **PRO** is invisible to the semantic interpretation component, but visible to the syntax, which means that it can be displaced, leaving a trace. The result could be LFs like $[\text{PRO}_1 [t_1 \text{ love himself}_i]]$, for example, which are interpreted as $\Box x [x \text{ love } x]$, treating the index on **PRO** as a binder (neglecting events).

⁷⁹. See also Lasnik 1995.

$$(91) \quad \Box P_{\langle et \rangle} \Box Q_{\langle et \rangle} [P+Q = \Box x \Box y \Box z [x = y+z \ \& \ P(y) \ \& \ Q(z)]]$$

(91) is motivated by the conjunctions in (92):

- (92) a. **Yoyo and Beverly are a cat and a rabbit**⁸⁰.
 b. **Der Schreibtisch war aus Holz und Metall.**
 The desk was of wood and metal
 The desk was made of wood and metal.

92(a) is true just in case one of Yoyo and Beverly is a cat, and the other one a rabbit. 92(b) is true just in case some of the desk is made of wood, and the rest is plastic.

Returning to (89), (89) is a statement about the marriage wishes of the two women. Suppose the wish of the first woman is a wish to marry Willie Brigham. This is a wish whose content is the property *P* of marrying Willie Brigham. Now take the wish of the second woman and assume that it is a wish to marry Spencer Hubbard. That wish is a wish whose content is the property *Q* of marrying Spencer Hubbard. The content of a wish states a necessary condition for the possible worlds in which it is fulfilled. The wish *s*₁, for example, is only fulfilled in worlds in which the possessor of the wish marries Willie Brigham. And the wish *s*₂ is only fulfilled in worlds in which the possessor' of *s*₂ marries Spencer Hubbard. The state *s*₁+*s*₂, then, is a plural wish whose content is *P*+*Q*. In our case, *P*+*Q* is computed as follows, switching to an intensional framework that has quantification over possible worlds, but still neglecting time for convenience:

⁸⁰ . I am assuming that in their predicative uses, **a cat** and **a rabbit** denote properties of individuals.

- (93) a. $P = \Box y \Box w \Box e [\Box \text{marry}_w(\text{Willie Brigham})(e) \ \& \ \Box \text{agent}_w(y)(e)]$
 b. $Q = \Box z \Box w \Box e [\Box \text{marry}_w(\text{Spencer Hubbard})(e) \ \& \ \Box \text{agent}_w(z)(e)]$
 c. $P + Q = \Box x \Box w \Box y \Box z [x = y+z \ \& \ \Box e [\Box \text{marry}_w(\text{Willie Brigham})(e) \ \& \ \Box \text{agent}_w(y)(e)] \ \& \ \Box e [\Box \text{marry}_w(\text{Spencer Hubbard})(e) \ \& \ \Box \text{agent}_w(z)(e)]]$

If the two women's plural marriage wish has the content $P+Q$, it is only fulfilled in worlds in which they have the property $P+Q$. Those worlds correspond to the proposition p in (94):

- (94) $p = \Box w \Box y \Box z [\text{the 2 women} = y+z \ \& \ \Box e [\Box \text{marry}_w(\text{Willie Brigham})(e) \ \& \ \Box \text{agent}_w(y)(e)] \ \& \ \Box e [\Box \text{marry}_w(\text{Spencer Hubbard})(e) \ \& \ \Box \text{agent}_w(z)(e)]]$

Given that the 2 men are Willie Brigham + Spencer Hubbard, and that the relations ' $\Box \text{agent}_w$ ' and ' $\Box \text{marry}_w$ ' are cumulative for any given value for ' w ', p logically implies q :

- (95) $q = \Box w \Box e [\Box \text{marry}_w(\text{the 2 men})(e) \ \& \ \Box \text{agent}_w(\text{the 2 women})(e)]$

If the 2 women's plural wish s_1+s_2 is only fulfilled in worlds that have the property p , it is only fulfilled in worlds that have the property q . But then sentence (89) is true, given our scenario. In this case, lexical Cumulativity alone derived the correct interpretation. No movement or even a D-operator was needed at all.

The analysis of (89) I just went through has some interesting consequences. Look at (96):

- (96) a. Scenario: My parents are having a disagreement. My mother wants me to marry Dr. Heintz. My Father wants me to marry Dr. Dietz.
- b. **My parents want me to marry those two doctors.**

There is something very odd about 96(b). We seem to be talking polygamy. For some reason, then, 96(b) just can't seem to get across what (97) can:

- (97) **Each of my parents wants me to marry a different one of those 2 doctors.**

Our analysis of **want** explains what's wrong with 96(b), once it is adapted to the case where **want** embeds a proposition. In that analysis, 96(b) is about a plural wish of my parents and characterizes its joint content. If p is the proposition that I marry Dr. Heintz, and q is the proposition that I marry Dr. Dietz, my mother's wish is only fulfilled in p-worlds, my father's wish is only fulfilled in q-worlds, and the sum of their wishes is jointly fulfilled only in worlds in which both p and q are true, - interpreting the sum operation for propositions as conjunction. But then, according to 96(b), I should marry both of those men, as far as my parents' joint wishes are concerned. This conclusion does indeed follow from the facts of the case, but why on earth would we ever want to give a joint characterization of my parents' marriage wishes for me? Weren't they having a disagreement? Our analysis says that 96(b) misfires because it is odd to lump together two fairly incompatible wishes and then talk about their joint content. See what you get! It is like

taking an article from Scientific American and another one from Martha Stuart Living and then go ahead and give a concise joint characterization of what they are about.

The proposed analysis of (89) establishes that no polyadic distributivity operator is needed for sentences of this kind. Within an event semantics, lexical cumulativity alone can account for the observed cumulative reading. An event semantics is crucial for other cases as well. A central theme in Beck's and Sauerland's work is to show that cumulation can affect predicates derived by movement, and that the limits for the phrasal predicates that can be cumulated are set by the usual constraints on movement. However, without the resources of an event semantics, Beck and Sauerland's generalization cannot be maintained. Take (98) below:

(98) **He broke those 2 toys to upset those 2 children.**

Without events, the cumulative reading of (98) would have to be derived by cumulating the relation ' $\lambda x \lambda y$ [he broke x to upset y]'. Movement cannot plausibly derive this predicate, since **those two children** would have to be extracted from an adjunct:

(99) * **Who₁ did he break those 2 toys to upset t_1 ?**

In an event semantics, we can piece together the cumulative reading of (98) as follows, assuming cumulativity for the basic predicates ' λ break', ' λ mean to', and ' λ upset': If action a meant to P and action b meant to Q , then action $a+b$ meant to $P+Q$. Action a was breaking toy 1 and action 2 was breaking toy 2, hence action $a+b$ was breaking toy1+toy2. P is upsetting child 1, and Q is

upsetting child 2, hence $P+Q$ is upsetting child1+child2. Ergo: Breaking toy1+toy2 meant to upset child1+child2, and hence breaking those 2 toys meant to upset those 2 children.

Let me summarize where we stand. I looked at two powerful and intriguing examples that Beck and Sauerland offered as problems for Winter's account of phrasal cumulativity. I have shown that within an event semantics, those examples do not require an analysis that forces us to go beyond the boundaries set by Winter's Lexical Cumulativity Plus theory. More specifically, we haven't yet seen any evidence that polyadic cumulation operators that affect more than one non-event at a time are ever truly needed. Moreover, we found that Beck and Sauerland's important generalization about the kind of predicates that can be cumulated is only tenable if we assume an event-based theory. Adapted to an event semantics, then, a Lexical Cumulativity Plus theory along the lines of Winter looks like a very promising starting point.

There are two features of Winter's theory that need further scrutiny, however. First: Do atomic D-operators yield the correct account of phrasal cumulativity? If not, what kind of operators do? And second: What is it in the syntax that is responsible for pluralization of phrasal verbal projections? What are the carriers of the operators that bring about phrasal cumulativity? I will address those important issues in the next and final section of this chapter.

4.9 Agreement morphology as the source of phrasal cumulativity

In this section, it is argued that [plural] agreement features related to functional projections that are the landing sites for DP movement are the source for phrasal cumulativity. We thus

have good support for a particular Lexical Cumulativity Plus theory, hence for the Cumulativity Universal that inspired this chapter.

Sentences (72) and (73) above all had singular subjects since we wanted to observe the effects of initial cumulativity for verbs:

(72) What does this intern do?

- a. **She guards a parking lot.**
- b. **He cooks for an elderly lady.**
- c. **She waters a garden.**
- d. **He watches a baby.**
- e. **She cleans an office building.**

(73) a. **I dialed a wrong phone number for 5 minutes.**

- b. **She bounced a ball for 20 minutes.**
- c. **He kicked a wall for a couple of hours.**
- d. **She opened and closed a drawer for half an hour.**
- e. **I ran up and down a hill for half a day.**

In this section we want to observe the effects of verbal plural number agreement. The next step in our experiment, then, is to replace the singular subjects with plural ones.

(100) What do your interns do?

- a. **They guard a parking lot.**
- b. **They cook for an elderly lady.**
- c. **They water a garden.**
- d. **They watch a baby.**
- e. **They clean an office building.**

- (101) a. **They dialed a wrong phone number for 5 minutes.**
 b. **They bounced a ball for 20 minutes.**
 c. **They kicked a wall for a couple of hours.**
 d. **They opened and closed a drawer for half an hour.**
 e. **They ran up and down a hill for half a day.**

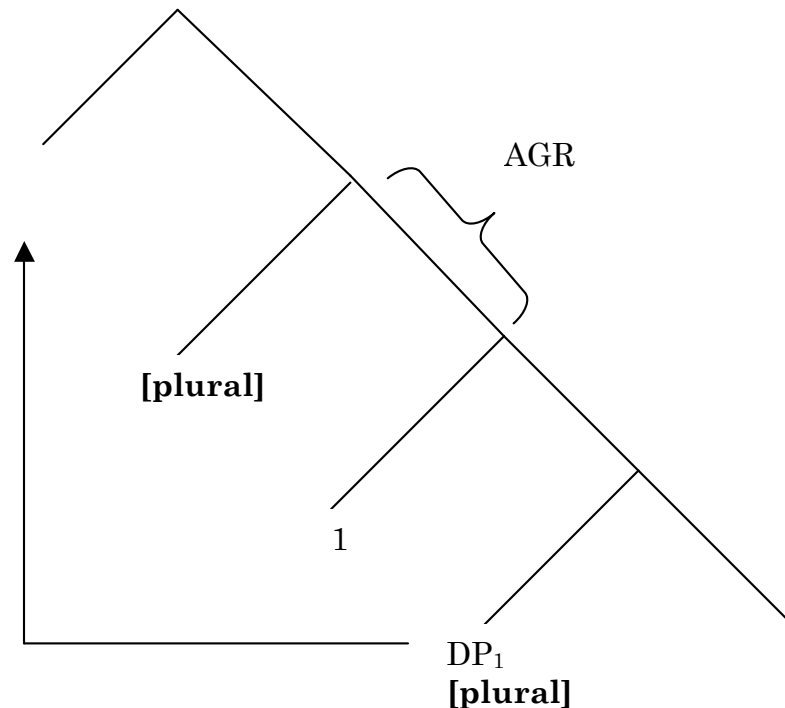
What happened? Take 100(e). 100(e) has an interpretation where each of the interns cleans a possibly different office building. But even on that interpretation, the sentence still requires that for each of them, there be an office building that she is in the habit of cleaning. Using scope talk for convenience, we have added an apparent intermediate reading for **an office building**, but we still do not seem to get anything amounting to a narrow scope interpretation. The effect is stronger in the examples of (101). 101(a) allows for different people dialing different wrong phone numbers, for example, but requires that each person dial the same wrong phone number for 5 minutes. Again, we have an apparent ‘intermediate’ scope effect, and a ‘narrow scope’ interpretation for the direct object is missing.

The main result of the test we just ran is that plural DPs bring in a distributivity operator. This is why we got an additional reading. Phrasal distributivity, then, is tightly linked to plural DPs, hence possibly to verbal plural agreement morphology, as we suspected earlier. Let’s be concrete about what that would mean in syntactic terms. I already spelled out what I take to be the available options in a passage at the beginning of this chapter:

When we ask about the meanings of verbs and VPs within the current framework of assumptions, we are talking about the meanings of ‘bare’ verbs and VPs, which are verbal projections that do not yet include any functional structure. I assume that the functional

projections of verbs are built step by step in the course of a syntactic derivation by introducing ('merging') functional heads with possibly meaningful features. This is compatible with the view that verbs enter a syntactic derivation fully inflected, as long as the features of those initial pieces of inflection are not meaningful themselves. Possible carriers of inflectional meaning would be matching features carried by functional heads. The question is now whether the number features of verbal functional heads are ever meaningful. Suppose they are. Bare verbs and VPs, - and in fact all verbal projections below the point where functional heads with number features come in, - should then have denotations that allow us to construct singular and plural denotations with the help of number operators. Alternatively, suppose that verbal number features are not meaningful. In that case, verbs and verbal projections should have denotations that, without any further modification by number operators, directly make the right contributions to the truth-conditions of the sentences they occur in. The important point is that in either case, number-neutral denotations are needed for bare verbs and VPs. It is those number neutral denotations that I will be concerned with. The chunks of a verb's extended projection that we will be examining are mostly located below the point where number features might leave their mark. We will mostly have to consider number-neutral denotations, then, that is, denotations that have not yet been affected by number operators, - if indeed they ever will be.

Suppose that there are meaningful number features within a verb's extended projection, and that it is those number features that introduce the operators that account for all irreducibly phrasal cases of cumulativity. Here is a way of fleshing out this proposal. Within Chomsky's Minimalist Program, Agreement projections are the landing sites for certain types of movement. Agreement projections are headed by features, and features can 'attract' DPs with matching features. What does it take for a DP to move? Minimally, the DP needs an index, and a matching binder index has to be present in a higher position. We can think of indices as privative features. Indices are



83. “Feature Scattering Principle: Each feature can head a projection.” Giorgi and Pianesi 1997, p. 15. Giorgi and Pianesi also argue for a fixed hierarchical order among the features. For our purposes, it is crucial that the number feature be above the index feature.

Here is what happens next. The DP moves into the specifier position of the ‘multi-headed’ AGR projection, leaving a co-indexed trace. The trace is bound by the binder index in AGR⁸⁴. The interpretable plural features are interpreted as pluralization operators, and the uninterpretable index of the DP is eliminated⁸⁵.

We have built a structure that is in principle interpretable. We also have an account of DP movement that can no longer build structures that would trigger the insertion of pluralization operators that can simultaneously affect more than one non-event argument. Recall that Beck and Sauerland 2000 argued for polyadic λ -operators to account for the cumulative readings of sentence (86) and (89):

(86) **These five teachers gave a bad mark to those 20 protesting students.**

(89) **The two women wanted to marry the two men.**

⁸⁴. Heim and Kratzer 1998. The present account differs slightly from the one in Heim and Kratzer, however. There, the index on the moved DP itself is ‘parsed’ as a λ -operator.

⁸⁵. I am not excluding the possibility that nominal number features might sometimes be uninterpretable, in dependent plurals, for example. They can then be eliminated via agreement.

We established that within an event semantics that has neo-Davidsonian association of the agent argument, the cumulative reading of (86) can be accounted for without positing a polyadic pluralization operator. As for (89), we convinced ourselves that lexical cumulativity alone could be held responsible for its cumulative reading. Our account of DP-movement, then, might constrain the configurations that can be created by movement in just the right way.

Given the proposed syntactic analysis, we also expect a tight link between the displacement of plural DPs and phrasal cumulativity. That pluralization operators can operate over predicates that are the result of non-overt movement is a recurrent claim in the semantic literature, starting at least with Roberts 1987, 1990. That phrasal cumulativity is constrained by the constraints on movement is documented in Sauerland 1998, Beck 2000, and Beck and Sauerland 2000. That a plural DP is necessary for phrasal cumulativity was shown by the difference between (72) and (73) on the one hand, and (100) and (101) on the other. We saw that distributive readings are absent in (72) and (73). In (100) and (101), distributive readings come into existence as soon as plural subjects are introduced.

If movement is needed for phrasal cumulativity, we should find inertia effects in out of the blue utterances. Suppose a distributive reading would require a DP to move beyond the point where it would have to move otherwise - for checking a case feature or a wh-feature, for example. In this case, we would expect a distributive reading to be dispreferred, unless there are contextual pressures for a particular interpretation. On the other hand, if a distributive reading can be picked up on the way to a landing site that a DP is headed to anyway, a distributive reading should be much easier to get. Examples (102)

to (104) show such inertia effects. For the (a)-examples, a distributive reading is much harder to perceive than for the (b)-examples. In the (a) cases, the plural DPs have no other reason to move over the subject apart from producing a distributive interpretation. They can check or receive their case in lower positions. In the absence of any contextual forces, then, there is no motivation for those DPs to move. They should prefer to stay put in lower positions. In the (b) example, the plural DP is relativized, hence has moved overtly over the subject. As expected, distributive readings are much easier to perceive:

- (102) a. **I want to climb a mountain in New Hampshire and Vermont.**
 b. **The states I want to climb a mountain in are New Hampshire and Vermont⁸⁶.**
- (103) a. **A student saw me do all the experiments.**
 b. **All the experiments that a student saw me do didn't work out.**
- (104) a. **A student wants me to read the papers in this pile.**
 b. **The papers that a student wants me to read are in this pile.**

The connection between phrasal cumulativity and movement of plural DPs is also shown by the following examples from German, where object shift creates distributive readings that are not available if the object is left *in situ*:

⁸⁶ . 102(b) is a variation of an example in Beck 2000.

- (105) a. **Ich hab' 10 Minuten lang zwei Hasen gestreichelt.**
 I have 10 minutes long two rabbit petted.
 'I petted a group of 2 rabbits for 10 minutes'.
- b. **Ich hab' zwei Hasen 10 Minuten (lang) gestreichelt.**
 I have two rabbits 10 minutes (long) petted.
 'I petted a group of 2 rabbits for 10 minutes'.
 'I petted 2 rabbits for 10 minutes each'.
- (106) a. **Ich hab' nach und nach 10 Bücher gelesen.**
 I have little by little 10 books read.
 'Little by little, I read a total of 10 books'.
- b. **Ich hab' 10 Bücher nach und nach gelesen.**
 I have 10 books little by little read
 'Little by little, I read a total of 10 books'.
 'I read each of 10 books little by little'.

There is considerable support for the hypothesis, then, that phrasal cumulativity involves movement of plural DPs.

Our final task is to find out what kind of pluralization operator is introduced by plural DPs. The main options to decide between are the atomic D-operator (adapted to our event semantics), or a cross-categorical $\square$ -operator.

Look at (107).

- (107) **They bounced a ball for 20 minutes.**

Using the D-operator, (107) winds up saying that for each of those (athletes?) there was a 20-minute interval during which one and the same ball kept being bounced. Using the *-operator instead, we would moreover cover cases where we have teams bouncing one and the same ball for 20-minutes, and we could also describe multiple events of bouncing a ball for 20 minutes. There might be regular 20-minute ball bouncing warm-up sessions at the beginning of basketball practice, for example. (107) can be understood to describe situations of this kind, and with the *-operator things come out right. Similar comments apply to habitual cases. Take (108):

(108) What do those interns do?

They watch a baby.

Minimally, habitual aspect contributes the information that we are dealing with a plurality of events of the kind described by the VP. Using the D-operator, (108) would convey that for each of the interns, there is a string of iterated actions of watching one and the same baby. The $\square$ -operator additionally allows for teams of interns, as well as for pluralities of strings of iterated actions of watching one and the same baby. It could be, for example, that during every 6 month internship, the babies to watch change monthly. (108) can describe such situations. Neglecting the habitual part, the pluralized predicate of (108) should have the denotation in (110), then, where the $\square$ -operator simultaneously affects the subject argument and the event argument⁸⁷. Literally, we would have a binary $\square$ -operator, then, but crucially, only one non-event argument is affected:

⁸⁷. See Beck 2002 for discussion of more cases of this kind in connection with pluractional markers.

(112) $\lambda x \lambda e \lambda y [\text{baby}(y) \ \& \ \text{watch}(y)(e) \ \& \ \text{agent}(x)(e)]$

Do the cases we have just looked at truly require the full resources of the λ -operator? The most tangible differences between the D-operator and the λ -operator have to do with intermediate grouping effects. If we use the D-operator to pluralize predicates, we predict that all non-trivial cover effects must be reducible to lexical cumulativity, or involve other mechanisms like dependent definites. When we looked at Schwarzschild's merchant, who wished the vegetables were light, we had an example of a non-trivial cover effect - distribution was to intermediate pluralities. But the predicate involved was just the adjective **light**, hence lexical cumulativity alone accounted for that case. We have to examine non-trivial cover effects, then, that come from predicates that are essentially phrasal. We saw such non-trivial cover effects with (107) and (108). What we still have to think about, however, is whether those apparent cover effects couldn't be produced by dependent definites.

Rather than pluralization of properties of events, there could be implicit universal quantification over events. Winter argues for an implicit universal quantifier over events or situations in the following case⁸⁸:

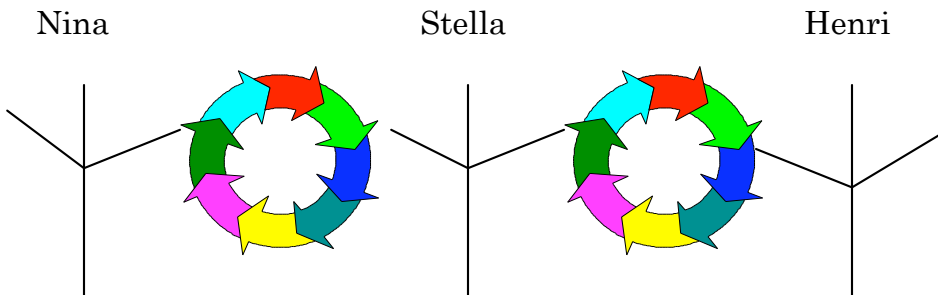
(114) "In each of the years 2000-2010, one grand opera will be commissioned by the municipal opera house. Each year, two composers chosen by a special committee will be asked to collaborate in writing a new opera."

The selected composers will earn \$ 5,000.

⁸⁸. Winter 2000, p. 64

In this context, there is implicit quantification over years between 2000 and 2010, and relying on a ‘dependent definite’ mechanism, we get an immediate cover reading: In each year between 2000 and 2010, the composers chosen in that year will receive \$5,000. A similar move is not possible for the intermediate cover readings of (107) and (108). In both cases, the subject is a referential pronoun, hence the dependent definite mechanism can’t be at work. We do seem to need the $\Box$ operator, then, as the source of phrasal cumulativity.

Are there cases where the $\Box$ operator would be too powerful? Winter presents potentially relevant cases, and here is one of them⁸⁹. Suppose three children, Stella, Nina, and Henri, are lined up in a row. Nina and Stella are holding one wheel, and Stella and Henri are holding another:



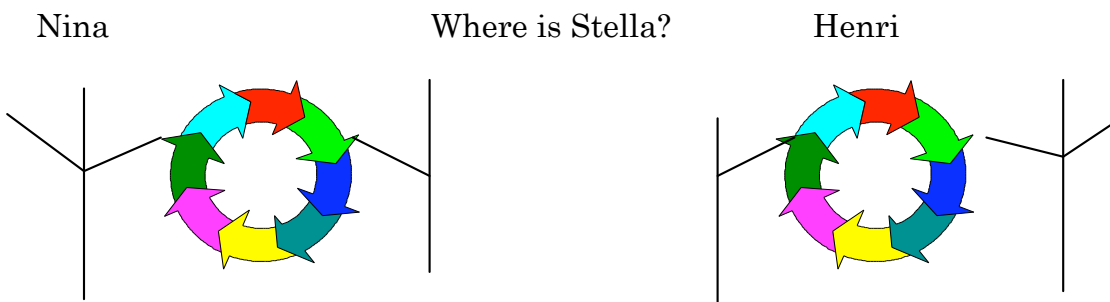
Winter observes that in such a situation, sentence (110) is “false or highly strange”:

⁸⁹. Winter 2000, p. 63. My apologies to Yoad Winter for having distorted his lovely picture. I don’t have a scanner and I am not an artist

(110) **The children are holding a wheel.**

(110) doesn't seem to be able to describe the situation depicted above, and we want to know why. Winter's answer is that non-lexical cases of distributivity are due to a D- operator that enforces atomic distribution. Having chosen the $\square$ -operator over its competitor, how could we explain why is (110) false or strange on Winter's scenario? Suppose the DP **the children** in (110) is interpreted *in situ*. In this case, Lexical Cumulativity is all we have, and that means that you can't get more than one wheel in all, as we have seen.

Suppose now that **the children** is interpreted in a raised position. This requires the presence of a $\square$ -operator pluralizing the sister constituent of the raised DP. If that sister constituent expresses a relation between individuals and events, it is that relation that is cumulated. (110) can now describe possibly multiple events of one or more children holding a single wheel. What it still can't do, however, is describe events of the kind depicted by Winter. There we have a single event where two wheels are being held. Winter's scene is not easily parsed as a plural event. We might force ourselves to partition Winter's scene into two subscenes, though, hence try to perceive it as a plural event. One with Nina and part of Stella holding a wheel, and one with Henri and the other part of Stella holding another wheel. We resist that way of dividing up the scene, but trying to be cooperative we might go as far as that:



(110) is true in this situation, but parsing the original scene that way is ‘strange’. Equally strange is allowing overlap by having all of Stella in both situations. Having opted for the $\sqcup$ -operator as the source of phrasal cumulativity, then, still allows us to explain Winter’s important observation about (110).

We have finally come to the end of what turned out to be a *tour de force* in verbal cumulativity. In this last section of chapter 4, I have defended a Lexical Cumulativity Plus theory that is a variation of Winter’s. The main variation concerns the nature of the operator that is responsible for phrasal cumulativity effects, differences coming from an event-based framework, and a commitment to a particular syntactic realization of pluralization operators. I have supported the view pioneered by Krifka and Landman that there is lexical cumulativity for verbs and thematic role predicates. I have argued furthermore that cumulation operators cannot be inserted freely, and that phrasal cumulativity comes from [plural] features that are needed to get plural DPs to move. As an extra benefit for the event based approach to the semantics of verbs, I have shown that lexical cumulativity implies no commitment to inflated NP meanings of the kind argued for in Landman’s work nor to quantification over covers, as proposed by Schwarzschild. The flexible part structure of events and states gives us all the necessary distinctions, a point made years ago by Barry Schein and exploited extensively in the work of Peter Lasnik.

There is very strong support for a Cumulativity Universal, then. With respect to the big plot of this book, this means that, most likely, there is no

such thing as a general thematic role ‘theme’ or ‘object’. It would violate a substantial universal for basic lexical meanings.

References: The Event Argument and the Semantics of Verbs

Note: The following references are all the references that have gone into this work so far, including those for earlier versions that have been superseded and chapters not yet posted.

- Abney, Steven Paul. 1987. The English Noun Phrase in its Sentential Aspects, MIT: Ph.D.
- Abusch, Dorit. 1993-1994. The Scope of Indefinites. *Natural Language Semantics* 2:83-135.
- Agricola, Erhard, Görner, Herbert, and Küfner, Ruth eds. 1979. *Wörter und Wendungen. Wörterbuch zum deutschen Sprachgebrauch*. VEB Bibliographisches Institut: Leipzig.
- Aissen, Judith L. 1987. *Tzotzil Clause Structure*: Studies in Natural Language and Linguistic Theory. Dordrecht: Reidel.
- Anderson, Stephen R. 1982. Where's Morphology. *Linguistic Inquiry* 13:571 - 612.
- Andrews, Avery D. 1971. Case Agreement of Predicate Modifiers in Ancient Greek. *Linguistic Inquiry* 11:127-151.
- Anscombe, G. E. M. 1957. *Intention*. Oxford: Basil Blackwell.
- Antinucci, Francesco, and Miller, Ruth. 1976. How Children Talk About What Happened. *Journal of Child Language* 3:167-189.
- Austin, J. L. 1979. *Philosophical Papers*. Oxford: Oxford University Press.
- Babby, L., and Brecht, R. 1975. The syntax of voice in Russian. *Language* 51:342-367.
- Babby, Leonard. 1980. *Existential Sentences and Negation in Russian*. Ann Arbor: Karoma.
- Bach, Emmon W. 1967. *Have* and *be* in English Syntax. *Language* 43:462-485.
- Bach, Emmon W. 1977. Five Chapters on Natural Language Metaphysics: University of Massachusetts.
- Bach, Emmon. 1979. Control in Montague Grammar. *Linguistic Inquiry* 10:515-531.
- Bach, Emmon. 1980. In Defense of Passive. *Linguistics and Philosophy* 3:297-342.
- Bach, Emmon. 1981. On Time, Tense, and Aspect: An Essay in English Metaphysics. In *Radical Pragmatics*, ed. Peter Cole, 63-81. New York: Academic Press.
- Bach, Emmon W. 1981. On Time, Tense, and Aspect. In *Radical Pragmatics*, ed. Peter Cole. New York: Academic Press.
- Bach, Emmon. 1986. The Algebra of Events. *Linguistics and Philosophy* 9:5-16.

- Baker, Carl L. 1979. Syntactic Theory and the Projection Problem. *Linguistic Inquiry* 10:533-581.
- Baker, Mark. 1985. Syntactic Affixation and English Gerunds. Paper presented at *West Coast Conference on Formal Linguistics*, UCLA.
- Baker, Mark. 1985. Incorporation: A Theory of Grammatical Function Changing, MIT: Ph.D.
- Baker, Mark. 1988. Theta Theory and the Syntax of Applicatives in Chichewa. *Natural Language & Linguistic Theory* 6:353-389.
- Baker, Mark C. 1988. *Incorporation. A Theory of Grammatical Function Changing*. Chicago: The University of Chicago Press.
- Baker, Mark, Johnson, Kyle, and Roberts, Ian. 1989. Passive Arguments Raised. *Linguistic Inquiry* 20:219-252.
- Baker, Mark. 1996. *The Polysynthesis Parameter*. Oxford: Oxford University Press.
- Baker, Mark, and Stewart, Osamuyimen T. 1999. On Double-Headedness and the Anatomy of the Clause. Ms. Rutgers University.
- Bamgbose, Ayo. 1986. Negation and Serial Verb Construction Types in Yoruba. In *Current Approaches to African Linguistics (vol. 3)*, ed. Gerrit J. Dimmendaal, 31-40. Dordrecht: Foris.
- Barker, Chris. 1991. Possessive Descriptions, Department of Linguistics, University of California at Santa Cruz: Ph.D.
- Barker, Chris. 1995. *Possessive Descriptions*. Stanford: CSLI Publications.
- Barss, Andrew. 1986. Chains and Anaphoric Dependence., MIT: Ph.D.
- Barss, Andrew, and Lasnik, Howard. 1986. A Note on Anaphora and Double Objects. *Linguistic Inquiry* 17:347-354.
- Barwise, Jon. 1981. Scenes and Other Situations. *The Journal of Philosophy* 78:369-397.
- Barwise, Jon, and Cooper, Robin. 1981. Generalized Quantifiers and Natural Language. *Linguistics and Philosophy* 4:159-219.
- Barwise, Jon, and Perry, John. 1983. *Situations and Attitudes*. Cambridge/Mass.: MIT Press.
- Bäuerle, Rainer. 1977. Tempus, Temporaladverb und die temporale Frage, Universität Konstanz (reproduced by SFB 99): Ph.D.
- Bäuerle, Rainer. 1978. Fugitive Propositions Again. *Analysis* 38:78 - 80.
- Bäuerle, Rainer. 1979. Tense Logics and Natural Language. *Synthese* 40:225-230.
- Bayer, Joseph. 1982. Zur Syntax und Semantik von wie-Sätzen und Dialogen mit wie-Fragen, Universität Konstanz: Ph. D.
- Bayer, Josef. 1984. COMP in Bavarian Syntax. *The Linguistic Review* 3:209-274.
- Beck, Sigrid. 1999. Plural Predication and Partitional Discourse. Paper presented at *12th Amsterdam Colloquium*, University of Amsterdam.

- Beck, Sigrid. 2000. Star Operators. Episode 1: Defense of the Double Star. In *UMOP 23: Issues in Semantics*, eds. Kiyomi Kusumoto and Elisabeth Villalta. Amherst/Mass.: GLSA, UMass Amherst.
- Beck, Sigrid, and Sauerland, Uli. 2000. Cumulation is needed: A Reply to Winter 2000. *Natural Language Semantics* 8(4):349-371.
- Beck, Sigrid. 2001. Reciprocals are Definites. *Natural Language Semantics* 9(1):1-69.
- Beck, Sigrid. 2002. Event-Based Cumulation. Ms. UConn.
- Beedham, Christopher. 1982. *The Passive Aspect in English, German and Russian*. Tübingen: Gunter Narr Verlag.
- Beghelli, Filippo. 1993. A Minimalist Approach to Quantifier Scope. Paper presented at *NELS 23*, University of Ottawa.
- Beghelli, Filippo, Ben-Shalom, Dorit, and Szabolcsi, Anna. 1993. When do Subjects and Objects Exhibit a Branching Reading. Paper presented at *WCCFL XII*.
- Beghelli, Filippo, Ben-Shalom, Dorit, and Szabolsci, Anna. (to appear). Variation, Distributivity, and the Illusion of Branching. In *Ways of Scope Taking*, ed. Anna Szabolsci.
- Belletti, Adriana. 1982. 'Morphological' Passive and Pro-Drop: The Impersonal Construction in Italian. *Journal of Linguistic Research* 2:1 - 34.
- Belletti, Adriana. 1988. The Case of Unaccusatives. *Linguistic Inquiry* 19:1-34.
- Belletti, Adriana, and Rizzi, Luigi. 1988. Psych-Verbs and q-Theory. *Natural Language & Linguistic Theory* 6:291-352.
- Bengl, Hans. 1958. *Griechische Wortkunde*. München: Bayerischer Schulbuchverlag.
- Benseler, Eduard. 1900. *Griechisch-Deutsches Schulwörterbuch*. Leipzig: B. G. Teubner.
- Benthem, Johan F.A.K. van. 1977. Tense Logic and Standard Logic. *Logique et Analyse, Nouvelle Série* 20e année.
- Benthem, Johan van. 1986. *Essays in Logical Semantics*. Dordrecht: D. Reidel Publishing Company.
- Benveniste, Emile. 1971. *Problems in General Linguistics*. Coral Gables, Florida: University of Miami Press.
- Berman, Stephen. 1987. Situation-Based Semantics for Adverbs of Quantification. In *UMOP 12*, eds. Jim Blevins and Anne Vainikka: GLSA, University of Massachusetts, Amherst.
- Besten, Hans den. 1985. The Ergative Hypothesis and Free Word Order in Dutch. In *Studies in German Grammar*, ed. J. Toman, 23-64. Dordrecht: Foris.
- Besten, Hans den, and Webelhuth, Gert. 1987. Remnant Topicalization and the Constituent Structure of VP in the Germanic SOV Languages. Paper presented at *GLOW Newsletter*, 18.

- Bierwisch, Manfred. 1983. Semantische und konzeptuelle Repräsentation lexikalischer Einheiten. In *Untersuchungen zur Semantik (= Studia Grammatika 22*, eds. R. Ruzicka and W. Motsch, 61-99. Berlin: Akademie Verlag.
- Bierwisch, Manfred. 1989. Event-Nominalizations. Proposals and Problems. In *Linguistische Studien, Reihe A 194*, 1-73. Berlin: Akademie Verlag.
- Binnick, Robert I. 1991. *Time and the Verb*. New York Oxford: Oxford University Press.
- Bittner, Maria. 1994. Crosslinguistic Semantics. *Linguistics and Philosophy* 17:53-108.
- Bittner, Maria. 1999. Concealed Causatives. *Natural Language Semantics* 7.
- Blatz, Friedrich. 1896. *Neuhochdeutsche Grammatik (mit Berücksichtigung der historischen Entwicklung der deutschen Sprache)*.vol. Zweiter Band. Satzlehre (Syntax). Karlsruhe: J. Lang's Verlagsbuchhandlung und Buchdruckerei.
- Blatz, Friedrich. 1900. *Neuhochdeutsche Grammatik (mit Berücksichtigung der historischen Entwicklung der Deutschen Sprache)*.vol. Erster Band. Einleitung. Lautlehre. Wortlehre. Karlsruhe: J. Lang's Verlagsbuchhandlung.
- Bobaljik, Jonathan David, and Jonas, Dianne. 1996. Subject Positions and the Role of TP. *Linguistic Inquiry* 27:195 - 236.
- Bolinger, Dwight. 1971. *The Phrasal Verb in English*. Cambridge/Mass.: Harvard University Press.
- Bonet, Eulàlia. 1990. Subjects in Catalan. In *Papers on Wh-Movement*, eds. Lisa L. S. Cheng and Hamida Demirdash. Cambridge/Mass.: MIT Working Papers in Linguistics.
- Bonomi, Andrea, and Casalegno, Paolo. 1993. Only: Association with Focus in Event Semantics. *Natural Language Semantics* 2(1):1-45.
- Borer, Hagit. 1984. The Projection Principle and Rules of Morphology. Paper presented at *NELS*, University of Massachusetts at Amherst.
- Borer, Hagit ed. 1986. *Syntax and Semantics 19: The Syntax of Pronominal Clitics*. Orlando: Academic Press.
- Borer, Hagit, and Grodzinsky, Yosef. 1986. Syntactic Cliticization and Lexical Cliticization: The Case of Hebrew Dative Clitics. In *Syntax and Semantics 19. The Syntax of Pronominal Clitics*, ed. Hagit Borer, 175 - 217. Orlando: Academic Press.
- Borer, Hagit. 1988. On the morphological parallelism between compounds and constructs. In *Morphology Yearbook 1*, ed. Geert Booij et al., 45 - 65. Dordrecht: Foris.
- Borer, Hagit. 1989. Anaphoric AGR. In *The Null Subject Parameter*, eds. O. Jaeggli and K. J. Safir, 69 - 109. Dordrecht: Kluwer Academic Publishers.
- Borer, Hagit. 1990. Derived Nominals and the Causative-Inchoative Alternation: Two Case Studies in Parallel Morphology: University of California at Irvine.

- Borer, Hagit. 1990. V+ing : It Walks like an Adjective, It Talks like an Adjective. *Linguistic Inquiry* 21:95-103.
- Borer, Hagit. 1991. The Causative-Inchoative Alternation: A Case Study in Parallel Morphology. *The Linguistic Review* 8:119-158.
- Borer, Hagit, and Wexler, Ken. 1992. Bi-Unique Relations and the Maturation of Grammatical Principles. *Natural Language and Linguistic Theory* 10:147-189.
- Borer, Hagit. forthcoming. *Determining Structure. An Exo-Skeletal Trilogy*. Oxford: Oxford University Press.
- Borkin, Ann. 1973. To Be and Not To Be. Paper presented at *Papers from the Ninth Regional Meeting of the Chicago Linguistic Society*, Chicago.
- Bornemann, Eduard, and Risch, Ernst. 1978 (2nd edition). *Griechische Grammatik*. Frankfurt / Main: Diesterweg.
- Bowers, John. 1990. A Structural Theory of Predication: Manuscript, Cornell University.
- Bowers, John. 1991. The Syntax and Semantics of Nominals. Paper presented at *SALT 1*, Cornell University.
- Bowers, John. 1993. The Syntax of Predication. *Linguistic Inquiry* 24:591-656.
- Bowers, John. 1997. A binary analysis of resultatives. In *Texas Linguistic Forum 38: The Syntax and Semantics of Predication*, eds. R. C. Blight and M. J. Moosally, 43-58. Austin/Texas: Department of Linguistics, University of Texas.
- Bowers, John. 2001. Predication. In *The Handbook of Contemporary Syntactic Theory*, eds. Mark Baltin and Chris Collins, 299 - 333. Oxford: Blackwell Publishers.
- Brekke, M. 1988. The Experiencer Constraint. *Linguistic Inquiry* 19:169-180.
- Bresnan, Joan. 1982. The Passive in Lexical Theory. In *The Mental Representation of Grammatical Relations*, ed. Joan Bresnan. Cambridge: MIT Press.
- Bresnan, Joan. 1982. Control and Complementation. *Linguistic Inquiry* 13:343-434.
- Bresnan, Joan, and Kanerva, Jonni M. 1989. Locative Inversion in Chichewa: A case Study of Factorization in Grammar. *Linguistic Inquiry* 20:1-50.
- Bresnan, Joan, and Moshi, Lioba. 1990. Object Asymmetries in Comparative Bantu Syntax. *Linguistic Inquiry* 21:147-185.
- Bresnan, Joan. 1994. Locative Inversion and Universal Grammar. *Language* 70:72-131.
- Brugmann, Karl. 1913 (4th edition). *Griechische Grammatik*. München: C. H. Beck'sche Verlagsbuchhandlung.
- Buchholz, Oda, and Fiedler, Wilfried. 1987. *Albanische Grammatik*. Leipzig: VEB Enzyklopädie.
- Burzio, Luigi. 1986. *Italian Syntax. A Government-Binding Approach*. Dordrecht: D. Reidel Publishing Company.

- Burzio, Luigi. 1991. The morphological basis of anaphora. *Journal of Linguistics* 27:81-105.
- Burzio, Luigi. 1998. Anaphora and Soft Constraints. In *Is the Best Good Enough? Optimality and Competition in Syntax*, eds. Pilar Barbosa, Danny Fox, Paul Hagstrom, Martha McGinnis and David Pesetsky, 93 - 113. Cambridge/Mass.: The MIT Press.
- Bybee, Joan L. 1985. *Morphology. A Study of the Relation Between Meaning and Form*. Amsterdam/Philadelphia: John Benjamins Publishing Company.
- Bybee, Joan, Perkins, Revere, and Pagliuca, William. 1994. *The Evolution of Grammar: Tense, Aspect and Modality in the Languages of the World*. Chicago: Chicago University Press.
- Camaj, Martin. 1969. *Lehrbuch der Albanischen Sprache*. Wiesbaden: Otto Harrassowitz.
- Camaj, Martin. 1984. *Albanian Grammar*. Wiesbaden: Otto Harrassowitz.
- Carey, Kathleen. 1996. From Resultativity to Current Relevance: Evidence from the History of English and Modern Castilian Spanish. In *Conceptual Structure, Discourse and Language*, ed. Adele E. Goldberg, 31-48. Stanford: CSLI Publications.
- Carlson, Gregory N. 1977. Reference to Kinds in English, University of Massachusetts at Amherst: Ph. D.
- Carlson, Gregory N. 1984. Thematic Roles and their role in semantic interpretation. *Linguistics* 22:259-279.
- Carrier, Jill, and Randall, Janet H. 1992. The Argument Structure and Syntactic Structure of Resultatives. *Linguistic Inquiry* 23:173-234.
- Carstairs-McCarthy, Andrew. 1998. Paradigmatic Structure: Inflectional Paradigms and Morphological Classes. In *The Handbook of Morphology*, eds. Andrew Spencer and Arnold M. Zwicky, 322 - 334. Oxford: Blackwell Publishers.
- Castañeda, Hector-Neri. 1967. Comments on Donald Davidson's 'The logical form of action sentences'. ed. Rescher, 104-112.
- Cattell, Ray. 1984. *Syntax and Semantics 17. Composite Predicates in English*. Sydney: Academic Press.
- Chierchia, Gennaro. 1985. Formal Semantics and the Grammar of Predication. *Linguistic Inquiry* 16:417-443.
- Chierchia, Gennaro, and Jacobson, Pauline. 1985. Local and Long Distance Control. Paper presented at *NELS 16*, McGill University.
- Chierchia, Gennaro, and Turner, Ray. 1988. Semantics and Property Theory. *Linguistics and Philosophy* 11:261-302.
- Chierchia, Gennaro. 1989. Structured Meanings, Thematic Roles and Control. In *Properties, Types and Meaning. Volume II: Semantic Issues*, eds. Gennaro Chierchia, Barbara H. Partee and Raymond Turner, 131-166. Dordrecht: Kluwer Academic Publishers.

- Chierchia, Gennaro. 1989. A Semantics for Unaccusatives and its syntactic consequences. Ithaca: Cornell University.
- Chierchia, Gennaro. 1991. Anaphors and Attitudes De Se. In *Language in Context*, eds. R. Bartsch, J. van Benthem and P. van Emde Boas. Dordrecht: Foris.
- Chierchia, Gennaro. 1992. Anaphora and Dynamic Binding. *Linguistics and Philosophy* 15:111-183.
- Chierchia, Gennaro. 1995. *Dynamics of Meaning*. Chicago: The University Press of Chicago.
- Chierchia, Gennaro. 1998. Reference to kinds across languages. *Natural Language Semantics* 6(4):339-405.
- Choe, Jae Woong. 1987. Anti-Quantifiers and a Theory of Distributivity, University of Massachusetts Ph.D. dissertation.
- Chomsky, Noam. 1981. *Lectures on Government and Binding*. Dordrecht: Foris.
- Chomsky, Noam. 1986. *Barriers*. Cambridge/Mass.: The MIT Press.
- Chomsky, Noam. 1991. Some Notes on Economy of Derivation and Representation. In *Principles and Parameters in Comparative Grammar*, ed. Robert Freidin, 417-454. Cambridge/Mass.: MIT Press.
- Chomsky, Noam. 1993. A Minimalist Program for Linguistic Theory. In *The View from Building 20. Essays in Linguistics in Honor of Sylvain Bromberger*, eds. Kenneth Hale and Samuel J. Keyser, 1-52. Cambridge/Mass.: The MIT Press.
- Chomsky, Noam. 1995. *The Minimalist Program*. Cambridge/Mass.: The MIT Press.
- Chomsky, Noam. 1995. Bare Phrase Structure. In *Government and Binding Theory and the Minimalist Program*, ed. Gert Webelhuth, 382-439. Oxford: Blackwell Publishers.
- Chomsky, Noam. 2000. Minimalist Inquiries: The Framework. In *Step by Step. Essays in Honor of Howard Lasnik*, eds. Roger Martin, David Michaels and Juan Uriagereka, 89-155. Cambridge/Mass.: The MIT Press.
- Chomsky, Noam. 2001. Derivation by Phase. In *Ken Hale. A Life in Language*, ed. Michael Kenstowicz, 1-52. Cambridge/Mass.: The MIT Press.
- Chung, Sandra. 1976. On the Subject of Two Passives in Indonesian. In *Subject and Topic*, ed. Charles N. Li, 57-98. New York San Francisco London: Academic Press.
- Chung, Sandra. 1978. *Case Marking & Grammatical Relations In Polynesian*. Austin & London: University of Texas Press.
- Chung, Sandra, and McCloskey, James. 1987. Government, Barriers, and Small Clauses in Modern Irish. *Linguistic Inquiry* 18:173 -237.
- Cinque, Guglielmo. 1988. On *Si* Constructions and the Theory of *Arb*. *Linguistic Inquiry* 19:521-581.

- Cinque, Guglielmo. 1989. Ergative Adjectives and the Lexicalist Hypothesis. *Natural Language and Linguistic Theory* 8:1-40.
- Clahsen, Harald, Rothweiler, Monika, Woest, Andreas, and Marcus, Gary. 1992. Regular and Irregular Inflection in German Child Language. *Cognition* 45:225-255.
- Clahsen, Harald, Marcus, Gary F., Bartke, Susanne, and Wiese, Richard. 1996. Compounding and Inflection in German Child Language. In *Yearbook of Morphology*, eds. G. Booij and J. van Marle. Dordrecht: Kluwer.
- Cole, P., Harbert, W., Hermon, G., and Sridhar, S. N. 1981. On the Acquisition of Subjecthood. *Language* 56:719-743.
- Collins, Christopher Thad. 1993. Topics in Ewe Syntax, MIT: PH.D.
- Collins, Chris, and Thráinsson, Höskuldur. 1996. VP-Internal Structure and Object Shift in Icelandic. *Linguistic Inquiry* 27:391-444.
- Collins, Chris. 1997. Argument Sharing in Serial Verb Constructions. *Linguistic Inquiry* 28:461-497.
- Collins, Chris. 1997. *Local Economy*. Cambridge/Mass.: The MIT Press.
- Collins, Chris. 2001. Aspects of Plurality in ≠ Hoan. *Language* 77:456-476.
- Comrie, Bernard. 1976. *Aspect. An Introduction to the Study of Verbal Aspect and Related Problems*: Cambridge Textbooks in Linguistics. Cambridge: Cambridge University Press.
- Corver, Norbert. 1990. The Syntax of Left Branch Extraction, Tilburg: PhD.
- Corver, Norbert. 1991. The Internal Syntax and Movement Behavior of the Dutch 'Wat voor'-Construction. *Linguistische Berichte* 133:190-228.
- Corver, Norbert, and Delfitto, Denis. 1999. On the nature of pronoun movement. In *Clitics in the Languages of Europe*, ed. Henk van Riemsdijk, 799 - 861. Berlin / New York: Mouton de Gruyter.
- Cresswell, Max J. 1973. *Logics and Languages*. London: Methuen.
- Cresswell, Max J. 1974. Adverbs and Events. *Synthese* 28:455-481.
- Cresswell, Max. 1990. *Entities and Indices*. Dordrecht: Kluwer Academic Publishers.
- Czepluch, Hartmut. 1982. Case theory and the dative construction. *The Linguistic Review* 2:1 - 38.
- Czepluch, Hartmut. 1987. Lexikalische Argumentstruktur und syntaktische Projektionen: zur Beschreibung grammatischer Relationen. *Linguistische Berichte* 6:3 - 36.
- Dalrymple, Mary. 1991. Against Syntactic Reconstruction. Palo Alto /Cal.: Xerox Corporation.
- Danto, Arthur C. 1965. Basic Actions. *American Philosophical Quarterly* 2:141-148.
- Davidson, Donald. 1963. Actions, Reasons, and Causes. *The Journal of Philosophy* 60:682-700.

- Davidson, Donald. 1967. The Logical Form of Action Sentences. In *The Logic of Decision and Action*, ed. Nicholas Rescher, 81-95. Pittsburgh: University of Pittsburgh Press.
- Davidson, Donald. 1967. Causal Relations. *Journal of Philosophy* 64.
- Davidson, Donald. 1980. *Essays on Actions and Events*. Oxford: Clarendon Press.
- Davies, William D. 1986. *Choctaw Verb Agreement and Universal Grammar: Studies in Natural Language and Linguistic Theory*. Dordrecht: D. Reidel Publishing Company.
- Davis, Lawrence. 1970. Individuation of Action. *Journal of Philosophy* 67:520-530.
- Davis, Lawrence H. 1994. Action (1). In *A Companion to the Philosophy of Mind*, ed. Samuel Guttenplan, 111-117. Oxford: Blackwell Publishers.
- Déchine, Rose-Marie. 1989. An account of Serial Verbs in Haitian Creole: Argument Structure Coalescence. In *Current Approaches to African Linguistics (vol. 5)*, eds. Paul Newman and Robert D. Botne, 237-251. Dordrecht: Foris.
- Depuydt, Leo. 1993. *Conjunction Contiguity Contingency. On Relationships between Events in the Egyptian and Coptic Verbal Systems*. New York / Oxford: Oxford University Press.
- Diesing, Molly. 1988. Bare plurals and the stage/individual contrast. Paper presented at *Genericity in Natural Language*, Tübingen.
- Diesing, Molly. 1990. The Syntactic Roots of Semantic Partition, University of Massachusetts at Amherst: Ph.D.
- Diesing, Molly. 1992. *Indefinites*. Cambridge/Mass.: MIT Press.
- Diesing, Molly. 1992a. *Indefinites*. Cambridge/Mass.: MIT Press.
- Diesing, Molly. 1992b. Bare Plural Subjects and the Derivation of Logical Representations. *Linguistic Inquiry* 23 (3):353-380.
- Diesing, Molly, and Jelinek, Eloise. 1995. Distributing Arguments. *Natural Language Semantics* 3.
- Diesing, Molly. 1997. Yiddish VP Order and the Typology of Object Movement in Germanic. *Natural Language and Linguistic Theory* 15(2):369-427.
- Dikken, Marcel den. 1995. *Particles. On the Syntax of Verb-Particle, Triadic, and Causative Constructions*. New York Oxford: Oxford University Press.
- Dixon, Robert M. W. 1980. *The Languages of Australia*: Cambridge Language Surveys. Cambridge: Cambridge University Press.
- Dixon, R.M.W. 1988. *A grammar of Bouma Fijian*. Chicago and London: The University of Chicago Press.
- Dowty, David R. 1978. Governed Transformations as Lexical Rules in a Montague Grammar. *Linguistic Inquiry* 9:393-426.
- Dowty, David R. 1979. *Word Meaning and Montague Grammar*. Dordrecht: Ridel Publishing Company.

- Dowty, David. 1985. On Recent Analyses of the Semantics of Control. *Linguistics and Philosophy* 8:291-331.
- Dowty, David. 1987. Collective predicates, distributive predicates and *all*. Paper presented at *Eastern States Conference on Linguistics. ESCOL 3*.
- Dowty, David. 1989. On the Semantic Content of the Notion of 'Thematic Role'. In *Properties, Types and Meaning. Volume II: Semantic Issues*, eds. Gennaro Chierchia, Barbara H. Partee and Raymond Turner. Dordrecht/Boston/London: Kluwer Academic Publishers.
- Dowty, David. 1991. Thematic Proto-Roles and Argument Selection. *Language* 67:547-619.
- Dryer, Matthew S. 1983. Indirect Objects in Kinyarwanda Revisited. In *Studies in Relational Grammar 1*, ed. David M. Perlmutter. Chicago and London: The University of Chicago Press.
- Dubinsky, Stanley, and Simango, Silvester Ron. 1996. Passive and Stative in Chichewa: Evidence for Modular Distinctions. *Language* 72:749-781.
- Eisele, John. 1990. Aspectual Classification of Verbs in Cairene Arabic. In *Perspectives on Arabic Linguistics II. Papers from the Second Annual Symposium on Arabic Linguistics*, eds. Mushira Eid and John McCarthy, 192-233. Amsterdam: John Benjamins Publishing Company.
- Emonds, Joseph. 1972. Evidence that Indirect Object Movement is a Structure Preserving Rule. *Foundations of Language* 8:546-561.
- Emonds, Joseph. 1976. *A Transformational Approach to English Syntax*. New York: Academic Press.
- Emonds, Joseph. 1985. *A Unified Theory of Syntactic Categories*. Dordrecht: Foris.
- Emonds, Joseph E. 1987. The Invisible Category Principle. *Linguistic Inquiry* 18:613-632.
- Enç, Mürvet. 1981. Tense without Scope: An Analysis of Nouns as Indexicals, University of Wisconsin, Madison: Ph.D.
- Engdahl, Elisabet. 1980. The Syntax and Semantics of Questions in Swedish, University of Massachusetts at Amherst: Ph.D.
- Engdahl, Elisabet. 1986. *Constituent Questions. The Syntax and Semantics of Questions with Special Reference to Swedish*. Dordrecht: D. Reidel Publishing Company.
- Engel, Ulrich. 1988. *Deutsche Grammatik*. Heidelberg: Julius Groos Verlag.
- Fabb, Nigel Alexander John. 1984. Syntactic Affixation, Massachusetts Institute of Technology: Ph.D.
- Fabb, Nigel. 1988. English Suffixation is Constrained only by Selectional Restrictions. *Natural Language and Linguistic Theory* 6(4):527-539.
- Fagan, Sarah M. B. 1988. The English Middle. *Linguistic Inquiry* 19:181-203.
- Fagan, Sarah M. B. 1992. *The syntax and semantics of middle constructions. A study with special reference to German*: Cambridge Studies in Linguistics 60. Cambridge: Cambridge University Press.

- Fanselow, Gisbert. 1985. What is a Possible Complex Word? In *Studies in German Grammar*, ed. Jindrich Toman, 289-318. Dordrecht: Foris.
- Fanselow, Gisbert. 1988. Aufspaltung von NPn und das Problem der 'freien' Wortstellung. *Linguistische Berichte* 114:91-113.
- Fanselow, Gisbert. 1991. *Minimale Syntax*. vol. 32: Groninger Arbeiten zur germanistischen Linguistik. Groningen: Germanistisch Instituut, Rijksuniversiteit Groningen.
- Feldman, Harry. 1978. Passivizing on Datives in Greek. *Linguistic Inquiry*:499-502.
- Fillmore, Charles J. 1968. The Case for Case. In *Universals in Linguistic Theory*, eds. Emmon Bach and Robert Harms, 1-88. New York Chicago San Francisco: Holt, Rinehart and Winston.
- Fintel, Kai von. 1989. Theticity in Generative Grammar. In *Papers on Quantification*, eds. Emmon Bach, Angelika Kratzer and Barbara Partee. NSF Grant BNS 8719999, Department of Linguistics, UMass Amherst.
- Fintel, Kai von. 1990. Licensing of Clausal Specifiers in German. Paper presented at *Proceedings of the First Meeting of the Formal Linguistics Society of Midamerica (FLSM1)*, University of Wisconsin at Madison.
- Fintel, Kai von. 1992. The Modal-Existential Construction in Germanic Syntax. Paper presented at *FLSM III*, Northwestern University.
- Fintel, Kai von. 1994. Restrictions on Quantifier Domains, UMass Amherst: Ph.D.
- Fintel, Kai von. 1995. A Minimal Theory of Adverbial Quantification: manuscript, MIT.
- Fleischer, Wolfgang. ⁴ 1976. *Wortbildung der deutschen Gegenwartssprache*. Leipzig: VEB Bibliographisches Institut.
- Fodor, Jerry A. 1970. Three reasons for not deriving "kill" from "cause to die". *Linguistic Inquiry* 1:429-438.
- Fodor, Janet D. 1979. *The Linguistic Description of Opaque Contexts*. New York: Garland (1970 MIT Dissertation).
- Fodor, Janet D., and Sag, Ivan. 1982. Referential and Quantificational Indefinites. *Linguistics and Philosophy* 5:355-398.
- Fodor, Jerry A., and Lepore, Ernie. 1998. The Emptiness of the Lexicon: Reflections on James Pustejovsky's *The Generative Lexicon*. *Linguistic Inquiry* 29:269 - 288.
- Fodor, Jerry, and Lepore, Ernest. 1999. Impossible Words? *Linguistic Inquiry* 30:445-452.
- Fox, Danny. 2000. *Economy and Semantic Interpretation*. Cambridge/ Mass.: The MIT Press.
- Freidin, Robert, and Sprouse, Rex A. 1991. Lexical Case Phenomena. In *Principles and Parameters in Comparative Grammar*, ed. Robert Freidin, 392-416. Cambridge/Mass.: MIT Press.
- Fruechtel, Ludwig. 1959 (fifth edition). *Griechische Grammatik*. München: Bayerischer Schulbuchverlag.

- Fu, Jingqi, Roeper, Thomas, and Borer, Hagit. 1998. The VP within Process Nominals: Evidence from Adverbs and the VP Anaphor *Do-So*. University of Massachusetts at Amherst.
- Gallin, D. 1975. *Intensional and Higher-Order Modal Logic*. Amsterdam: North-Holland.
- Gary, Judith, and Keenan, Edward. 1977. On Collapsing Grammatical Relations In Universal Grammar. In *Syntax and Semantics 8: Grammatical Relations*, eds. Peter Cole and Jerrold Sadock. New York: Academic Press.
- Gawron, Jean Mark. 1986. Situations and Prepositions. *Linguistics and Philosophy* 9:327-382.
- Geenhoven, Veerle van. 1998. *Semantic Incorporation and Indefinite Descriptions*. Stanford: CSLI Publications.
- Geenhoven, Veerle van. 2000. On (early) VP interpretation: the case of *for*-adverbials. Paper presented at *Semantics Meets Acquisition*, Nijmegen.
- Giannakidou, Anastasia, and Merchant, Jason. 1998. Why Giannis can't scrub his plate clean: On the absence of resultative secondary predication in Greek. Paper presented at *3rd International Conference on Greek Linguistics*, University of Athens, Athens.
- Gil, David. 1982. Distributive Numerals, UCLA Ph.D. thesis.
- Gil, David. 1988. Georgian reduplication and the domain of distributivity. *Linguistics* 26:1039-1065.
- Ginet, Carl. 1990. *On Action*. Cambridge: Cambridge University Press.
- Giorgi, Alessandra, and Longobardi, Giuseppe. 1991. *The Syntax of Noun Phrases. Configuration, Parameters and Empty Categories*. Cambridge Studies in Linguistics, 57. Cambridge: Cambridge University Press.
- Giorgi, Alessandra, and Pianesi, Fabio. 1997. *Tense and Aspect. From Semantics to Morphosyntax*. Oxford: Oxford University Press.
- Goodman, Nelson. 1954. *Fact, Fiction, and Forecast*. Cambridge / Mass.: Harvard University Press.
- Green, Giorgia. 1974. *Semantic and Syntactic Regularity*. Bloomington: Indiana University Press.
- Gregor, Bernd. 1983. *Genuszuordnung. Das Genus englischer Lehnwörter im Deutschen*. Tübingen: M. Niemeyer.
- Grewendorf, Günther. 1983. Reflexivierung in deutschen A.c.I.-Konstruktionen - kein transformations-grammatisches Dilemma mehr., 120 - 196. Groningen: Groninger Arbeiten zur Linguistik (GAGL).
- Grewendorf, Günther. 1989. *Ergativity in German*. Dordrecht: Foris.
- Grimshaw, Jane, and Selkirk, Elisabeth. 1976. Infinitival Noun Phrases in Italian: University of Massachusetts at Amherst.
- Grimshaw, Jane, and Mester, Armin. 1988. Light Verbs and Q-Marking. *Linguistic Inquiry* 19:205-232.
- Grimshaw, Jane. 1990. *Argument Structure*. Cambridge/Mass.: MIT Press.
- Grimshaw, Jane. 1991. Extended Projection: Brandeis University.

- Gropen, Jess, Pinker, Steven, Hollander, Michelle, Goldberg, Richard, and Wilson, Ronald. 1989. The Learnability and Acquisition of the Dative Alternation in English. *Language* 65:203-257.
- Gruber, Jeffrey S. 1965. Studies in Lexical Relations, MIT: Ph.D. dissertation.
- Guttenplan, Samuel ed. 1994. *A Companion to the Philosophy of Mind*. Oxford: Basil Blackwell.
- Hackl, Martin. 2000. Comparative Quantifiers, Department of Linguistics, MIT: Ph.D. Dissertation.
- Haegeman, Liliane. 1986. INFL, COMP and Nominative Case Assignment in Flemish. In *Features and Projections*, eds. Pieter Muysken and Henk van Riemsdijk, 123-137. Dordrecht: Foris.
- Haider, Hubert. 1984. Was zu haben ist und was zu sein hat. *Papiere zur Linguistik* 30:21-34.
- Hale, Kenneth, and Keyser, Samuel J. 1987. A View from the Middle: Lexicon Project Working Papers No. 7, Center for Cognitive Science, MIT.
- Hale, Kenneth, and Keyser, Samuel J. 1988. Explaining and Constraining the English Middle. In *Studies in Generative Approaches to Aspect*, ed. C. Tenny: Lexicon Project Working Papers No. 24, Center for Cognitive Science, MIT.
- Hale, Kenneth. 1992. Subject Obviation, Switch Reference, and Control. In *Control and Grammar*, eds. Richard K. Larson, Sabine Iatridou, Utpal Lahiri and James Higginbotham, 51 - 77. Dordrecht: Kluwer Academic Publishers.
- Hale, Kenneth, and Keyser, Samuel Jay. 1993. On Argument Structure and the Lexical Expression of Syntactic Relations. In *The View from Building 20*, eds. Kenneth Hale and Samuel Jay Keyser, 53-109. Cambridge/Mass.: MIT Press.
- Hale, Ken, and Keyser, Samuel J. 1999. A Response to Fodor and Lepore, "Impossible Words?" *Linguistic Inquiry* 30:453-466.
- Hale, Ken, and Keyser, Samuel Jay. 2002. *Prolegomena to a Theory of Argument Structure*. Cambridge/Mass.: The MIT Press.
- Halle, Morris, and Marantz, Alec. 1993. Distributed Morphology. In *The View from Building 20. Essays in Linguistics in Honor of Sylvain Bromberger*, eds. Kenneth Hale and Samuel Jay Keyser, 111-176. Cambridge/Mass.: The MIT Press.
- Harley, Heidi B. 1995. Subjects, Events and Licensing, Linguistics, MIT: Ph.D.
- Harris, Alice C. 1981. *Georgian Syntax*: Cambridge Studies in Linguistics. Cambridge: Cambridge University Press.
- Harris, Alice C. 1984. Case Marking, Verb Agreement, and Inversion in Udi. In *Studies in Relational Grammar 2*, eds. David M. Perlmutter and Carol G. Rosen, 243-258. Chicago: The University of Chicago Press.

- Hauser, Marc D. 2000. *The Evolution of Communication*. Cambridge/Mass.: The MIT Press.
- Hazout, Ilan. 1990. Verbal Nouns: Theta-Theoretic Studies in Hebrew and Arabic, University of Massachusetts at Amherst: Ph.D.
- Heim, Irene. 1982. The Semantics of Definite and Indefinite Noun Phrases, University of Massachusetts: Ph.D.
- Heim, Irene. 1990. E-Type Pronouns and Donkey Anaphora. *Linguistics and Philosophy* 13:137-178.
- Heim, Irene. 1992. Anaphora and Semantic Interpretation: A Reinterpretation of Reinhart's Approach: MIT.
- Heim, Irene, and Kratzer, Angelika. 1998. *Semantics in Generative Grammar*. Oxford: Basil Blackwell.
- Heimann, Axel. 1991. Argumentstruktur: Arbeitspapiere des Sonderforschungsbereichs 340: Sprachtheoretische Grundlagen für die Computerlinguistik, Universitäten Stuttgart und Tübingen, IBM Deutschland GmbH.
- Helbig, Gerhard, and Buscha, Joachim. 1979. *Deutsche Grammatik. Ein Handbuch für den Ausländerunterricht*. Leipzig: VEB Verlag Enzyklopädie.
- Hendrick, Randall. 1990. Breton Pronominals, Binding, and Barriers. In *The Syntax of the Modern Celtic Languages. Syntax and Semantics 23*, ed. Randall Hendrick, 121 - 165. San Diego: Academic Press.
- Henzen, Walter. 1957. *Deutsche Wortbildung: Sammlung kurzer Grammatiken Germanischer Dialekte*. Tübingen: Max Niemeyer Verlag.
- Herman, Gabriella. 1985. *Syntactic Modularity: Studies in Generative Grammar*. Dordrecht: Foris.
- Hestvik, Arild. 1995. Reflexives and Ellipsis. *Natural Language Semantics* 3:211 - 237.
- Hetzer, Armin. 1978. *Lehrbuch der vereinheitlichten albanischen Schriftsprache*. Hamburg: Helmut Buske Verlag.
- Higginbotham, James. 1983. The Logic of Perceptual Reports: An Extensional Alternative to Situation Semantics. *The Journal of Philosophy* 80:100-127.
- Higginbotham, James. 1985. On Semantics. *Linguistic Inquiry* 16:547-593.
- Higginbotham, James. 1987. Indefiniteness and Predication. In *The Representation of (IN)definiteness*, eds. Eric J. Reuland and Alice G.B. ter Meulen, 42-70. Cambridge, Massachusetts: MIT Press.
- Higginbotham, James. 1989. Elucidations of Meaning. *Linguistics and Philosophy* 12:465-517.
- Higginbotham, James. 2000. On Events in Linguistic Semantics. In *Speaking of Events*, eds. James Higginbotham, Fabio Pianesi and Achille C. Varzi, 49 - 79. New York and Oxford: Oxford University Press.
- Higgins, F. Roger. 1990. By Mishap out of Control: University of Massachusetts at Amherst.

- Hintikka, Jaakko. 1986. The Semantics of *A Certain*. *Linguistic Inquiry* 17:331-336.
- Hirschbühler, Paul. 1988. The middle and pseudo-middle in French. In *Advances in Romance Linguistics*, eds. David Birdsong and Jean-Pierre Monreuil, 97-111. Dordrecht: Foris.
- Hoekstra, Teun. 1984. *Transitivity: Grammatical Relations in Government-Binding Theory*. Dordrecht: Foris.
- Hoekstra, Teun. 1988. Small Clause Results. *Lingua* 74:101-139.
- Hoffman, Mika Christine. 1991. The Syntax of Argument-Structure-Changing Morphology, Massachusetts Institute of Technology: PhD.
- Höhle, Tilman N. 1976. Die Aktiv-Passiv Relation im Deutschen. Grundlagen einer lexikalistischen Syntaxtheorie: Universität Köln.
- Höhle, Tilman N. 1978. *Lexikalistische Syntax: Die Aktiv-Passiv-Relation und andere Infinitkonstruktionen im Deutschen*. Tübingen: Max Niemeyer Verlag.
- Höhle, Tilman N. 1982. Über Komposition und Derivation. *Zeitschrift für Sprachwissenschaft* 1:76-112.
- Holisky, Ann Dee. 1978. Stative verbs in Georgian, and elsewhere. *International Review of Slavic Linguistics* 3:139-162.
- Holisky, Ann Dee. 1979. On lexical aspect and verb classes in Georgian. Paper presented at *Chicago Linguistic Society. Parasession on the Elements/ Non-Slavic Languages of the USSR*.
- Holisky, Ann Dee. 1981. *Aspect and Georgian medial verbs*. New York: Caravan.
- Holmberg, Anders. 1985. Icelandic Word Order and Binary Branching. *Nordic Journal of Linguistics* 8:161-195.
- Holmberg, Anders, and Nikanne, Urpo. 1994. Expletives and Subject Positions in Finnish. Paper presented at *NELS 24*.
- Hoop, Helen de. 1992. Case Configuration and Noun Phrase Interpretation, Rijksuniversiteit Groningen: PhD.
- Hopper, P., and Thompson, S. eds. 1982. *Syntax and Semantics 15: Studies in Transitivity*. New York: Academic Press.
- Horne, Laurence R. 1980. Affixation and the Unaccusative Hypothesis. Paper presented at *Chicago Linguistic Society*.
- Hout, Angeliek van. 1996. Event Semantics of Verb Frame Alternations: A Case Study of Dutch and its Acquisition, Department of Linguistics, Tilburg University: Ph.D.
- Hovav, Malka Rappaport, and Levin, Beth. 2001. An Event Structure Account of English Resultatives. *Language* 77:766 - 797.
- Huang, C.-T. James. 1982. Logical Relations in Chinese and the Theory of Grammar, MIT: Ph.D.
- Hubbard, Philip L. 1985. *The Syntax of the Albanian Verb Complex: Outstanding Dissertations in Linguistics*. New York: Garland (1980 University of California at San Diego Ph.D. dissertation).

- Hung, Henrietta J. 1988. *The Structure of Derived Nouns and Verbs in Malagasy: A Syntactic Account*. McGill University.
- Hyams, Nina. 1986. The Interpretation and Acquisition of Italian Impersonal *si*. In *Syntax and Semantics 19. The Syntax of Pronominal Clitics*, ed. Hagit Borer, 219 - 240. Orlando: Academic Press.
- Jackendoff, Ray. 1977. *X' Syntax: A Study of Phrase Structure*. Cambridge, Massachusetts: The MIT Press.
- Jackendoff, Ray. 1983. *Semantics and Cognition*. Cambridge, Massachusetts: The MIT Press.
- Jackendoff, Ray S. 1987. The Status of Thematic Relations in Linguistic Theory. *Linguistic Inquiry* 18:369-411.
- Jackendoff, Ray. 1990. *Semantic Structures*. Cambridge, Massachusetts: The MIT Press.
- Jacobson, Pauline. to appear. Raising Without Movement. In *Control and Grammatical Theory*, eds. Richard Larson and James Higginbotham (?). Dordrecht: Kluwer.
- Jaeggli, Osvaldo A. 1986. Three Issues in the Theory of Clitics: Case, Doubled NPs, and Extraction. In *Syntax and Semantics 19. The Syntax of Pronominal Clitics*, ed. Hagit Borer, 15-42. Orlando: Academic Press.
- Jaeggli, Osvaldo. 1986. Passive. *Linguistic Inquiry* 17:587-622.
- Jaeggli, Osvaldo, and Hyams, Nina. 1989. On the Independence and Interdependence of Syntactic and Morphological Properties: English Aspectual *come* and *go*: ms., UCLA.
- Jespersen, Otto. 1927 (reprinted 1961). *A Modern English Grammar on Historical Principles, Part III (Syntax)*. London Copenhagen: George Allen & Unwin LTD Ejnar Munksgaard.
- Jespersen, Otto. 1940 (reprinted 1961). *A Modern English Grammar on Historical Principles. Part V. Syntax*. London Copenhagen: George Allen & Unwin Ejnar Munksgaard.
- Johns, Alana. 1992. Deriving Ergativity. *Linguistic Inquiry* 23:57-87.
- Johnson, Kyle. 1991. Object Positions. *Natural Language and Linguistic Theory* 9:577-636.
- Jones, Charles. 1985. Two BE+infinitival Constructions. Paper presented at *West Coast Conference on Formal Linguistics*, UCLA.
- Joseph, Brian D., and Zwicky, Arnold M. eds. 1990. *When Verbs Collide: Papers from the 1990 Ohio State Mini-Conference on Serial Verbs*. Department of Linguistics, The Ohio State University.
- Joseph, Brian D., and Smirniotopoulos, Jane C. 1993. The Morphosyntax of the Modern Greek Verb as Morphology and not Syntax. *Linguistic Inquiry* 24:388-398.
- Kadmon, Nirit. 1987. On Unique and Non-Unique Reference and Asymmetric Quantification, University of Massachusetts at Amherst: Ph.D.
- Kähler, Hans. 1983, 3rd edition. *Grammatik der Bahasa Indonésia*. Wiesbaden: Otto Harrassowitz.

- Kamp, Hans. 1971. Formal Properties of 'now'. *Theoria* 37:227-273.
- Kamp, Hans. 1981. A Theory of Truth and Semantic Interpretation. In *Mathematical Centre, Tract 135*, eds. Jeroen Groenendijk, Theo Janssen and Martin Stokhof, 277-322. Amsterdam.
- Kamp, J.W.A. 1981. A Theory of Truth and Semantic Representation. In *Formal Methods in the Study of Language*, eds. J. Groenendijk, T. Janssen and M. Stokhof. Amsterdam: Mathematical Centre.
- Kamp, Hans, and Partee, Barbara. 1995. Prototype Theory and Compositionality. *Cognition* 57:129 - 191.
- Kathol, Andreas. 1989. Adverbial Quantification and Scrambled Objects. In *Papers on Quantification*, eds. Emmon Bach, Angelika Kratzer and Barbara Partee. NSF Report, Department of Linguistics, University of Massachusetts at Amherst.
- Kay, Paul, and Fillmore, Charles J. 1999. Grammatical Constructions and Linguistic Generalizations: *The What's X doing Y? Construction*. *Language* 75:1-33.
- Kayne, Richard. 1981. Unambiguous Paths. In *Levels of Syntactic Representation*, eds. Robert May and Jan Koster, 143- 183. Dordrecht: Foris.
- Kayne, Richard S. 1984. *Connectedness and Binary Branching*: Studies in Generative Grammar 16. Dordrecht: Foris.
- Keenan, Edward L. 1976. Remarkable Subjects in Malagasy. In *Subject and Topic*, ed. Charles N. Li, 247-301. New York San Francisco London: Academic Press.
- Keenan, Edward L., and Faltz, Leonard. 1978. *Logical Types for Natural Language*: UCLA Occasional Papers in Linguistics 3: Department of Linguistics, UCLA.
- Keenan, Edward L. 1984. Semantic Correlates of the Ergative/Absolutive Distinction. *Linguistics* 22:197-223.
- Keenan, Edward L., and Faltz, Leonard M. 1985. *Boolean Semantics for Natural Language*. Dordrecht: D. Reidel Publishing Company.
- Keenan, Edward L. 2000. Morphology is Structure: A Malagasy Test Case. In *Formal Issues in Austronesian Linguistics*, eds. Ileana Paul, Vivianne Phillips and Lisa Travis, 27-47. Dordrecht: Kluwer.
- Kemmer, Suzanne. 1993. *The Middle Voice*. Amsterdam/Philadelphia: John Benjamins Publishing Company.
- Kenny, Anthony. 1963. *Action, Emotion and Will*. London: Routledge & Kegan Paul.
- Keyser, Samuel Jay, and Roeper, Thomas. 1984. On the Middle and Ergative Constructions in English. *Linguistic Inquiry* 15:381-416.
- Keyser, Samuel Jay, and Roeper, Thomas. 1992. Re: The Abstract Clitic Hypothesis. *Linguistic Inquiry* 23:89-125.
- Kimenyi, Alexandre. 1980. *A Relational Grammar of Kinyarwanda*. Berkeley: University of California Press.

- King, Jeffrey. 1988. Are Indefinite Descriptions Ambiguous. *Philosophical Studies* 53:417-440.
- Kiparsky, Paul. 1998. Partitive Case and Aspect. In *The Projection of Arguments*, eds. Miriam Butt and Wilhelm Geuder, 265-307. Stanford: CSLI Publications.
- Klaiman, M.H. 1988. Affectedness and control: a typology of voice systems. In *Passive and Voice*, ed. Masayoshi Shibatani, 25 - 83. Amsterdam: John Benjamins.
- Klaiman, M. H. 1991. *Grammatical Voice*: Cambridge Studies in Linguistics. Cambridge: Cambridge University Press.
- Klein, Ewan, and Sag, Ivan. 1985. Type Driven Translation. *Linguistics and Philosophy* 8:163-201.
- Koopman, Hilda, and Sportiche, Dominique. 1991. The Position of Subjects. *Lingua* 85:211-258.
- Kratzer, Angelika. 1978. *Semantik der Rede. Kontexttheorie - Modalwoerter - Konditionalsaeetze*. Kronberg: Scriptor.
- Kratzer, Angelika. 1989. Stage-Level and Individual-Level Predicates: NSF Grant Report, BNS 8719999, UMASS.
- Kratzer, Angelika. 1989. An Investigation of the Lumps of Thought. *Linguistic and Philosophy* 12:607-653.
- Kratzer, Angelika. 1990. How Specific is a Fact. Paper presented at *Conference on Theories of Partial Information*. Center for Cognitive Science, Austin Texas.
- Kratzer, Angelika. 1994. On External Arguments. In *Functional Projections*, eds. Elena Benedicto and Jeffrey Runner, 103-130. Amherst/Mass.: GLSA, UMass Amherst.
- Kratzer, Angelika. 1995. Stage-Level and Individual-Level Predicates. In *The Generic Book*, eds. Gregory N. Carlson and Jeffrey Pelletier. Chicago: Chicago University Press.
- Kratzer, Angelika. 1996. Severing the External Argument from its Verb. In *Phrase Structure and the Lexicon*, eds. Johan Rooryck and Laurie Zaring, 109-137. Dordrecht: Kluwer.
- Kratzer, Angelika. 1998. More Structural Analogies Between Pronouns and Tenses. Paper presented at *SALT VIII*, MIT.
- Kratzer, Angelika. 1998. Scope or Pseudoscope? Are There Wide-Scope Indefinites? In *Events and Grammar*, ed. Susan Rothstein, 163-196. Dordrecht: Kluwer Academic Publishers.
- Kratzer, Angelika. 1998. *Aspect in Adjectival Passives*: Paper presented at WCCFL 17, February 20, 1998: University of British Columbia, Vancouver.
- Kratzer, Angelika. 2000. Telicity and the Meaning of Objective Case. Paper presented at *International Round Table 'The Syntax of Tense and Aspect'*, Université de Paris 7.

- Krifka, Manfred. 1986. Nominalreferenz und Zeitkonstitution. Zur Semantik von Massentermen, Pluraltermen und Aspektklassen, Ludwig - Maximilians -Universität München: Ph.D.
- Krifka, Manfred. 1987. Nominal Reference and Temporal Constitution. Paper presented at *Proceedings of the Sixth Amsterdam Colloquium*, Amsterdam.
- Krifka, Manfred. 1989. Nominal Reference, Temporal Constitution and Quantification in Event Semantics. In *Semantics and Contextual Expression*, eds. R. Bartsch, J. van Benthem and P. van Emde Boas, 75-116. Dordrecht: Foris.
- Krifka, Manfred. 1990. Four Thousand Ships Passed Through the Lock: Object-Induced Measure Functions on Events. *Linguistics and Philosophy* 13:487-520.
- Krifka, Manfred. 1992. Thematic Relations as Links Between Nominal Reference and Temporal Constitution. In *Lexical Matter*, eds. Ivan Sag and Anna Szabolsci, 29-53. Stanford: CSLI.
- Krifka, Manfred. 1995. Common Nouns: A Contrastive Analysis of Chinese and English. In *The Generic Book*, eds. Gregory N. Carlson and Francis Jeffry Pelletier, 398-411. Chicago: The University of Chicago Press.
- Krifka, Manfred. 1996. Pragmatic Strengthening. Paper presented at *Semantics and Linguistic Theory VI*, Rutgers University.
- Krifka, Manfred. 1998. The Origins of Telicity. In *Events in Grammar*, ed. Susan Rothstein, 197-235. Dordrecht: Kluwer Academic Publishers.
- Kroch, Anthony S. 1974. The Semantics of Scope of English, MIT: Ph.D.
- Kuroda, S.-Y. 1979. *The (W)hole of the Doughnut*. Ghent -Antwerp-Brussels: E. Story-Scientia P.V.B.A.
- Lakoff, George. 1965. On the Nature of Syntactic Irregularity, Indiana University: Ph.D.
- Lakoff, George. 1972. Linguistics and Natural Logic. In *Semantics of Natural Language*, eds. Donald Davidson and Gilbert Harman, 545-665. Dordrecht: Reidel.
- Landman, Fred. 1987. Groups, Part 1 and Part 2. *Linguistics and Philosophy* 12:559-605, 723-744.
- Landman, Fred. 1992. The Progressive. *Natural Language Semantics*.
- Landman, Fred. 1996. Plurality. In *The Handbook of Contemporary Semantic Theory*, ed. Shalom Lappin, 425-457. Oxford: Blackwell.
- Landman, Fred. 1998. Plurals and Maximalization. In *Events in Grammar*, ed. Susan Rothstein, 237-271. Dordrecht: Kluwer Academic Publishers.
- Landman, Fred. 2000. *Events and Plurality. The Jerusalem Lectures*. Dordrecht: Kluwer Academic Publishers.
- Larson, Richard. Adjuncts and Implicit Arguments in Situation Semantics. *Linguistics and Philosophy*.
- Larson, Richard K. 1988. On the Double Object Construction. *Linguistic Inquiry* 19:335-391.

- Larson, Richard K., and Levebvre, Claire. 1990. Predicate Clefting in Haitian Creole. Paper presented at *North East Linguistic Society 21*, Université Du Québec à Montréal.
- Lasersohn, Peter. 1988. A Semantics for Groups and Events, Ohio State University: Ph.D.
- Lasersohn, Peter. 1990. Group Action and Spatio-Temporal Proximity. *Linguistics and Philosophy* 13:179-206.
- Lasersohn, Peter. 1991. *A Semantics for Groups and Events*. New York: Garland.
- Lasersohn, Peter. 1992. Generalized Conjunction and Temporal Modification. *Linguistics and Philosophy* 15(4):381-410.
- Lasersohn, Peter. 1995. *Plurality, Conjunction and Events*. Dordrecht: Kluwer Academic Publishers.
- Lasersohn, Pater. 1998. Events in the semantics of collectivizing adverbials. In *Events in Grammar*, ed. Susan Rothstein, 273-292. Dordrecht: Kluwer Academic Publishers.
- Lasnik, Howard. 1988. Subjects and the Q-Criterion. *Natural Language and Linguistic Theory* 6:1-17.
- Lebeaux, David. 1984. Nominalizations, Argument Structure, and the Organization of the Grammar: University of Massachussetts unpublished manuscript.
- Lebeaux, David. 1986. The Interpretation of Derived Nominals. Paper presented at *CLS (Papers from the General Session at the Twnty-Second Regional Meeting of the Chicago Linguistics Society)*.
- Legendre, Géraldine. 1997. Secondary Predication and Functional Projections in French. *Natural Language and Linguistic Theory* 15:43 - 87.
- Levin, Beth, and Rappaport, Malka. 1986. The Formation of Adjectival Passives. *Linguistic Inquiry* 17:623-662.
- Levin, Beth, and Rapoport, Tova R. 1988. Lexical Subordination. In *Papers from the 24th Regional Meeting of the Chicago Linguistic Society*, eds. Lynne MacLeod, Gary Larson and Diane Brentari, 275 - 289. Chicago: Chicago Linguistic Society.
- Levin, Beth, and Rappaport, Malka. 1988. Non-event -er Nominals: A Probe into Argument Structure. *Linguistics* 26:1067-1083.
- Levin, Beth. 1993. *English Verb Alternations. A Preliminary Investigation*. Chicago: Chicago University Press.
- Levin, Beth, and Hovav, Malka Rappaport. 1995. *Unaccusativity. At the Syntax-Lexical Semantics Interface*. Cambridge/Mass.: The MIT Press.
- Levin, Beth, and Hovav, Malka Rappaport. 1996. Lexical Semantics and Syntactic Structure. In *The Handbook of Contemporary Semantic Theory*, ed. Shalom Lappin, 487-507. Oxford: Blackwell.
- Levin, Beth. 1999. Objecthood: An event structure perspective. Paper presented at *Chicago Linguistic Society 35*.

- Lewis, David. 1968. Counterpart Theory and Quantified Modal Logic. *The Journal of Philosophy* 65:113-126.
- Lewis, David. 1972. General Semantics. In *Semantics of Natural Language*, eds. Donald Davidson and Gilbert Harman, 169-218. Dordrecht: Reidel.
- Lewis, David. 1973. Causation. *Journal of Philosophy* 70:556-567.
- Lewis, David. 1979. Attitudes *De Dicto* and *De Se*. *The Philosophical Review* 88:513-543.
- Lewis, David. 1986. *Philosophical Papers, Volumes I and II*. Oxford: Oxford University Press.
- Lewis, David. 1986. *On the Plurality of Worlds*. Oxford: Blackwell.
- Lewis, David K. 1986. Causal Explanation. In *Philosophical Papers, Volume II*, 214-240. Oxford: Oxford University Press.
- Lewis, David. 1991. *Parts of Classes*. Oxford: Blackwell.
- Li, Charles N. ed. 1976. *Subject and Topic*. New York San Francisco London: Academic Press.
- Li, Yen-hui Audrey. 1990. *Order and Constituency in Mandarin Chinese: Studies in Natural Language and Linguistic Theory*. Dordrecht: Kluwer Academic Publishers.
- Lieber, Rochelle. 1980. On the Organization of the Lexicon, MIT: Ph.D.
- Lieber, Rochelle. 1981. Morphological Conversion within a Restrictive Theory of the Lexicon. In *The Scope of Lexical Rules*, eds. M. Moortgat, H. v. d. Hulst and T. Hoekstra. Dordrecht: Foris.
- Lieber, Rochelle. 1992. *Deconstructing Morphology*. Chicago: The University of Chicago Press.
- Link, Godehard. 1983. The logical analysis of plurals and mass terms: A lattice-theoretic approach. In *Meaning, Use, and Interpretation of Language*, ed. R. Bäuerle et al., 302-323. Berlin: De Gruyter.
- Link, Godehard. 1984. Hydras: On the logic of relative constructions with multiple heads. In *Varieties of Formal Semantics. Proceedings of the Fourth Amsterdam Colloquium*, eds. Fred Landman and Frank Veltman, 245-257. Dordrecht: Foris.
- Link, Godehard. 1991. Plural. In *Semantik /Semantics. An International Handbook of Contemporary Research*, eds. Arnim von Stechow and Dieter Wunderlich, 418-440. Berlin: de Gruyter.
- Link, Godehard. 1998. *Algebraic Semantics in Language and Philosophy*. Stanford: CSLI Publications.
- Longobardi, Giuseppe. 1994. Reference and Proper Names: A Theory of N-Movement in Syntax and Logical Form. *Linguistic Inquiry* 25(4):609-665.
- Ludlow, Peter, and Neale, Stephen. 1991. Indefinite Descriptions: In Defense of Russell. *Linguistics and Philosophy* 14:171-202.
- Mahajan, Anoop. 1990. Clitic Doubling, Object Agreement and Specificity. Paper presented at *North East Linguistic Society 21*, Université du Québec à Montréal.

- Majajan, Anoop Kumar. 1990. The A/A-Bar Distinction and Movement Theory, Massachusetts Institute of Technology: Ph.D.
- Maling, Joan, and Kim, Soowon. 1992. Case Assignment in the Inalienable Possession Construction in Korean. *Journal of East Asian Linguistics* 1:37-68.
- Manzini, Rita. 1983. On Control And Control Theory. *Linguistic Inquiry* 14:421-446.
- Manzini, Maria Rita. 1986. On Italian *si*. In *Syntax and Semantics 19. The Syntax of Pronominal Clitics*, ed. Hagit Borer, 241 - 262. Orlando: Academic Press.
- Marantz, Alec. 1982. On the Acquisition of Grammatical Relations. *Linguistische Berichte* 80-82:32-69.
- Marantz, Alec. 1984. *On the Nature of Grammatical Relations*. Cambridge/Mass.: MIT Press.
- Martin, Jane R. 1972. Basic Actions and Simple Actions. *American Philosophical Quarterly* 9:59-68.
- Massey, Victoria Walker. 1991. Compositionality and Constituency in Albanian, University of North Carolina at Chapel Hill: Ph.D.
- Mathiot, Madeleine. 1967. The Cognitive Significance of the Category of Nominal Number in Papago. In *Studies in Southwestern Ethnolinguistics (Studies in General Anthropology III)*, eds. D. Hymes and W. Bittle, 197-237. The Hague: Mouton.
- Matthewson, Lisa. 2000. On Distributivity and Pluractionality. Paper presented at *SALT 10*, Cornell University, Ithaca, New York.
- May, Robert. 1977. The Grammar of Quantification, MIT: PhD.
- May, Robert. 1985. *Logical Form. Its Structure and Derivation*. Cambridge/Mass.: MIT Press.
- McCarthy, John J., and S.Prince, Alan. 1998. Prosodic Morphology. In *The Handbook of Morphology*, eds. Andrew Spencer and Arnold M. Zwicky, 283-305. Oxford: Blackwell Publishers.
- McCawley, James. 1968. Concerning the Base Component of a Transformational Grammar. *Foundations of Language* 4:243-269.
- McCloskey, James, and Hale, Kenneth. 1984. On the Syntax of Person-Number Inflection in Modern Irish. *Natural Language and Linguistic Theory* 1:487 - 533.
- McCloskey, Jim. 1997. Subjecthood and Subject Positions. In *Elements of Grammar. Handbook in Generative Grammar*, ed. Liliane Haegeman, 197 - 235. Dordrecht: Kluwer Academic Publishers.
- McConnell-Ginet, Sally. 1982. Adverbs and Logical Form: A Linguistically Realistic Theory. *Language* 58:144-184.
- Mchombo, Sam. 1993. A formal analysis of the stative construction in Bantu. *Journal of African Languages and Linguistics* 14:5-28.

- McLaughlin, Brian P. 1994. Epiphenomenalism. In *A Companion to The Philosophy of Mind*, ed. Samuel Guttenplan, 277-288. Oxford: Basil Blackwell.
- Menge, Hermann. 2000. *Lehrbuch der lateinischen Syntax und Semantik*. Darmstadt: Wissenschaftliche Buchgesellschaft.
- Milsark, Gareth. 1974. Existential Sentences in English, MIT: Ph.D.
- Milsark, Gareth L. 1988. "Singl-ing". *Linguistic Inquiry* 19:611-634.
- Mitchell, Jonathan. 1986. The Formal Semantics of Point of View, University of Massachusetts at Amherst: Ph.D.
- Mitchell, Erika. 1991. Case and the Finnish Object. In *Non-Canonical Case-assignment*, eds. Almeida J. Toribio and Wayne E. Harbert, 193-228: Cornell Working Papers in Linguistics 9.
- Mithun, Marianne. 1991. Active/Agentive Case Marking and its Motivation. *Language* 67:510-546.
- Mithun, Marianne. 1999. *The Languages of Native North America*. Cambridge: Cambridge University Press.
- Mittwoch, Anita. 1982. On the Difference between *Eating* and *Eating Something* : Activities versus Accomplishments. *Linguistic Inquiry* 13.
- Mohanan, K. P., and Mohanan, Tara. 1998. Strong and Weak Projection: Lexical Reflexives and Reciprocals. In *The Projection of Arguments*, eds. Miriam Butt and Wilhelm Geuder, 165-194. Stanford: CSLI Publications.
- Moltmann, Friederike. 1991. Measure Adverbials. *Linguistics and Philosophy* 14:629-660.
- Moltmann, Friederike. 1996. *Parts and Wholes in Semantics*. Oxford: Oxford University Press.
- Montague, Richard. 1974. *Formal Philosophy*. New Haven: Yale University Press.
- Moortgaat, Michael. 1984. A Fregean Restriction on Metarules. Paper presented at *Proceedings of NELS 14*.
- Morzycki, Marcin. Forthcoming. UMass Amherst: Ph.D.
- Müller, Ana. 2000. The Expression of Genericity in Brazilian Portuguese. In *UMOP 23: Issues in Semantics*, eds. Kiyomi Kusumoto and Elisabeth Villalta, 137-154. Amherst/Mass.: GLSA.
- Muskens, Reinhard. 1989. Meaning and Partiality, University of Amsterdam: Ph.D.
- Myers, Scott. 1984. Zero-Derivation and Inflection. Paper presented at *MIT Working Papers in Linguistics 7: Papers from the January 1984 MIT Workshop in Morphology, Department of Linguistics and Philosophy, MIT Cambridge/Mass.*
- Nakamura, Masanori. 1998. Reference Set, Minimal Link Condition, and Parametrization. In *Is the Best Good Enough? Optimality and Competition in Syntax*, eds. Pilar Barbosa, Danny Fox, Paul Hagstrom, Martha McGinnis and David Pesetsky, 291 - 313. Cambridge/Mass.: The MIT Press.

- Napoli, Donna Jo. 1973. The Two Si's of Italian, Harvard University: PhD.
- Nedjalkov, V. P., and Jaxontov, S. J. 1988. The Typology of Resultative Constructions. In *Typology of Resultative Constructions*, ed. V. P. Nedjalkov, 3-62. Amsterdam: John Benjamins.
- Nedjalkov, Vladimir P. ed. 1988. *Typology of Resultative Constructions. Typological Studies in Language 12*. Amsterdam: John Benjamins Publishing Company.
- Neeleman, Ad. 1994. *Complex Predicates*: OTS Dissertation Series. Utrecht: OTS-Publications.
- Neeleman, Ad. 1997. PP-Complements. *Natural Language and Linguistic Theory* 15:89-137.
- Oehrle, Richard. 1976. The Grammatical Status of the English Dative Alternation, MIT: Ph.D.
- Ojeda, Almerindo E. 1992. The Semantics of Number in Arabic. In *SALT II: Proceedings of the Second Conference on Semantics and Linguistic Theory*, eds. Chris Barker and David Dowty, 303-325. Columbus Ohio: Department of Linguistics, Ohio State University.
- Ojeda, Almerindo. 1998. Collectives and Distributives in Papago. *Natural Language Semantics* 6:245-270.
- Pape, W. 1845. *Deutsch - Griechisches Wörterbuch zum Schulgebrauch*. Braunschweig: Friedrich Vieweg.
- Paramasivam, K. 1979. Effectivity and Causativity in Tamil. *Journal of Dravidian Linguistics* 8:71-151.
- Parsons, Terence. 1972. Some Problems Concerning the Logic of Grammatical Modifiers. In *Semantics of Natural Language*, eds. Donald Davidson and Gilbert Harman, 127-141. Dordrecht: D. Reidel Publishing Company.
- Parsons, Terence. 1980. Modifiers and quantifiers in natural language. *Canadian Journal of Philosophy* Suppl. Vol. 6:29-60.
- Parsons, Terence. 1985. Underlying events in the logical analysis of English. In *Actions and Events: Perspectives on the Philosophy of Donald Davidson*, eds. Ernest LePore and Brian McLaughlin. Oxford: Blackwell.
- Parsons, Terence. 1990. *Events in the Semantics of English. A Study in Subatomic Semantics*. Cambridge/Mass.: MIT Press.
- Parsons, Terence. 1995. Thematic Relations and Arguments. *Linguistic Inquiry* 26:635-662.
- Partee, Barbara. 1973. Some Structural Analogies Between Tenses and Pronouns in English. *The Journal of Philosophy* 70:601-609.
- Partee, Barbara H. 1986. Noun Phrase Interpretation and Type-Shifting Principles. In *Studies in Discourse Representation Theory and the Theory of Generalized Quantifiers*, eds. Jeroen Groenendijk and Martin Stokhof, 115-143. Dordrecht: Foris.

- Partee, Barbara. 1989. Binding Implicit Variables in Quantified Contexts. Paper presented at *Papers from the 25th Regional Meeting of the Chicago Linguistic Society. Part One: The General Session*.
- Partee, Barbara. 1997. Genitives: A case study. In *Handbook of Logic and Language*, eds. Johan van Benthem and Alice ter Meulen, 464 - 470. Amsterdam / Cambridge, Mass.: Elsevier/The MIT Press.
- Partee, Barbara H., and Borschev, Vladimir. 1998. Integrating lexical and formal semantics: Genitives, relational nouns, and type shifting. In *Proceedings of the Second Tbilisi Symposium on Language, Logic and Computation*, eds. R. Cooper and T. Gamkrelidze, 229-241. Tbilisi: Center on Language, Logic, Speech, Tbilisi State University.
- Paul, Hermann. 1919. *Deutsche Grammatik*. vol. III, Teil IV: Syntax (Erste Hälfte). Halle an der Saale: Max Niemeyer.
- Paul, Hermann. 1920. *Deutsche Grammatik*. vol. Band 5, Wortbildungslehre. Halle (Saale): Max Niemeyer.
- Payne, John. 1979. Transitivity and Intransitivity in the Iranian Languages of the U.S.S.R. Paper presented at *Conference on Non-Slavic Languages of the USSR*, University of Chicago.
- Payne, Doris L., and Payne, Thomas E. 1990. Yagua. In *Handbook of Amazonian Languages, Volume 2*, eds. Desmond C. Derbyshire and Geoffrey K. Pullum, 249-474. Berlin & New York: Mouton de Gruyter.
- Perlmutter, David. 1978. Impersonal Passives and the Unaccusative Hypothesis, 157-189: Berkeley Linguistic Society IV, University of California.
- Perlmutter, David M. 1983. Personal vs. Impersonal Constructions. *Natural Language and Linguistic Theory* 1:141-200.
- Perlmutter, David M. 1984. Working 1s and Inversion in Italian, Japanese, and Quechua. In *Studies in Relational Grammar 2*, eds. David M. Perlmutter and Carol G. Rosen, 292-330. Chicago: The University of Chicago Press.
- Perlmutter, David M., and Postal, Paul M. 1984. Impersonal Passives and Some Relational Laws. In *Studies in Relational Grammar 2*, eds. David M. Perlmutter and Carol G. Rosen. Chicago: The University of Chicago Press.
- Pesetsky, David. 1982. Paths and Categories, MIT: PhD.
- Pesetsky, David. 1985. Morphology and Logical Form. *Linguistic Inquiry* 16:193-248.
- Pesetsky, David. 1989. Language-particular processes and the Earliness Principle: ms., MIT.
- Pesetsky, David. 1990. Experiencer Predicates and Universal Alignment Principles. *MIT manuscript*.
- Pesetsky, David. 1995. *Zero Syntax*. Cambridge/Mass.: MIT Press.

- Pesetsky, David, and Torrego, Esther. 2000. The Syntax of Tense and the Nature of Case. Paper presented at *International Round Table 'The Syntax of Tense and Aspect'*, Université de Paris 7.
- Pesetsky, David, and Torrego, Esther. 2001. T-to-C Movement. In *Ken Hale: A Life in Language*, ed. Michael Kenstowicz, 355-426. Cambridge/Mass.: The MIT Press.
- Phillips, Colin. 1996. Order and Structure, MIT: Ph. D. Dissertation.
- Pinker, Steven. 1989. *Learnability and Cognition. The Acquisition of Argument Structure*. Cambridge, Massachusetts: The MIT Press.
- Pinker, Steven. 1999. *Words and Rules. The Ingredients of Language*. New York: Perennial.
- Piñon, Christopher. 1999. Durative Adverbials for result states. *Paper presented at WCCFL 18, University of Arizona at Tucson, 1999*.
- Pollock, Jean-Yves. 1989. Verb Movement, UG and the Structure of IP. *Linguistic Inquiry* 20:365-424.
- Portner, Paul. 1992. Situation Theory and the Semantics of Propositional Expressions, University of Massachusetts: PhD.
- Postal, Paul. 1971. *Cross-Over Phenomena*. New York: Holt, Rinehart and Winston.
- Postal, Paul M. 1986. *Studies of Passive Clauses*. Albany: State University of New York Press.
- Press, Ian. 1986. *A Grammar of Modern Breton*. Berlin: Mouton de Gruyter.
- Proudfoot, Anna, and Cardo, Francesco. 1997. *Modern Italian Grammar*. London: Routledge.
- Pütz, Herbert. 1975. *über die Syntax der Pronominalform 'es' im modernen Deutsch*: Studien zur deutschen Grammatik 3. Tübingen: Gunter Narr.
- Pylkkänen, Liina. 2002. Introducing Arguments, Department of Linguistics, MIT.
- Ramchand, Gillian. 1997. *Aspect and Predication. The Semantics of Argument Structure*. Oxford: Clarendon Press.
- Rapp, Irene. 1997. *Partizipen und semantische Struktur. Zu passivischen Konstruktionen des 3. Status*. Tübingen: Stauffenburg Verlag.
- Rappaport, Malka, and Levin, Beth. 1986. What to do with Theta Roles: Lexicon Project Working Papers 11, MIT.
- Rappaport-Hovav, Malka, and Levin, Beth. 1998. Building Verb Meanings. In *The Projection of Arguments: Lexical and Compositional Factors*, eds. Miriam Butt and Willi Geuder, 97-134. Stanford: CSLI.
- Reinhart, Tanya. 1976. The Syntactic Domain of Anaphora, MIT: Ph.D.
- Reinhart, Tanya. 1987. Specifier and Operator Binding. In *The Representation of (In)definiteness*, eds. Eric J. Reuland and Alice G.B. ter Meulen, 130-167. Cambridge/Mass.: MIT Press.
- Reinhart, Tanya, and Reuland, Eric. 1993. Reflexivity. *Linguistic Inquiry* 4:657-720.
- Reinhart, Tanya. 1995. Quantifier Scope: Tel Aviv University.

- Reis, Marga. 1976. Reflexivierung in deutschen A.c.I. - Konstruktionen. Ein transformationsgrammatisches Dilemma. *Papiere zur Linguistik* 9:5 - 82.
- Rescher, Nicholas ed. 1967. *The Logic of Decision and Action*. Pittsburgh: University of Pittsburgh Press.
- Riemschneider, Kaspar K. 1984. *Lehrbuch des Akkadischen*. Leipzig: VEB Verlag Enzyklopädie.
- Riemsdijk, Henk van. 1989. Movement and Regeneration. In *Linguistic Theory and Dialectology*, ed. P. Bennincà. Dordrecht: Foris.
- Riemsdijk, Henk van ed. 1999. *Clitics in the Languages of Europe*. Berlin / New York: Mouton de Gruyter.
- Rivero, María - Luisa. 1990. The Location of Non-Active Voice in Albanian and Modern Greek. *Linguistic Inquiry* 21:135 -146.
- Rivero, María - Luisa. 1992. Adverb Incorporation and the Syntax of Adverbs in Modern Greek. *Linguistics and Philosophy* 15:289-331.
- Rizzi, Luigi. 1986. On Chain Formation. In *Syntax and Semantics 19: The Syntax of Pronominal Clitics*, ed. Hagit Borer, 65-95. Orlando: Academic Press.
- Rizzi, Luigi. 1986. Null Objects in Italian and the Theory of *pro*. *Linguistic Inquiry* 17.
- Rizzi, Luigi. 1990. *Relativized Minimality*. Cambridge/Mass.: The MIT Press.
- Roberts, Craige. 1987. Modal Subordination, Anaphora, and Distributivity, UMass: Ph.D.
- Roberts, Ian G. 1987. *The Representation of Implicit and Dethematized Subjects*: Linguistic Models. Dordrecht: Foris.
- Roberts, Craige. 1990. *Modal Subordination, Anaphora, and Distributivity*. New York: Garland.
- Roeper, Thomas, and Siegel, Muffy E. A. 1978. A Lexical Transformation for Verbal Compounds. *Linguistic Inquiry* 9:199 - 260.
- Roeper, Thomas. 1983. Implicit Arguments and the Lexicon: University of Massachusetts at Amherst.
- Roeper, Thomas. 1987. Implicit Arguments and the Head-Complement Relation. *Linguistic Inquiry* 18:267-310.
- Roeper, Thomas. to appear. Explicit Syntax in the Lexicon: the Representation of Nominalizations. In *Semantics and the Lexicon*, ed. James Pustejovsky. Dordrecht: Kluwer Academic Publishers.
- Rooryck, Johan, and Wyngaerd, Guido Vanden. 1998. The Self as Other. A minimalist approach to *zich* and *zichzelf* in Dutch. Paper presented at *NELS 28*.
- Rooth, Mats E. 1985. Association with Focus, University of Massachusetts at Amherst: Ph.D.
- Rooth, Mats. 1995. Indefinites, Adverb of Quantification and Focus Semantics. In *The Generic Book*, eds. Gregory N. Carlson and Francis Jeffry Pelletier. Chicago: Chicago University Press.
- Rosen, Carol G. 1984. The Interface between Semantic Roles and Initial

- Grammatical Relations. In *Studies in Relational Grammar 2*, eds. David M. Perlmutter and Carol G. Rosen, 38-77. Chicago and London: The University of Chicago Press.
- Rosen, Carol, and Wali, Kashi. 1989. Twin Passives, Inversion and Multistratality in Marathi. *Natural Language & Linguistic Theory* 7:1-50.
- Rosen, Carol. 1990. Rethinking Southern Tiwa: The Geometry of a Triple-Agreement Language. *Language* 66:669-713.
- Ross, John R. 1972. Act. In *Semantics of Natural Language*, eds. Donald Davidson and Gilbert Harman, 70-126. Dordrecht: D. Reidel Publishing Company.
- Rothstein, Susan ed. 1998. *Events in Grammar*. Dordrecht: Kluwer Academic Publishers.
- Rothstein, Susan. 1999. Fine-Grained Structure in the Eventuality Domain: The Semantics of Predicative Adjective Phrases and *Be*. *Natural Language Semantics* 7:347-420.
- Rothstein, Susan. 2001. *Predicates and their Subjects*. Dordrecht: Kluwer Academic Publishers.
- Roux, Cecile le. 1988. On the Interface of Morphology and Syntax, Stellenbosch Papers in Linguistics 18: Ph.D.
- Rullmann, Hotze. 1989. Indefinite Subjects in Dutch. In *Papers on Quantification*, eds. Emmon Bach, Angelika Kratzer and Barbara Partee. University of Massachusetts at Amherst: NSF Grant BNS 8719999 Report.
- Ruys, Eduard G. 1992. The Scope of Indefinites, University of Utrecht: Ph.D.
- Saarinen, Esa. 1979. Backwards-Looking Operators in Tense Logic and in Natural Language. In *Essays on Mathematical and Philosophical Logic*, eds. Jaakko Hintikka, Ilkka Niiniluoto and Esa Saarinen. Dordrecht: Reidel.
- Safir, Ken. 1991. Evaluative Predicates and the Representation of Implicit Arguments. In *Principles and Parameters in Comparative Grammar*, ed. Robert Freidin, 99 - 131. Cambridge/Mass.: The MIT Press.
- Safir, Ken. 1992? The Syntactic Projection of Lexical Thematic Structure. *Natural Language and Linguistic Theory* 5:561-601.
- Sasse, Hans-Jürgen. 1987. The thematic/categorical distinction revisited. *Linguistics* 25:511-580.
- Sauerland, Uli. 1998. Plurals, Derived Predicates and Reciprocals. In *The Interpretative Tract*, eds. Uli Sauerland and Orin Percus, 177-204. Cambridge/Mass.: MIT Working Papers in Semantics.
- Scha, Remko. 1981. Distributive, Collective and Cumulative Quantification. In *Formal Methods in the Study of Language, Part 2, Mathematical Center Tracts 136*, eds. Jeroen Groenendijk, Theo Janssen and Martin Stokhoff, 483-512. Amsterdam: University of Amsterdam.

- Scha, Remko J.H. 1984. Distributive, Collective and Cumulative Quantification. In *Truth, Interpretation, and Information*, eds. Jeroen Groenendijk, Theo Janssen and Martin Stokhof, 131-158. Dordrecht: Foris.
- Scha, R., and Stallard, D. 1988. Multi-Level Plurals and Distributivity. Paper presented at *26th Meeting of the ACL*, Buffalo, N.Y.
- Schachter, Paul. 1976. The Subject in Philippine Languages: Topic, Actor, Actor-Topic, or None of the Above? In *Subject and Topic*, ed. Charles N. Li, 491- 518. New York San Francisco London: Academic Press.
- Schafer, Robin. 1995. Negation and Verb Second in Breton. *Natural Language and Linguistic Theory* 13:135 - 172.
- Schein, Barry. 1986. Event Logic and the Interpretation of Plurals, MIT Ph.D. dissertation.
- Schein, Barry. 1993. *Plurals and Events*. Cambridge/Mass.: MIT Press.
- Schiebe, Traugott. 1972. "Man". Eine grammatische Skizze. In *Germanistische Beiträge. Gert Melbourn zum 60. Geburtstag am 21. 5. 1972, dargebracht von Kollegen und Schülern des Deutschen Instituts der Universität Stockholm.*, 303-313.
- Schönfinkel, Moses. 1924. Über die Bausteine der mathematischen Logik. *Mathematische Annalen* 92:305-316.
- Schwarzschild, Roger. 1990. Against Groups. In *Proceedings of the Seventh Amsterdam Colloquium*, eds. M. Stokhof and L. Torenvliet, 475-493. Amsterdam: Institute for Logic, Language and Information, Universiteit van Amsterdam.
- Schwarzschild, Roger. 1991. On the Meaning of Definite Plural Noun Phrases, University of Massachusetts: Ph.D.
- Schwarzschild, Roger. 1992. Types of Plural Individuals. *Linguistic and Philosophy* 15:641-676.
- Schwarzschild, Roger. 1993-94. Plurals, Presuppositions, and Sources of Distributivity. *Natural Language Semantics* 2:201-248.
- Schwarzschild, Roger. 1996. *Pluralities*. Dordrecht: Kluwer Academic Publishers.
- Schwarzschild, Roger. 2000. Singleton Indefinites. Ms., *Rutgers University*.
- Schwyzer, Eduard, and Debrunner, Albert. 1950. *Griechische Grammatik auf der Grundlage von Karl Brugmanns Griechischer Grammatik*: Handbuch der Altertumswissenschaft. München: C.H. Beck'sche Verlagsbuchhandlung.
- Schwyzer, Eduard. 1950 (2nd edition). *Griechische Grammatik. Auf der Grundlage von Karl Brugmanns Griechischer Grammatik*.vol. 2, Syntax und Syntaktische Stilistik. München: C.H. Beck'sche Verlagsbuchhandlung.
- Sciullo, Anna-Maria Di, and Williams, Edwin. 1987. *On the Definition of Word*. Cambridge/Mass.: The MIT Press.

- Searle, John R. 1994. Intentionality (1). In *A Companion to the Philosophy of Mind*, ed. Samuel Guttenplan, 379-386. Oxford: Basil Blackwell.
- Sebba, Mark. 1987. *The Syntax of Serial Verbs*: Creole Language Library, Volume 2. Amsterdam: John Benjamins Publishing Company.
- Seiter, William J. 1980. *Studies in Niuean Syntax*: Outstanding Dissertations in Linguistics. New York & London: Garland Publishing, Inc.
- Selkirk, Elisabeth O. 1982. *The Syntax of Words*: Linguistic Inquiry Monographs. Cambridge/Mass.: MIT Press.
- Sher, Gila. 1990. Ways of Branching Quantifiers. *Linguistics and Philosophy* 13(4):393-422.
- Shibatani, Masayoshi ed. 1988. *Passive and Voice. Typological Studies in Language 16*. Amsterdam: John Benjamins Publishing Company.
- Shibatani, Masayoshi. 1990. *The languages of Japan*: Cambridge Language Surveys. Cambridge: Cambridge University Press.
- Shlonsky, Ur. 1987. Null and Displaced Subjects, MIT: Ph.D. dissertation.
- Shlonsky, Ur. 1990. Pro in Hebrew Subject Inversion. *Linguistic Inquiry* 21:263 - 275.
- Siewierska, Anna. 1984. *The Passive. A Comparative Linguistic Analysis*.: Croom Helm Linguistics Series. London Sydney Dover, New Hampshire: Croom Helm.
- Sigurdhsson, Halldór Arman. 1991. Icelandic Case-Marked PRO and the Licensing of Lexical Arguments. *Natural Language & Linguistic Theory* 9:327-363.
- Sihler, Andrew L. 1995. *New Comparative Grammar of Greek and Latin*. New York Oxford: Oxford University Press.
- Simpson, Jane. 1983. Resultatives. In *Papers in Lexical Functional Grammar*, eds. L. Levin, M. Rappaport and A. Zaenen. Bloomington: Indiana University Linguistics Club.
- Smith, Carlota S. 1972. On Causative Verbs and Derived Nominals in English. *Linguistic Inquiry* 3:136-138.
- Smith, Carlota. 1983. A Theory of Aspectual Choice. *Language* 59:479-501.
- Smith, Carlota S. 1991. *The Parameter of Aspect*: Studies in Linguistics and Philosophy. Dordrecht: Kluwer Academic Publishers.
- Smyth, Herbert Weir. 1984. *Greek Grammar*. Cambridge/Massachusetts: Harvard University Press.
- Snyder, William. 2001. On the Nature of Syntactic Variation: Evidence from Complex Predicates and Complex Word-Formation. *Language* 77:324 - 342.
- Speas, Margaret J. 1990. *Phrase Structure in Natural Language*: Studies in Natural Language and Linguistic Theory. Dordrecht: Kluwer Academic Publishers.

- Speas, Margaret. 1997. Optimality Theory and Syntax: Null Pronouns and Control. In *Optimality Theory. An Overview*, eds. Diana Archangeli and D. Terence Langendoen, 171 - 199. Oxford: Blackwell Publishers.
- Spelke, Elizabeth S. 2003. What makes us so smart? Core Knowledge and Natural Language. In *Language in Mind. Advances in the Study of Language and Thought.*, eds. Dedre Gentner and Susan Goldin-Meadow, 277-311. Cambridge/Mass.: The MIT Press.
- Sportiche, Dominique. 1988. Conditions on Silent Categories: UCLA.
- Sridhar, S. N. 1976. Dative Subjects, Rule Government, and Relational Grammar. *Studies in the Linguistic Sciences* 6:130-150.
- Sridhar, S. N. 1979. Dative Subjects and the Notion of Subjects. *Lingua* 49:1-28.
- Stechow, Arnim von. 1991. Syntax und Semantik. In *Semantik / Semantics. Ein internationales Handbuch der zeitgenössischen Forschung. An International Handbook of Contemporary Research*, eds. Arnim von Stechow and Dieter Wunderlich, 90 -148. Berlin: de Gruyter.
- Stechow, Arnim von. 1996. The Different Readings of *Wieder 'Again'*: A Structural Account. *Journal of Semantics* 13:87-138.
- Sternefeld, Wolfgang. 1998. Reciprocity and Cumulative Interpretation. *Natural Language Semantics* 6:303-337.
- Stiebels, Barbara. 1996. *Lexikalische Argumente und Adjunkte. Zum semantischen Beitrag von verbalen Präfixen und Partikeln*. Berlin: Akademie Verlag.
- Stowell, Tim. 1978. What was there before there was there. Paper presented at *Papers from the Fourteenth Regional Meeting of the Chicago Linguistic Society*, Chicago.
- Stowell, Tim. 1983. Subjects Across Categories. *The Linguistic Review* 2(3):285-312.
- Stowell, Tim. 1991. The Alignment of Arguments in Adjective Phrases, ed. Susan Rothstein.
- Streitberg, Wilhelm. 1891. Perfective und imperfective Actionsart im Germanischen. *Beiträge zur Geschichte der deutschen Sprache und Literatur* 15:70-177.
- Streitberg, Wilhelm. 1920 (6th edition, reprinted 1981). *Gotische Syntax*. Heidelberg: Carl Winter Universitätsverlag.
- Stump, Gregory T. 1984. Agreement vs. Incorporation in Breton. *Natural Language and Linguistic Theory* 2:289-348.
- Swart, Henriette de. 1991. Adverbs of Quantification: A Generalized Quantifier Approach., University of Groningen Ph.D. Dissertation.
- Sybesma, Rint. 1992. Causatives and accomplishments: The case of Chinese *ba*, University of Leiden Ph.D. dissertation.
- Szabolcsi, Anna. 1983. The Possessor that Ran Away from Home. *The Linguistic Review*:89-102.

- Takezawa, Koichi. 1984. Perfective *have* and the Bar Notation. *Linguistic Inquiry* 15:675-687.
- Taub, Alison. 1989. Collective Predicates, Aktionsarten and All. In *Papers on Quantification*, eds. Emmon Bach, Angelika Kratzer and Barbara Partee. NSF Report, Department of Linguistics, University of Massachusetts at Amherst.
- Tenny, Carol Lee. 1987. Grammaticalizing Aspect and Affectedness, Massachusetts Institute of Technology: Ph.D.
- Terada, Michiko. 1990. Incorporation and Argument Structure In Japanese, University of Massachusetts: Ph.D.
- Thraínsson, Höskuldur. 1986. V1, V2, V3 in Icelandic. In *Verb Second Phenomena in Germanic Languages*, eds. Hubert Haider and Martin Prinzhorn, 169-201. Dordrecht: Foris.
- Timberlake, Alan. 1976. Subject Properties in the North Russian Passive. In *Subject and Topic*, ed. Charles N. Li, 545-570. New York San Francisco London: Academic Press.
- Toman, Jindrich. 1986. A (Word -) Syntax for Participles. *Linguistische Berichte*:367 - 408.
- Tomasello, Michael. 1999. *The Cultural Origins of Human Cognition*. Cambridge/Mass.: Harvard University Press.
- Travis, Lisa. 1984. Parameters and Effects of Word Order Variation, MIT: Ph.D.
- Truckenbrodt, Hubert. 1992. Clitics and the Chain Condition. Cambridge ? Mass.: MIT.
- Tunstall, Susanne L. in progress. Interpreting Quantifiers: Semantics and Processing, University of Massachusetts: Ph. D.
- Ullmer-Ehrich, Veronika. 1977. *Zur Syntax und Semantik von Substantivierungen im Deutschen*. Kronberg: Scriptor.
- Urbina, Jon Ortiz de. 1989. *Parameters in the Grammar of Basque*: Studies in Generative Grammar. Dordrecht: Foris.
- Vainikka, Anne M. 1989. Deriving Syntactic Representations in Finnish, University of Massachusetts: Ph.D.
- Vainikka, Anne, and Levy, Yonata. 1999. Empty Subjects in Finnish and Hebrew. *Natural Language and Linguistic Theory* 17:613 - 671.
- Vendler, Zeno. 1957. Verbs and Times. *The Philosophical Review* 66:143-160.
- Vendler, Zeno. 1967. *Linguistics in Philosophy*. Ithaca: Cornell University Press.
- Vendler, Zeno. 1984. Agency and Causation. *Midwest Studies in Philosophy* 9:371-384.
- Verkuyl, Henk J. 1972. *On the Compositional Nature of the Aspects*: Foundations of Language, Supplementary Series, Vol. 15. Dordrecht: D. Reidel Publishing Co.
- Vlach, Frank. 1973. 'Now' and 'Then', a Formal Study of Tense Anaphora, UCLA: Ph.D.

- Vlach, Frank. 1983. On Situation Semantics for Perception. *Synthese* 54:129-152.
- Volterra, Virginia. 1976. A Few Remarks on the Use of the Past Participle in Child Language. *Italian Linguistics* 2:149-157.
- Voorst, Jan van. 1988. *Event Structure: Amsterdam Studies in the Theory and History of Linguistic Science*. Amsterdam Philadelphia: John Benjamins Publishing Company.
- Wackernagel, Jacob. 1957 (1926). *Vorlesungen über Syntax mit besonderer Berücksichtigung von Griechisch, Lateinisch und Deutsch. Zweite Reihe*. Basel: Birkhäuser Verlag.
- Wackernagel, Jacob. 1981 (1926, 1950). *Vorlesungen über Syntax mit besonderer Berücksichtigung von Griechisch, Lateinisch und Deutsch. Erste Reihe*. Basel Boston Stuttgart: Birkhäuser Verlag.
- Wasow, Thomas, and Roeper, Thomas. 1972. On the Subjects of Gerunds. *Foundations of Language* 8.
- Wasow, Thomas. 1977. Transformations and the Lexicon. In *Formal Syntax*, eds. Peter Culicover, Thomas Wasow and Adrian Akmajian, 327-360. New York: Academic Press.
- Webelhuth, Gert. 1986. Some Data on the Verb-Object Relation in German. *Linguistic Inquiry* 17:772-776.
- Webelhuth, Gert. 1989. Syntactic Saturation Phenomena and the Modern Germanic Languages, University of Massachusetts at Amherst: Ph.D.
- Wechsler, Stephen. 1995. *The Semantic Basis of Argument Structure*. Stanford: CSLI Publications.
- Wehrli, Eric. 1986. On Some Properties of Clitic *se*. In *Syntax and Semantics 19. The Syntax of Pronominal Clitics*, ed. Hagit Borer, 263 - 283. Orlando: Academic Press.
- Wiese, Richard. 1996. *The Phonology of German*. Oxford: Clarendon Press.
- Wilkins, Wendy ed. 1988. *Syntax and Semantics 21: Thematic Relations*. San Diego: Academic Press.
- Williams, Esdwin. 1974. Rule Ordering in Syntax, Massachusetts Institute of Technology: Ph.D.
- Williams, Edwin. 1975. Small Clauses in English. In *Syntax and Semantics 4*, ed. John Kimball, 249-273. New York: Academic Press.
- Williams, Edwin. 1980. Predication. *Linguistic Inquiry* 11:203-238.
- Williams, Edwin. 1981. Argument Structure and Morphology. *The Linguistic Review* 1:81-114.
- Williams, Edwin. 1981. On the Notions 'Lexically Related' and 'Head of a Word'. *Linguistic Inquiry* 12:245-274.
- Williams, Edwin. 1985. PRO and Subject of NP. *Natural Language and Linguistic Theory* 3:297-315.
- Williams, Edwin. 1987. Implicit Arguments, the Binding Theory, and Control. *Natural Language and Linguistic Theory* 5.

- Williams, Edwin. 1994. *Thematic Structure in Syntax*. Cambridge/Mass.: The MIT Press.
- Wilmanns, W. 1899. *Deutsche Grammatik. Zweite Abteilung: Wortbildung*. Berlin und Leipzig: Walter de Gruyter.
- Wilmanns, W. 1906. *Deutsche Grammatik. Dritte Abteilung: Flexion. 1. Hälfte: Verbum*. Strassburg: Verlag von Karl J. Trübner.
- Winter, Yoad. 1998. Flexible Boolean Semantics. Coordination, Plurality and Scope in Natural Language, Universiteit Utrecht: Ph.D.
- Winter, Yoad. 2000. Distributivity and Dependency. *Natural Language Semantics* 8:27-69.
- Wunderlich, Dieter. 1997. Cause and the Structure of Verbs. *Linguistic Inquiry* 28:27-68.
- Wurzel, Ulrich. 1990. The Mechanism of Inflection: Lexicon Representations, Rules, and Irregularities. In *Contemporary Morphology*, eds. Wolfgang U. Dressler, Hans C. Luschützky, Oskar E. Pfeiffer and John Rennison, 203 - 216. Berlin: de Gruyter.
- Zabbal, Youri. 2002. The Semantics of Number in the Arabic Noun Phrase, Department of Linguistics, University of Calgary: M.A.
- Zagona, Karen. 1988. *Verb Phrase Syntax: A Parametric Study of English and Spanish*: Studies in Natural Language and Linguistic Theory. Dordrecht: Kluwer Academic Publishers.
- Zimmermann, Thomas Ede. 1989. Intensional Logic and Two-Sorted Type Theory. *The Journal of Symbolic Logic* 54:65-77.
- Zimmermann, Thomas Ede. 1992. On the Proper Treatment of Opacity in Certain Verbs. *Natural Language Semantics*.
- Zimmermann, Malte. 2000. The Occasional Sailor Construction in English and German. Paper presented at *SALT 10*, Cornell University, Ithaca, New York.
- Zubizarreta, Maria Luisa. 1985. The Relation between Morphophonology and Morphosyntax: The Case of Romance Causatives. *Linguistic Inquiry* 16:247-289.
- Zubizarreta, Maria Luisa. 1987. *Levels of Representation in the Lexicon and in the Syntax*. Dordrecht: Foris.
- Zucchi, Alessandro. 1988. *The Language of Propositions and Events. Issues in the Syntax and Semantics of Nominalization*. University of Massachusetts at Amherst: Ph.D.
- Zucchi, Alessandro. 1993. *The Language of Propositions and Events*. Dordrecht: Kluwer Academic Publishers.
- Zucchi, Alessandro. 1999. Incomplete Events, Intensionality and Perfective Aspect. *Natural Language Semantics* 7(2):179-215.
- Zucchi, Sandro, and White, Michael. 2001. Twigs, Sequences and the Temporal Constitution of Predicates. *Linguistics and Philosophy* 24(2):223-270.